Where the Answer Takes Shape: Shape-Change Peaks Mark Computational Boundaries in LLMs

Anonymous ACL submission

Abstract

Steering, patching and probing act on a chosen layer, yet the layer at which a model's answer becomes controllable is usually found by sweeping. Within a task, we group the residual stream by the correct answer and measure how the shape of the answer-class centroids changes between adjacent layers. Across six instruction-tuned models from five families (7–14B parameters) and ten arithmetic and retrieval tasks, these shapes change little at most layers and sharply at a few, which we call shape-change peaks. These peaks coincide with representational events such as the writing of the answer, and locate three causal boundaries: the handoff of control from the input token to the answer positions, the onset of steering, and the layer at which steering-vector transfer rises most. For each of these boundaries, 71–85% of model–task pairs fall within one layer of a peak, against about 32% for randomly placed peaks ($p < 0.001$). For cyclic arithmetic, a simple model explains why: before the peak, answer-grouped geometry is a blurred image of the inputs. Because the signal needs only answer labels, it locates these boundaries without any intervention.

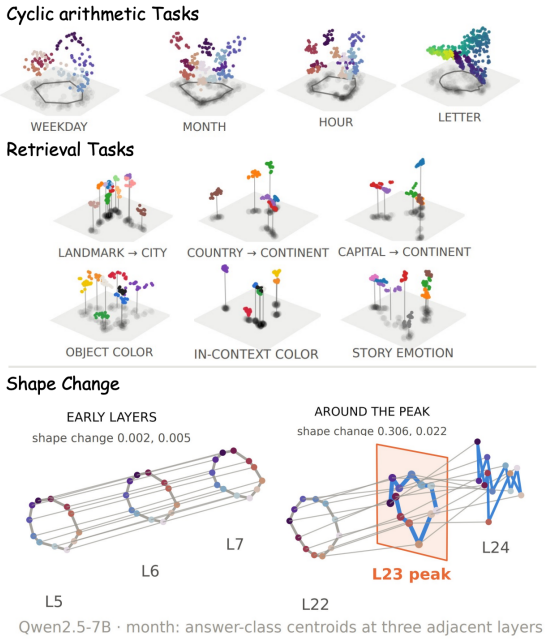


Figure 1: **Ten tasks and the shape-change signal.** Top: answer-position activations of Qwen2.5-7B at the last layer, colored by the correct answer. Bottom: month centroids at three adjacent layers. They barely change early on but reorganize at the shape-change peak (layer 23, $\Delta = 0.306$).

1 Introduction

Most tools for reading and controlling language models act at a single layer. Activation patching swaps hidden states at some chosen layers (Meng et al., 2022; Zhang and Nanda, 2024; Heimersheim and Nanda, 2024).

Activation steering adds a direction to the residual stream at some chosen layers (Turner et al., 2023; Li et al., 2023; Zou et al., 2023; Rimsky et al., 2024). Probes and lenses read representations one layer at a time (Alain and Bengio, 2016; Belinkov, 2022; nostalgebraist, 2020; Belrose et al., 2023). The gap is that these layers are usually found by sweeping (Turner et al., 2023; Rimsky et al., 2024; Tan et al., 2024; Chen et al., 2025), which requires an expensive intervention at each layer; or predicted from how well the hidden state already reads out as the answer (Billa, 2026), which can miss an answer that is already in the representation but not yet readable as an output token.

Representational geometry is a natural place to look for this gap. Concepts in language models are represented as directions, polytopes and low-dimensional manifolds (Park et al., 2024, 2025b; Engels et al., 2025). Such geometry has mostly been studied at a single layer (Hu et al., 2026b,a; Wurgaft et al., 2026). Studies that do compare across layers measure the similarity of representations pooled over all tokens, which typically changes smoothly from layer to layer (Kornblith

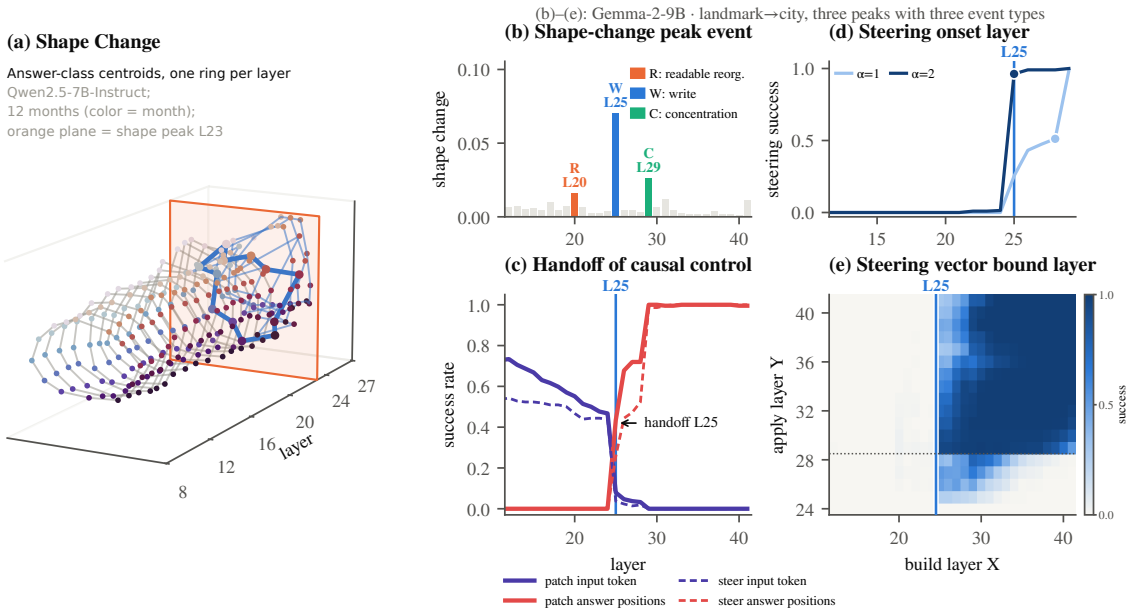


Figure 2: **Shape-change peaks mark causal boundaries.** (a) Qwen2.5-7B, month arithmetic. (b–e) Gemma-2-9B, landmark→city. (b) Adjacent-layer shape change; the three highest peaks are a readable reorganization (R), a write (W) and a concentration (C). (c) Patching (solid) and steering (dashed): control hands off from the input token to the answer positions at the peak. (d) Steering onset (dots) moves to earlier peaks as strength α grows. (e) Vectors built before the write peak fail at every layer (X: build layer, Y: apply layer). Vertical line: the peak (L25).

et al., 2019; Razzhigaev et al., 2024a; Bhattacharya and Kolli, 2026); they do not track the shape of the answer-class configuration. We ask both questions at once: *within a single task, how does the geometry of the answer change across layers, and do the largest changes mark where the computation happens?*

For each model and task, we group the residual stream at the answer position by the correct answer, and take the *shape* formed by the answer-class centroids. Between adjacent layers we measure the Procrustes distance between these shapes, which is invariant to translation, rotation, reflection and scale (Williams et al., 2021). This measure of *shape change* needs only answer labels: no probe, no unembedding, and no knowledge of how the task is computed. We apply it to six instruction-tuned models from five families (7–14B parameters) and ten tasks. These shapes change little at most layers and sharply at a few, which we call *shape-change peaks* (Figure 2a,b).

We show that shape-change peaks mark the computational boundaries of a task (Figure 2); Table 1 summarizes the evidence across all model–task pairs. Our contributions are:

- **A label-only signal and what it detects** (Section 3). Adjacent-layer Procrustes shape change of answer-class centroids needs no probe, unembedding or intervention. Its peaks host representational events: most often the layer where the correct answer is written, and in retrieval a reorganization in which the answer is decodable but not yet causal, invisible to the logit lens.

- **Peaks locate three causal boundaries** (Sections 4–6). The handoff of causal control from the input token to the answer positions, the onset of steering, and the layer at which steering-vector transfer rises most all fall within one layer of a peak in 71–85% of pairs, against about 32% for circularly shifted peaks, across five model families and both task families (Table 1).

- **An explanation** (Section 7). A blurred-input model shows why centroid directions fail before a peak in cyclic arithmetic, and its predictions hold qualitatively on the activations.

- **Practical guidance.** The earliest of the three highest peaks is a near lower bound on where steering can work, and steering vectors should be built after the highest peak; the last layer is the most robust choice (mean transfer 0.91).

	Target layer (one per pair)	n	On a peak	Within ±1	Null	p	Mean dist.	Gemma	Llama	Mistral	Qwen-7B	Qwen-14B	Phi-4
§3	Largest rise of the identity probe	41	66%	**85%**	32%	<0.001	0.76	86%	100%	83%	78%	100%	71%
	Largest rise of patching (answer pos.)	59	54%	**81%**	32%	<0.001	0.90	70%	70%	100%	90%	70%	90%
	Largest rise of steering (answer pos.)	59	49%	**83%**	32%	<0.001	0.90	90%	80%	89%	90%	70%	80%
	Largest rise of the logit lens (p(answer))	58	33%	55%	31%	<0.001	2.41	20%	60%	75%	70%	30%	80%
§4	Handoff: input-token patching fails	53	55%	**83%**	31%	<0.001	0.81	67%	89%	88%	100%	78%	78%
§5	Steering onset, $\alpha = 1$	59	39%	**81%**	32%	<0.001	1.07	80%	70%	100%	90%	70%	80%
	Steering onset, $\alpha = 8$	59	36%	**71%**	31%	<0.001	1.25	50%	90%	89%	80%	30%	90%
§6	Largest rise of steering-vector transfer	58	67%	**83%**	32%	<0.001	0.91	78%	80%	100%	100%	70%	70%

Table 1: **Shape-change peaks locate the layer of every boundary we measure.** For each model–task pair we take one target layer and compare it with the nearest of the three highest shape-change peaks. *On a peak* and *Within* ±1: fraction of pairs at distance 0 and at most 1. *Null*: fraction within ±1 when the peaks of each pair are circularly shifted within the late layers; p: share of 10,000 shifts that reach the observed fraction. *Mean dist.*: mean distance to the nearest peak, in layers. Per-model columns give the within-±1 fraction (dark green: $\geq 90\%$; light green: 70–89%; red: $< 70\%$). The logit lens (gray row) is shown for contrast. Rises count only if they exceed 0.1 in one layer (0.15 for the probe), which leaves 41 pairs for the probe.

2 Setup

Models and tasks. We study six instruction-tuned models: Qwen2.5-7B-Instruct (Qwen Team et al., 2024), Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Gemma-2-9B-it (Gemma Team et al., 2024), and two larger models, Qwen2.5-14B-Instruct (Qwen Team et al., 2024) and Phi-4 (14B) (Abdin et al., 2024). Each prompt uses the model's chat template and a system instruction to answer with a single name; we decode greedily. The ten tasks fall into two families. *Cyclic arithmetic* asks for a start item plus an offset $k \in \{1, \ldots, 6\}$, over weekdays, months, clock hours and letters ("What day is three days after Monday?"). *Retrieval* maps a cue to an answer: landmark $\rightarrow$ city, country $\rightarrow$ continent, capital $\rightarrow$ continent (a two-hop question through the country), object $\rightarrow$ color, in-context color ("Grace likes purple. … Anna likes red. … What color does Anna like?"), and story $\rightarrow$ emotion. Every task has three question templates and covers all combinations of start item and offset, or all cues of each class, for 126–468 prompts per task. The model's answer is the answer class with the highest first-token probability. We analyze the 59 model–task pairs on which the model answers above chance; we exclude Mistral on hours (accuracy 0.06). Appendix B lists the cues, templates, accuracies and sample sizes.

Answer geometry and shape change. Let $h_\ell(x) \in \mathbb{R}^d$ be the residual stream after layer ℓ at the last token of prompt x. For each answer class c of a task, we take the centroid μ_ℓ^c of $h_\ell(x)$ over the prompts whose correct answer is c. Stacking the K centroids gives a configuration $C_\ell \in \mathbb{R}^{K \times d}$. We remove its mean and scale it to unit Frobenius norm, giving $\tilde{C}_\ell$. The *shape change* between adjacent layers is the Procrustes disparity

$$\Delta_\ell = \min_{R \in O(d),\, s>0} \left\| s\, \tilde{C}_{\ell-1} R - \tilde{C}_\ell \right\|_F^2 \qquad (1)$$
$$= 1 - \left\| \tilde{C}_{\ell-1}^\top \tilde{C}_\ell \right\|_*^2.$$

where $\| \cdot \|_*$ is the nuclear norm (Williams et al., 2021). Translation, rotation, reflection and scale are removed by construction. These are the components by which a concept's geometry is known to differ across tasks at a fixed layer (Hu et al., 2026b). Δ_ℓ is therefore what remains: the change in the *shape* of the answer class from one layer to the next. It requires only answer labels.

We take the three highest peaks of Δ_ℓ in the late layers (from 25% of depth on) and call the largest the *highest peak* (Appendix C). Interventions, the three boundaries we read off them (the handoff layer, the steering onset and transfer), and the event at each peak are defined in Appendix D.

3 What Happens at a Shape-Change Peak

To interpret a peak, we track two kinds of signals at the answer positions. *Causal* signals include patching and steering, at all answer positions or at the last position only, and on the full space or on the answer-class centroid space only. *Observational* signals include a linear identity probe for the answer, and the logit lens (nostalgebraist, 2020). We interpret a peak by which signals jump on the peak, asking in a fixed order (Figure 3a). It is a **concentration** (C) if subspace patching starts to change an answer that full patching already controlled; a **write** (W) if any other causal signal jumps; a **readable reorganization** (R) if the probe decodes the answer while the causal signals stay near zero; a **vocabulary alignment** (V) if only the logit lens rises; and **no event** (–) otherwise. The same rules apply to every late layer, so peaks can be compared with the layers in between. Appendix D gives all thresholds. The direction of every comparison in Table 2 holds under all 144 settings of the general thresholds and all 135 settings of the rule-specific ones (Appendix C).

3.1 Peaks host representational events

Peaks and the layers between them host different events (Table 2). The highest peak is a write in 61% of all model–task pairs, nearly three times the rate at other late layers (22%), and carries no event in only 8%, against 46%. Only one highest peak (Qwen2.5-14B on object color) is a vocabulary alignment alone: where the shape of answer classes changes most, the model reorganizes, writes, or concentrates the answer, rather than only making it readable in the output vocabulary. The largest rise of the probe, of patching and of steering falls within one layer of a peak in 85%, 81% and 83% of pairs, against 32% for randomly shifted peaks; the largest rise of the logit lens does so in only 55%. In Gemma on landmark→city, the three highest peaks carry three different events: a readable reorganization, a write and a concentration (Figure 2b).

3.2 Some peaks mark the answer before it becomes causal

A linear probe (Alain and Bengio, 2016) can decode the correct answer from a hidden state even when the model does not use it (Belinkov, 2022), whereas patching and steering register it only once it has a causal effect on the output, and the logit lens only once it is aligned with the output vocabulary (Belrose et al., 2023). Readable-reorganization peaks fall in this gap: 38 of the 177 top-3 peaks, in 28 of the 36 retrieval pairs, show a decodable correct answer (median probe accuracy 0.91); at 33 of them patching and steering change at most 5% of prompts and the logit lens assigns a probability below 0.001 (the exceptions are all in the two 14B models, with causal success up to 16% or a logit-lens probability up to 0.03) (e.g., Qwen on landmark→city, layer 19; Figure 3b). A causal effect follows a median of four layers later, and the logit lens six layers later. Only three arithmetic peaks are of this type, all second or third peaks of the 14B models: in retrieval, the probe decodes the correct answer a median of 9.5 layers before any causal signal reaches 0.5, whereas in arithmetic it never does so earlier (23 of 23 pairs). A probe alone shows only that the correct answer is decodable, often over many layers; the shape signal marks the layer at which the configuration of answer classes reorganizes, without training a probe or running an intervention. Section 5 shows that these peaks bound how early steering can work.

	W	R	C	V	–	n
Highest peak	.61	.20	.08	.02	.08	59
2nd/3rd peaks	.35	.22	.09	.07	.27	118
Other layers	.22	.15	.05	.10	.46	1324

Table 2: Events at shape-change peaks and at the other late layers, over 59 model–task pairs. W: write; R: readable reorganization; C: concentration; V: vocabulary alignment only; –: no event.

3.3 One-step and multi-step writing

Peaks also separate two ways of writing the answer. In Qwen, the answer reaches all answer positions at once: patching all of them and patching the last position alone take effect at the same layer in 9 of 10 tasks for Qwen2.5-7B and 8 of 10 for Qwen2.5-14B (Figure 3d). A single peak then marks the write, the handoff and the onset of steering together (Figure 2). In Llama, Mistral and Gemma, and in half of the Phi-4 tasks, the answer is first written at earlier answer positions, including the chat-template tokens, and reaches the last position later (Figure 3e). The last position lags by a median of 2.5 layers in Llama (up to 4), 1 in Mistral (up to 8), 0.5 in Phi-4 (up to 4) and 7 in Gemma (up to 12), and steering vectors in these models need more layers after the highest peak to transfer well

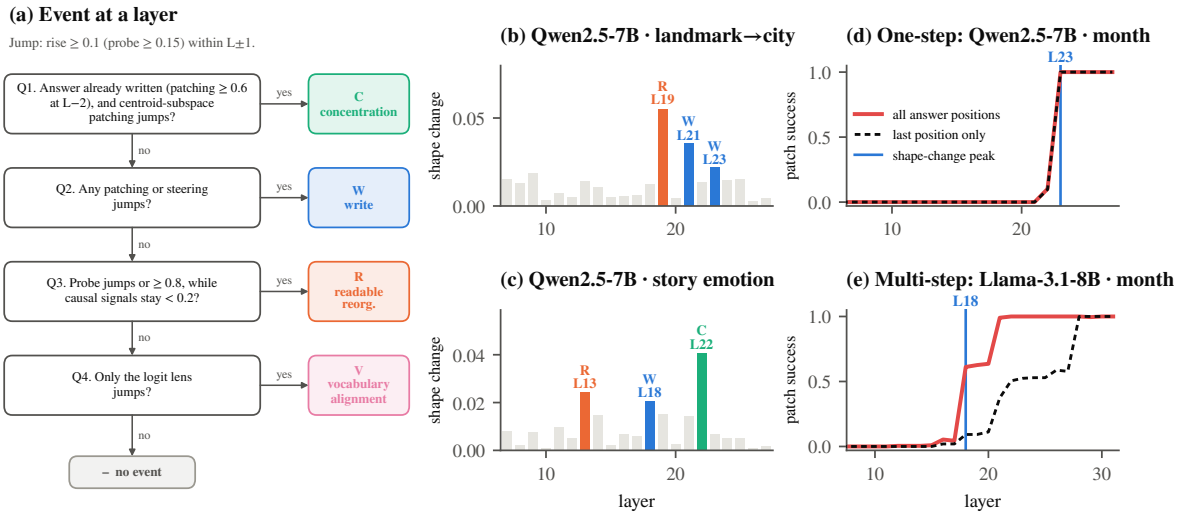


Figure 3: **What happens at a peak.** (a) Each late layer is named by the first question answered yes; thresholds as shown. (b, c) Qwen2.5-7B: adjacent-layer shape change, with the three highest peaks labeled by event type and layer. (d, e) One-step and multi-step writing: patching all answer positions (red) and only the last position (black, dashed). In Qwen the two take effect together at the highest peak; in Llama the last position lags by several layers.

(Section 6).

4 Peaks Mark the Handoff of Causal Control

Before the correct answer becomes causal at the answer positions, it is controlled by the input token that determines it (the start item, or the cue). We locate the layer at which this control passes from the input token to the answer positions. The *handoff layer* is the layer at which patching the input token stops working. We exclude the four emotion-story pairs, whose correct answer depends on the whole story rather than on a single input token, which leaves 53 of the 59 model–task pairs.

4.1 The handoff is abrupt

Control over the correct answer moves from the input token to the answer positions within one layer in 46 of 53 pairs: patching the input token falls from near ceiling to near zero as patching the answer positions rises (Figure 4a,b). Steering places the handoff within one layer of patching in 43 of the 51 pairs where it defines one, and random directions of equal norm change the output in at most 11% of prompts. In the models that write in several steps, steering the answer positions can lag patching by a few layers (Appendix F); Section 7 explains why the centroid direction that steering uses becomes informative only after the peak.

4.2 The handoff lands on a shape-change peak

The handoff layer lies within one layer of one of the three highest peaks in 44 of 53 pairs, and exactly on one in 29, against 16.6 and 5.9 when the peaks are circularly shifted ($p < 0.001$; Figure 4c and Table 1). The alignment holds in all six models and both task families, and uses nothing but the answer labels; in the remaining nine pairs the handoff lies two or three layers from a peak, except in Qwen2.5-14B on object color and in-context color (seven and six layers), where the input loses control gradually (Appendix F). The event at the highest peak tells on which side the handoff falls: a write is the handoff itself (within one layer in 28 of 35 pairs), a readable reorganization precedes it (8 of 8, by a median of 3.5 layers), and a concentration follows it (4 of 4). On Llama-3.1-8B cyclic arithmetic, the highest peak and the handoff match the layers identified by Feucht et al. (2026) and Arora et al. (2026) (Appendix F).

5 Steering Onset Lands on Peaks

Section 4 showed that causal control over the correct answer reaches the answer positions at a shape-change peak. We now ask from which layer steering these positions works, and whether a stronger push moves that layer earlier. Let v_Y be the difference between the target and original answer-class centroids at layer Y, computed on the other two templates. Steering adds αv_Y at every answer po-

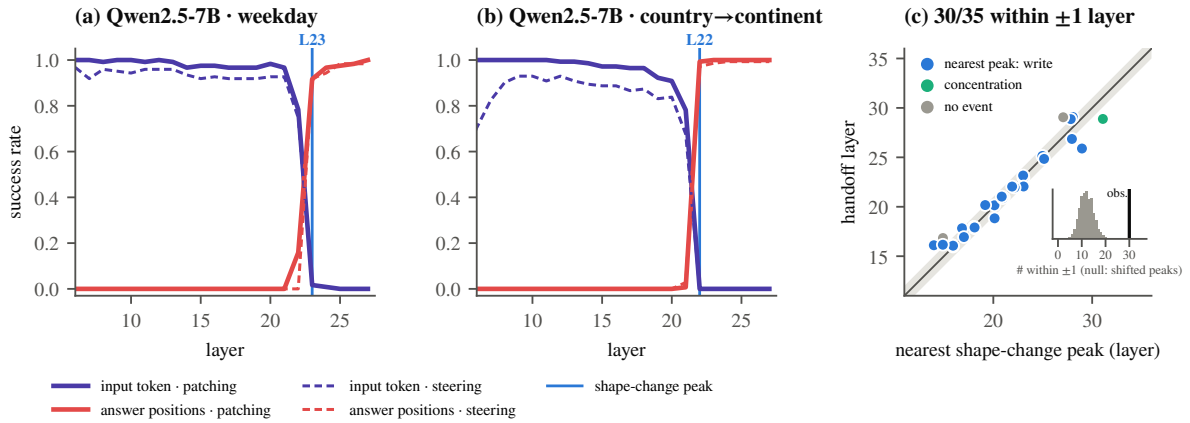


Figure 4: **Causal control hands off at a shape-change peak.** (a, b) Qwen2.5-7B: patching (solid) and steering (dashed) of the input token (violet) and the answer positions (red). Control moves from the input token to the answer positions at the highest peak (vertical line). (c) Handoff layer against the nearest of the three highest peaks for 53 model–task pairs, colored by event type; the gray band marks ±1 layer. Inset: the same count when peaks are circularly shifted.

sition, with $\alpha \in \{1, 2, 4, 8\}$; $\alpha = 1$ moves a state from its class centroid to the target centroid. The onset $o(\alpha)$ is the first layer at which at least half of the prompts switch to the target answer. We write λ_1 for the highest peak, $\lambda_{\min}$ for the earliest of the three highest, and λ_h for the handoff layer. To test $\lambda_{\min}$ as a lower bound as strictly as possible, we take the direction from three layers (Y, the last layer and the best-transferring layer) and report the earliest onset; random directions of equal norm switch at most 11% of prompts (Appendix G).

5.1 A staircase of onsets

At $\alpha = 1$, the onset lies no more than one layer before λ_1 in 56 of 59 pairs, and no more than one layer before $\lambda_{\min}$ in all 59; in all three exceptions the highest peak is not a write (two concentrations in Qwen2.5-7B, and the vocabulary-alignment peak of Qwen2.5-14B on object color), so the correct answer was written earlier. Stronger steering moves the onset earlier, along the peaks: at every α, 71–81% of onsets lie within one layer of one of the three highest peaks, against 32% for circularly shifted peaks ($p < 0.001$; Table 1). In Qwen, arithmetic onsets fall on the highest peak, while retrieval onsets fall on a write peak after the readable reorganization (Figure 5). $\lambda_{\min}$ is a near lower bound: even at $\alpha = 8$, at most 6 of 59 onsets fall more than one layer before it, and only two precede a readable-reorganization peak (2 of 12; Phi-4 on in-context color and Qwen2.5-14B on emotions), where the class structure is already in place for a strong push along v_Y to use. The onset also coincides with the handoff, although the two are measured with different interventions: at $\alpha = 2$, it lies within one layer of λ_h in 41 of 53 pairs, and more than one layer before it only in three pairs of the 14B models (two layers each). Steering the answer positions starts to work where causal control leaves the input token.

5.2 Comparison with logit-lens accessibility

Billa (2026) predict where difference-of-means steering succeeds from the logit-lens accuracy A_{lin}; we compute it as they define it and use the first layer with $A_{\text{lin}} > 0.1$ as a predictor (Table 3). As point predictors, the two signals are comparable: mean absolute errors are 2.5 and 2.1 layers for the onset at $\alpha = 1$, and 2.0 and 2.2 for the handoff; neither difference is significant. As lower bounds they differ. The onset precedes λ_1 by more than one layer in 3 of 59 pairs, but precedes the first layer with $A_{\text{lin}} > 0.1$ in 17 (exact McNemar test, $p = 0.001$); at $\alpha = 8$, $\lambda_{\min}$ is violated 6 times and A_{lin} 52 times. In 23 of 59 pairs, steering already works where $A_{\text{lin}} \leq 0.1$: the class structure is in place but not yet aligned with the output vocabulary, which the shape signal registers and the logit lens does not. The layer of maximal A_{lin} targets the strongest effect rather than the onset and points to the last layers (Appendix H).

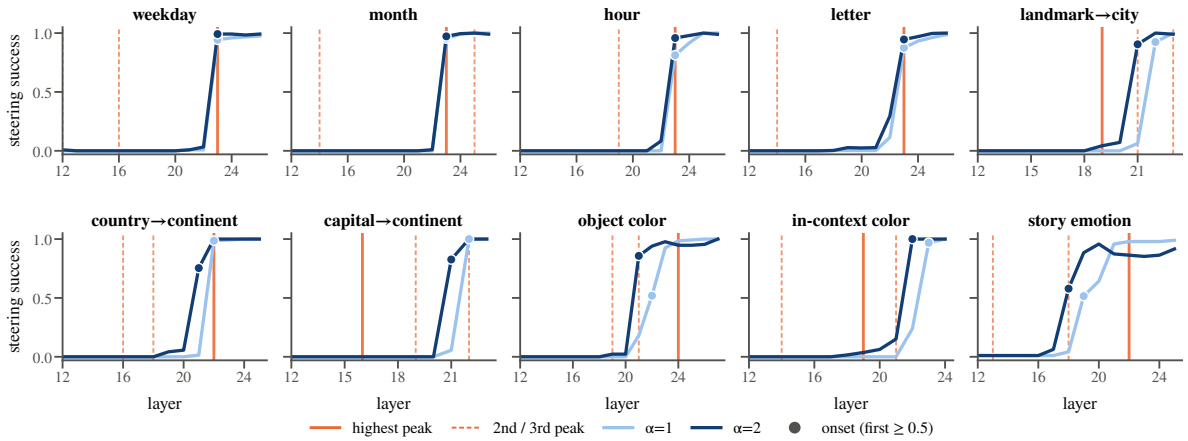


Figure 5: **Steering onset lands on shape-change peaks** (Qwen2.5-7B, all ten tasks). Steering success at the answer positions for $\alpha = 1$ and 2; dots mark the onset (first layer with success ≥ 0.5). Orange lines mark the three highest peaks (solid: highest). At $\alpha = 2$, all ten onsets lie within one layer of a peak.

Predictor	α=1	α=8	Handoff
λ_1	2.5 / 3	**3.0** / 34	**2.0** / 9
$\lambda_{\min}$	6.5 / **0**	3.1 / **6**	5.3 / **2**
A_{lin}>0.1	**2.1** / 17	4.8 / 52	2.2 / 21
$\arg\max A_{\text{lin}}$	9.2 / 57	13.4 / 59	10.3 / 53

Table 3: Shape-change peaks versus logit-lens accessibility (Billa, 2026). Each cell is the mean absolute error in layers / the number of pairs whose boundary lies more than one layer before the predictor; bold marks the best value of each metric. Onsets: 59 pairs; handoff: 53 pairs.

6 Peaks Bound Where Steering Vectors Can Be Built

A centroid-difference vector is often built at one layer and reused at others, and the similarity between such vectors decays with layer distance (Rimsky et al., 2024). We ask from which layer a vector v_X built at layer X carries over. For every pair of layers (X, Y), we rescale v_X to the norm of v_Y and add it at the answer positions of layer Y. Let $s(X \to Y)$ be the success of v_X applied at layer Y, and let $\mathcal{Y} = \{Y : s(Y \to Y) \geq 0.9\}$ be the layers where steering with the layer's own vector works. The transfer of X is

$$T(X) = \frac{1}{|\mathcal{Y}|} \sum_{Y \in \mathcal{Y}} s(X \to Y).$$

We use 58 model–task pairs; Gemma on emotions never reaches 0.9 (Appendix I).

6.1 Vectors built before the peak do not transfer

The highest peak separates vectors that transfer from vectors that do not (Figure 6). Vectors built in the five layers before λ_1 reach a mean transfer of 0.08 (median 0.01); vectors built at or after it reach 0.75, in all six models and both task families. The boundary is not where the correct answer appears: it is linearly decodable 7.6 layers, and its class structure reliable 6.4 layers, before transfer rises, and pre-peak layers with reliable class structure (split-half agreement ≥ 0.9) still transfer at only 0.20, against 0.82 after the peak.

6.2 Transfer rises at peaks

Transfer rises in steps at peaks rather than smoothly with X. Its largest one-layer increase falls exactly on one of the three highest peaks in 39 of 58 pairs, and within one layer of one in 48 (83%, against 32% for circularly shifted peaks, $p < 0.001$; Table 1). These steps fall on write peaks (32 pairs) or concentration peaks (6), and only once on a readable reorganization (Qwen2.5-14B on country→continent), whose class structure is not yet the one the model uses downstream. After λ_1, transfer keeps rising, from 0.39 at the peak to 0.8 a mean of 5.3 layers later; this takes 2.4 layers in Qwen2.5-7B and 3.8 in Qwen2.5-14B, which write in one step, against 4.9–8.1 in the other models (Section 3.3).

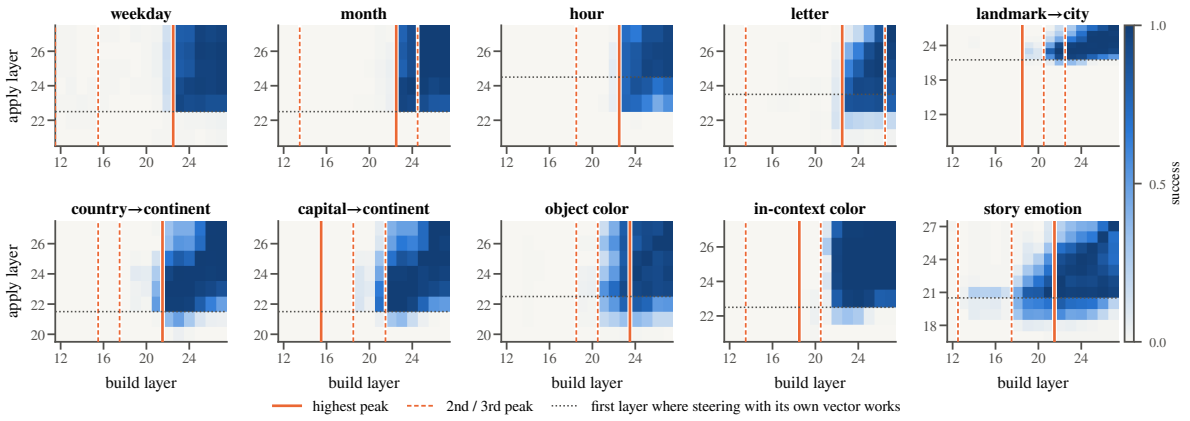


Figure 6: **Steering vectors transfer only when built at or after a shape-change peak** (Qwen2.5-7B, all ten tasks). Success of a vector built at one layer and applied at another. Orange lines mark the three highest peaks (solid: highest); below the dotted line, steering fails even with the layer's own vector. In all ten tasks, transfer rises most within one layer of a peak.

6.3 Transfer decays with cumulative shape change

$\Delta_\ell = \sin^2 \theta_\ell$, where the angular Procrustes distance θ_ℓ is a metric (Williams et al., 2021), so the shape difference between layers X and Y is bounded by $\sum_\ell \theta_\ell$, which grows with the accumulated shape change $\sum_\ell \Delta_\ell$. Accordingly, at a fixed layer distance, transfer decreases with $\sum_\ell \Delta_\ell$ in 54 of 58 pairs (median Spearman $\rho = -0.50$; Appendix I): shape change, not layer distance, governs how far a vector carries.

6.4 Choosing where to build a vector

A steering vector should be built after the highest peak; the last layer is the most robust choice (mean transfer 0.91). The peak is a conservative boundary: vectors built before it essentially never transfer (0.08). The first layer with $A_{\text{lin}} > 0.1$ lies closer to the transfer boundary (mean error 1.6 against 3.2 layers), but vectors built in the five layers before it still transfer at 0.19.

7 Why Centroid Directions Fail Before the Peak

Steering and transfer both fail when the centroid-difference direction is built before the highest peak (Sections 5 and 6). We explain why with a simple model of cyclic arithmetic, and test it on the stored activations.

The model. Take N items, a start item s and an offset k drawn uniformly from a set $\mathcal{K}$, and the correct answer $D = s + k \bmod N$. Before the answer is computed, suppose the state at the answer position is additive in the inputs,

$$h(s, k) = \mu + a(s) + b(k) + \varepsilon. \qquad (2)$$

Prompts that share the answer D have different start items $s = D - k$, so the centroid of class D averages their start-item features:

$$c(D) = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} a(D - k) = (a * p_{\mathcal{K}})(D), \quad (3)$$

the circular convolution of the start-item geometry with the offset distribution. Grouping by answer before the answer exists thus yields a *blurred image of the input*, and a centroid difference moves the representation of the inputs rather than of the answer. Once the answer is computed, a component $u(D)$ organized by the answer itself enters h; when it dominates, centroid differences point along the answer, and steering and transfer work. The switch between the two regimes changes the shape of the answer configuration, so the model predicts a shape-change peak where $u(D)$ takes over. Appendix J gives the derivation and its extension to other tasks.

Evidence. We fit Equation (2) at every layer of 29 cyclic-arithmetic pairs (our six models and four smaller ones; Appendix B) and split each answer centroid into the blurred additive part and the remaining answer-organized part. In the five layers before the highest peak, the additive model explains 83–92% of the variance. Before the peak, the blurred part carries a median of 76% of the

centroid energy; after it, the answer-organized part carries 66%, and the layer where the second overtakes the first lies within one layer of a peak in 19 of the 25 pairs where they cross (Appendix Figure 16). The weekday task also approximately matches the predicted form: with all six nonzero offsets, Equation (3) gives $c(D) = -a(D)/6$, a reversed, shrunken copy of the start-item configuration, and the measured cosine is 0.78 with a size ratio of 0.23 (predicted 1 and 0.17).

In cyclic arithmetic, before the peak, answer-grouped geometry reflects the inputs, so directions built there neither steer nor carry over; the peak marks the layer at which it starts to reflect the answer.

8 Conclusion

Grouping the residual stream by the correct answer and measuring how the shape of the answer-class centroids changes between layers reveals a few sharp peaks rather than smooth drift. Across six models from five families and ten tasks, these peaks mark where the computation happens: the correct answer is most often written there, and in retrieval it can be decodable at a peak before it becomes causal. The handoff of causal control, the onset of steering and the layer at which steering-vector transfer rises most all fall within one layer of a peak in most model–task pairs (Table 1); in cyclic arithmetic, this is because before the peak, answer-grouped geometry is a blurred image of the inputs. Since the signal needs only answer labels, it locates these boundaries without any intervention: steer no earlier than the earliest peak, and build vectors after the highest one.

Limitations

Scope. We study six instruction-tuned models of 7–14B parameters; only Section 7 also uses smaller models (1.5–4B; Appendix B). Our ten tasks have closed answer sets, and the answer is a single word. Models beyond 14B, base models, and open-ended generation remain to be tested. For one further model, Gemma-3-12B, the shape signal was below its measurement noise at every highest peak, so peaks could not be located; we exclude it and report it in Appendix B. One-step writing appears only in the two Qwen models (7B and 14B) among our models, so we report it as an observation about this family rather than a general category.

What a peak tells. A peak locates where the answer geometry reorganizes. Telling which kind of event occurs there (a write, a readable reorganization or a concentration) requires an intervention at that layer. Because the event types are defined with causal signals, we treat the relation between event type and the side of the handoff as a consistent description, and base the main claims on type-free permutation tests. Shape-change peaks give boundaries and lower bounds. For a point estimate of where vectors start to transfer, the first layer with $A_{\text{lin}} > 0.1$ is closer on average. In some pairs, two peaks of similar height compete for the first rank, which is why we state results in terms of the three highest peaks.

Interventions. We steer with centroid differences and patch with matched prompts. Other steering methods, such as sparse-autoencoder features or manifold-based steering, may have different onsets. The handoff is measured on the last token of the start item or cue. Decomposing the input supports this choice, but emotion stories have no single controlling token.

Theory. The blurred-input model is derived and tested for cyclic arithmetic, where the inputs combine additively before the peak. Its weekday prediction holds in form, but a small answer-organized component is already present before the peak. For retrieval tasks we give only the general variance argument.

Ethics Statement

Our method locates the layers at which a model's answer becomes controllable, which can make activation steering more efficient. Steering can be misused to manipulate model outputs or to weaken safety behavior. We study only closed-set factual and arithmetic tasks, release no steering vectors for harmful behavior, and do not evaluate safety-relevant steering. The signal itself is read-only: it requires no intervention and does not change the model. Our experiments use publicly available 7–14B models and modest compute.

Use of AI Assistants

We used an AI assistant (Claude, Anthropic) for three purposes: to turn our ideas and results into fluent English prose, to write and revise code for plots and statistical analyses, and to help monitor automated experiment runs. The research questions,

methods, experimental design and conclusions are our own. We checked every number, figure and claim against our experimental outputs, and we take full responsibility for the content of this paper.

Acknowledgments

References

Marah Abdin, Jyoti Aneja, Harkirat Behl, et al. 2024. Phi-4 technical report. *arXiv preprint arXiv:2412.08905*.

Guillaume Alain and Yoshua Bengio. 2016. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*.

Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, and 8 others. 2025. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*.

Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. 2019. Intrinsic dimension of data representations in deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Georgii Aparin and Tatiana Gaintseva. 2026. A geometric account of activation steering through angle-norm decomposition. *arXiv preprint arXiv:2606.06735*.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Gurbir Arora, Toni J. B. Liu, Jiajun Bao, Raphaël Sarfati, and Christopher J. Earls. 2026. Causal and interpretable structures in LLM compositional tasks. *arXiv preprint arXiv:2609.35970*.

Yu Bai, Heyan Huang, Cesare Spinoso-Di Piano, Marc-Antoine Rondeau, Sanxing Chen, Yang Gao, and Jackie Chi Kit Cheung. 2026. Identifying and analyzing performance-critical tokens in large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Yonatan Belinkov. 2022. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219.

Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. 2023. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*.

Sunit Bhattacharya and Ravi Shankar Kolli. 2026. An analysis of residual-stream geometry across transformer depth. *arXiv preprint arXiv:2607.18348*.

Jayadev Billa. 2026. Predicting where steering vectors succeed. *arXiv preprint arXiv:2604.15557*.

Eden Biran, Daniela Gottesman, Sohee Yang, Mor Geva, and Amir Globerson. 2024. Hopping too late: Exploring the limitations of large language models on multi-hop queries. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14113–14130.

Joschka Braun, Carsten Eickhoff, David Krueger, Seyed Ali Bahrainian, and Dmitrii Krasheninnikov. 2025. Understanding (un)reliability of steering vectors in language models. *arXiv preprint arXiv:2505.22637*.

Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. 2025. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*.

Emily Cheng, Diego Doimo, Corentin Kervadec, Iuri Macocco, Jade Yu, Alessandro Laio, and Marco Baroni. 2025. Emergence of a high-dimensional abstraction phase in language transformers. In *International Conference on Learning Representations (ICLR)*.

Róbert Csordás, Christopher D. Manning, and Christopher Potts. 2025. Do language models use their depth efficiently? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Patrick Queiroz Da Silva, Hari Sethuraman, Dheeraj Rajagopal, Hannaneh Hajishirzi, and Sachin Kumar. 2025. Steering off course: Reliability challenges in steering language models. *arXiv preprint arXiv:2504.04635*.

Quy-Anh Dang and Chris Ngo. 2026. Selective steering: Norm-preserving control through discriminative layer selection. *arXiv preprint arXiv:2601.19375*.

Elad David and Max Fomin. 2026. Context is king: How in-context specification shapes the geometry of concepts. *arXiv preprint arXiv:2607.24425*.

Frances Ding, Jean-Stanislas Denain, and Jacob Steinhardt. 2021. Grounding representation similarity with statistical testing. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Diego Doimo, Alessandro Serra, Alessio Ansuini, and Alberto Cazzaniga. 2024. The representation landscape of few-shot learning and fine-tuning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Joshua Engels, Eric J. Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. 2025. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations (ICLR)*.

Kawin Ethayarajh. 2019. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 55–65.

Javier Ferrando and Elena Voita. 2024. Information flow routes: Automatically interpreting language models at scale. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17432–17445.

Sheridan Feucht, Tal Haklay, Usha Bhalla, Daniel Wurgaft, Can Rager, Raphaël Sarfati, Jack Merullo, Thomas McGrath, Owen Lewis, Ekdeep Singh Lubana, Thomas Fel, and Atticus Geiger. 2026. Arithmetic in the wild: Llama uses base-10 addition to reason about cyclic concepts. *arXiv preprint arXiv:2605.01148*.

Soham Gadgil, Chris Lin, and Su-In Lee. 2026. Where to steer: Input-dependent layer selection for steering improves LLM alignment. *arXiv preprint arXiv:2604.03867*.

Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. 2025. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. 2021. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems*, volume 34.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, et al. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235.

Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. 2024. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 15466–15490.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. 2025. The unreasonable ineffectiveness of the deeper layers. In *International Conference on Learning Representations (ICLR)*.

Wes Gurnee, Emmanuel Ameisen, Isaac Kauvar, Julius Tarng, Adam Pearce, Chris Olah, and Joshua Batson. 2026. When models manipulate manifolds: The geometry of a counting task. *arXiv preprint arXiv:2601.04480*.

Shahar Haim and Daniel C. McNamee. 2026. Emergent causal-geometric dynamics across depth in large language models. *arXiv preprint arXiv:2602.04931*.

Stefan Heimersheim and Neel Nanda. 2024. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*.

Roee Hendel, Mor Geva, and Amir Globerson. 2023. In-context learning creates task vectors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9318–9333.

James Henry. 2026a. The concept allocation zone: Tracking how concepts form across transformer depth. *arXiv preprint arXiv:2605.24856*.

James Henry. 2026b. Geometric evolution maps: Extracting stable concept probes from transformer residual streams. *arXiv preprint arXiv:2605.25848*.

Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2024. Linearity of relation decoding in transformer language models. In *International Conference on Learning Representations (ICLR)*.

Zhimin Hu, Lanhao Niu, and Sashank Varma. 2026a. Language models represent and transform concepts with shared geometry. *arXiv preprint arXiv:2607.04525*.

Zhimin Hu, Lanhao Niu, and Sashank Varma. 2026b. The representational geometry of number. *arXiv preprint arXiv:2602.06843*.

Shawn Im and Sharon Li. 2025. A unified understanding and evaluation of steering methods. *arXiv preprint arXiv:2502.02716*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.

Xinyan Jiang, Wenjing Yu, Di Wang, and Lijie Hu. 2026. Global evolutionary steering: Refining activation steering control via cross-layer consistency. *arXiv preprint arXiv:2603.12298*.

Ole Jorgensen, Dylan Cope, Nandi Schoots, and Murray Shanahan. 2023. Improving activation steering in language models with mean-centring. *arXiv preprint arXiv:2312.03813*.

Abhinav Joshi, Divyanshu Bhatt, and Ashutosh Modi. 2025. Geometry of decision making in language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Subhash Kantamneni and Max Tegmark. 2025. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*.

Dhruva Karkada, Daniel J. Korchinski, Andres Nava, Matthieu Wyart, and Yasaman Bahri. 2026. Symmetry in language statistics shapes the geometry of model representations. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*.

Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. 2008. Representational similarity analysis – connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2:4.

Vedang Lad, Jin Hwa Lee, Wes Gurnee, and Max Tegmark. 2025. The remarkable robustness of LLMs: Stages of inference? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, and 8 others. 2025. On the biology of a large language model. *Transformer Circuits Thread*.

Xukai Liu, Ye Liu, Jipeng Zhang, Yanghai Zhang, Kai Zhang, and Qi Liu. 2026. Layer-order inversion: Rethinking latent multi-hop reasoning in large language models. *arXiv preprint arXiv:2601.03542*.

Gianfranco Lombardo, Giuseppe Trimigno, and Stefano Cagnoni. 2026. A geometric perspective on next-token prediction in large language models: Three emerging phases. *arXiv preprint arXiv:2605.09011*.

Nicole Ma and Nick Rui. 2026. Where's the plan? locating latent planning in language models with lightweight mechanistic interventions. *arXiv preprint arXiv:2605.07984*.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35.

Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. 2024. Language models implement simple Word2Vec-style vector arithmetic. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5030–5047.

Alexander Modell. 2026. Probing for representation manifolds in superposition. *arXiv preprint arXiv:2605.18537*.

Alexander Modell, Patrick Rubin-Delanchy, and Nick Whiteley. 2025. The origins of representation manifolds in large language models. *arXiv preprint arXiv:2505.18235*.

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. 2023. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*.

Yaniv Nikankin, Anja Reusch, Aaron Mueller, and Yonatan Belinkov. 2025. Arithmetic without algorithms: Language models solve math with a bag of heuristics. In *International Conference on Learning Representations*.

nostalgebraist. 2020. interpreting GPT: the logit lens. LessWrong.

Narmeen Oozeer, Shivam Raval, Philip Quirke, Manikandan Ravikiran, Jeff Phillips, Shriyash Upadhyay, and Amirali Abdullah. 2026. Riemannian-manifold steering: Geometry-aware generative autoencoders for label-free steering. *arXiv preprint arXiv:2605.24942*.

Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, and Hidenori Tanaka. 2025a. ICLR: In-context learning of representations. In *International Conference on Learning Representations (ICLR)*.

Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. 2025b. The geometry of categorical and hierarchical concepts in large language models. In *International Conference on Learning Representations (ICLR)*.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2024. The linear representation hypothesis and the geometry of large language models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Aditya Pola and Vineeth N. Balasubramanian. 2025. Where does an LLM begin computing an instruction? *arXiv preprint arXiv:2511.10694*.

Qwen Team, An Yang, Baosong Yang, Beichen Zhang, et al. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. 2017. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems (NIPS)*.

Anton Razzhigaev, Matvey Mikhalchuk, Elizaveta Goncharova, Nikolai Gerasimenko, Ivan Oseledets, Denis Dimitrov, and Andrey Kuznetsov. 2024a. Your transformer is secretly linear. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5376–5384.

Anton Razzhigaev, Matvey Mikhalchuk, Elizaveta Goncharova, Ivan Oseledets, Denis Dimitrov, and Andrey Kuznetsov. 2024b. The shape of learning: Anisotropy and intrinsic dimensions in transformer-based models. In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 868–874.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. 2024. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522.

Raphaël Sarfati, Eric Bigelow, Daniel Wurgaft, Siddharth Boppana, Jack Merullo, Atticus Geiger, Owen Lewis, Tom McGrath, and Ekdeep Singh Lubana. 2026. The shape of beliefs: Geometry, dynamics, and interventions along representation manifolds of language models' posteriors. *arXiv preprint arXiv:2602.02315*.

Suzanna Sia, David Mueller, and Kevin Duh. 2024. Where does in-context learning happen in large language models? In *Advances in Neural Information Processing Systems*, volume 37.

Oscar Skean, Md Rifat Arefin, Dan Zhao, Niket Patel, Jalal Naghiyev, Yann LeCun, and Ravid Shwartz-Ziv. 2025. Layer by layer: Uncovering hidden representations in language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Alessandro Stolfo, Vidhisha Balachandran, Safoora Yousefi, Eric Horvitz, and Besmira Nushi. 2025. Improving instruction-following in language models through activation steering. In *International Conference on Learning Representations (ICLR)*.

Alessandro Stolfo, Yonatan Belinkov, and Mrinmaya Sachan. 2023. A mechanistic interpretation of arithmetic reasoning in language models using causal mediation analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7035–7052.

Nishant Subramani, Nivedita Suresh, and Matthew E. Peters. 2022. Extracting latent steering vectors from pretrained language models. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 566–581.

Qi Sun, Marc Pickett, Aakash Kumar Nain, and Llion Jones. 2025. Transformer layers as painters. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 25219–25227.

Daniel Tan, David Chanin, Aengus Lynch, Brooks Paige, Dimitrios Kanoulas, Adrià Garriga-Alonso, and Robert Kirk. 2024. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. 2019. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4593–4601.

Federico Tiblias, Irina Bigoulaeva, Jingcheng Niu, Simone Balloccu, and Iryna Gurevych. 2026. Hypothesis-driven feature manifold analysis in LLMs via supervised multi-dimensional scaling. *Transactions on Machine Learning Research (TMLR)*.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. 2024. Function vectors in large language models. In *International Conference on Learning Representations (ICLR)*.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*.

Lucrezia Valeriani, Diego Doimo, Francesca Cuturello, Alessandro Laio, Alessio Ansuini, and Alberto Cazzaniga. 2023. The geometry of hidden representations of large transformer models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. 2020. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33.

Hieu M. Vu and Tan M. Nguyen. 2025. Angular steering: Behavior control via rotation in activation space. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023a. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations*.

Lean Wang, Lei Li, Damai Dai, Deli Chen, Hao Zhou, Fandong Meng, Jie Zhou, and Xu Sun. 2023b. Label words are anchors: An information flow perspective for understanding in-context learning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9840–9855.

Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. 2024. Do llamas work in English? on the latent language of multilingual transformers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15366–15394.

Alex H. Williams, Erin Kunz, Simon Kornblith, and Scott W. Linderman. 2021. Generalized shape metrics on neural representations. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Daniel Wurgaft, Can Rager, Matthew Kowal, Vasudev Shyam, Sheridan Feucht, Usha Bhalla, Tal Haklay, Eric Bigelow, Raphaël Sarfati, Thomas McGrath, Owen Lewis, Jack Merullo, Noah Goodman, Thomas Fel, Atticus Geiger, and Ekdeep Singh Lubana. 2026. Manifold steering reveals the shared geometry of neural network representation and behavior. *arXiv preprint arXiv:2605.05115*.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations*.

An Yang et al. 2025a. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Sohee Yang, Elena Gribovskaya, Nora Kassner, Mor Geva, and Sebastian Riedel. 2024. Do large language models latently perform multi-hop reasoning? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10210–10229.

Sohee Yang, Nora Kassner, Elena Gribovskaya, Sebastian Riedel, and Mor Geva. 2025b. Do large language models perform latent multi-hop reasoning without exploiting shortcuts? In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3971–3992.

Zeping Yu, Yonatan Belinkov, and Sophia Ananiadou. 2025. Back attention: Understanding and enhancing multi-hop reasoning in large language models. *arXiv preprint arXiv:2502.10835*.

Fred Zhang and Neel Nanda. 2024. Towards best practices of activation patching in language models: Metrics and methods. In *International Conference on Learning Representations*.

Long Zhang, Dai-jun Lin, and Wei-neng Chen. 2026. The geometric price of discrete logic: Context-driven manifold dynamics of number representations. *arXiv preprint arXiv:2603.23577*.

Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. 2023. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, and 2 others. 2023. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*.

Appendix Contents

A Related Work 14
B Data: tasks, cues, templates and accuracies 17
C Shape change and peaks: definition, stability, thresholds 18
D Interventions and event types 19
E Events at shape-change peaks 20
F Handoff of Causal Control 20
G Strength sweep 23
H Comparison with logit-lens accessibility 24
I Cross-layer transfer 25
J The blurred-input model 26
K Results without low-accuracy pairs . . 27
L Figures for the other models 28

A Related Work

Representational geometry across depth. A long line of work compares representations between layers using invariant similarity measures, including SVCCA (Raghu et al., 2017), CKA (Kornblith et al., 2019), representational similarity analysis (Kriegeskorte et al., 2008), and shape metrics such as orthogonal Procrustes distance, which is a proper metric on configurations (Williams et al., 2021). Of these, Procrustes offers the most balanced sensitivity to functionally relevant changes (Ding et al., 2021). Applied to LLMs, these tools mostly show smooth, depth-generic profiles. Intrinsic dimension and anisotropy rise and fall with depth (Ansuini et al., 2019; Valeriani et al., 2023; Ethayarajh, 2019; Razzhigaev et al., 2024b). A high-dimensional "abstraction phase" appears in early-to-middle layers (Cheng et al., 2025). Intermediate layers often give the best features (Skean et al., 2025). Adjacent layers are related by a nearly linear map (Razzhigaev et al., 2024a). A complementary literature proposes generic stages of infer-

ence, from probing pipelines (Tenney et al., 2019) to layer deletion and reordering (Lad et al., 2025; Gromov et al., 2025; Sun et al., 2025; Csordás et al., 2025), layer-wise decoding (nostalgebraist, 2020; Belrose et al., 2023; Wendler et al., 2024), and readout-subspace trajectories (Lombardo et al., 2026). Closer to our question, several studies report an abrupt mid-network transition toward answer- or decision-organized geometry (Doimo et al., 2024; Joshi et al., 2025), but they do not test whether the transition is causally meaningful. The closest methodological precedent is Bhattacharya and Kolli (2026), who decompose adjacent-layer transitions of residual-stream token clouds into a Procrustes rotation and a non-rigid residual. They find smooth curves that peak only at the final layer, and they explicitly make no causal claims. We use the same family of metrics but apply it to a different object: the configuration of *answer-class centroids* at the answer position. This object is task-specific and reveals sparse, mid-depth peaks that statistics pooled over all tokens and positions average out. We then show causally that these peaks mark where answers are written, where steering starts to work, and where steering vectors stop transferring.

Concept geometry and feature manifolds. Concepts in LLMs are often linearly represented (Park et al., 2024; Marks and Tegmark, 2024). Categorical and hierarchical concepts form polytopes and orthogonal subspaces (Park et al., 2025b). Periodic and numeric concepts live on multi-dimensional manifolds: circles for days and months (Engels et al., 2025), Fourier and helical features for modular and integer arithmetic (Nanda et al., 2023; Zhong et al., 2023; Kantamneni and Tegmark, 2025), and manifolds whose shape follows the symmetries of language statistics (Karkada et al., 2026; Modell et al., 2025). Most of this work characterizes geometry at a single, chosen layer. When shape is compared across tasks or contexts, again usually at one fixed depth, the typical finding is that the relational geometry is preserved and only its position moves. Hu et al. (2026b), using the same Procrustes disparity we use, find nearly identical number geometry across five tasks at 75% depth while the subspace shifts. Hu et al. (2026a) report context-specific displacements of word manifolds with preserved relational structure. Other work shows that context can reshape or distort these manifolds (David and Fomin, 2026; Zhang et al., 2026; Park et al., 2025a; Tiblias et al., 2026). Manifold-aware steering likewise operates at one late layer. Wurgaft et al. (2026) fit splines through answer-class centroids at layer 28 of Llama-3.1-8B, chosen "to ensure that concept geometries are fully computed"; related work steers along manifolds of beliefs, time, or output distributions (Sarfati et al., 2026; Modell, 2026; Oozeer et al., 2026). Gurnee et al. (2026) show that attention heads actively transform count manifolds across layers. We hold the task fixed and vary depth. Within a task, the shape of the answer-class configuration is far from invariant: it changes in a few large steps. This tells practitioners from which layer centroid-based (linear or manifold) steering can be expected to work.

Activation steering and the choice of layer. Additive steering methods add a direction to the residual stream at a chosen layer (Subramani et al., 2022; Turner et al., 2023; Li et al., 2023; Zou et al., 2023; Rimsky et al., 2024). The direction is typically a difference of class means, which is causally more faithful than probe directions (Marks and Tegmark, 2024; Arditi et al., 2024; Jorgensen et al., 2023) and optimal under a squared-error objective (Im and Li, 2025). In practice the layer is chosen by sweeping (Turner et al., 2023; Rimsky et al., 2024; Tan et al., 2024; Chen et al., 2025; Stolfo et al., 2025) or by hand-crafted filters (Arditi et al., 2024). Task and function vectors work best in intermediate layers and fall off sharply late (Hendel et al., 2023; Todd et al., 2024). Steerability varies widely across concepts and models (Tan et al., 2024; Da Silva et al., 2025). Several recent works try to predict it: from the coherence of activation differences (Braun et al., 2025), from per-input learned layer selectors (Gadgil et al., 2026), and from class-separation criteria (Dang and Ngo, 2026). Most closely related is the linear accessibility profile of Billa (2026). It is the logit-lens accuracy A_{lin} of the hidden state at each layer. On binary concepts, peak A_{lin} predicts steering efficacy ($\rho = 0.86$–0.91) and the best steering layer ($\rho = 0.63$–0.92). Billa (2026) further argue that where a concept is decodable nonlinearly but A_{lin} is still low, difference-of-means steering fails. We compare the two signals directly (Appendix H), computing A_{lin} exactly as defined there on all our model–task pairs. As point predictors of steering onset at natural strength and of the handoff layer, shape-change peaks and the first layer with $A_{\text{lin}} > 0.1$ are comparable (mean absolute errors within 0.5 layers, no significant differ-

ence). For the cross-layer transfer boundary, A_{lin} is more accurate. The two signals differ as lower bounds. At natural strength, steering already works before the highest shape peak in only 2 of 39 pairs, but before A_{lin} exceeds 0.1 in 11 (paired exact test, $p = 0.012$). In 16 of 39 pairs, centroid-difference steering is effective at a layer where $A_{\text{lin}} \leq 0.1$. We interpret these as cases where the answer-class structure is already in place but not yet aligned with the unembedding, as in the "readable reorganization" events of Section 3. In our multi-class setting, therefore, low output alignment does not imply that difference-of-means steering fails. The best-layer rule of Billa (2026) (the layer of maximal A_{lin}) targets peak efficacy and points to late layers, so it does not address onset. Strength is another confound: raw coefficients are not comparable across layers, which motivates norm-relative or angular parameterizations (Li et al., 2023; Vu and Nguyen, 2025; Aparin and Gaintseva, 2026). Steering vectors are sometimes observed to transfer to nearby layers, with similarity decaying with layer distance (Rimsky et al., 2024), and cross-layer pooling has been proposed to stabilize directions (Jiang et al., 2026). Our contribution is not a new steering method. It is a geometric account of *onset* and *transfer*. Measured with norm-relative strength, the earliest effective steering layer lands on shape-change peaks, moving to an earlier peak as strength grows. Vectors built before a peak barely transfer (mean 0.07). At a fixed layer distance, transfer decreases with the cumulative shape change in between. Unlike the logit-lens profile, our signal does not use the unembedding. It therefore detects answer structure that is not yet aligned with the output vocabulary, which output-based diagnostics cannot separate from absence.

Information flow and the input-to-answer handoff. Causal mediation and interchange interventions (Vig et al., 2020; Geiger et al., 2021, 2025), causal tracing (Meng et al., 2022), and path patching (Wang et al., 2023a) localize where information is used. Best practices recommend clean counterfactual inputs over noise corruption (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024), which we follow. A recurring finding is that attention moves information from input tokens to the final position in a specific layer window, after which MLPs at the final position write the answer. This has been shown for factual recall (Geva et al., 2023; Merullo et al., 2024; Hernandez et al., 2024), for arithmetic (Stolfo et al., 2023; Nikankin et al., 2025), and with inspection or attribution tools (Ghandeharioun et al., 2024; Ferrando and Voita, 2024). For the cyclic tasks we study, several papers place this step at specific layers:

- Engels et al. (2025) observe that the start day or month is copied to the final token at layers 15–17 in Mistral-7B, "which is why interventions drop off there".
- Feucht et al. (2026) show that Llama-3.1-8B copies the offset to the last token by about layer 15 and computes a base-10 sum with a handful of MLP neurons at layer 18; the circular geometry of the output concept appears only around layers 22–25.

Multi-hop questions add further layer structure (Yang et al., 2024; Biran et al., 2024; Yang et al., 2025b; Yu et al., 2025; Lindsey et al., 2025; Ameisen et al., 2025), though answers may become decodable before bridge entities (Liu et al., 2026). Several works locate a point after which the input is no longer needed:

- Sia et al. (2024) find this point for in-context translation.
- Pola and Balasubramanian (2025) find a patching "onset" after which swapping instruction keywords stops changing the answer.
- Ma and Rui (2026) describe a "handoff" of causal control from a rhyme word to a newline token.

Intermediate positions such as label words and template tokens can act as relays (Wang et al., 2023b; Bai et al., 2026), as distinct from attention sinks (Xiao et al., 2024). Most directly, Haim and McNamee (2026) steer the input tokens versus the final token with centroid differences in Llama-3.1-8B, Mistral-7B and Qwen2.5-7B on month, weekday and clock arithmetic. They find a sharp crossover at roughly two-thirds depth, while the geometric quantities they track (participation ratio, angular distances) reorganize gradually. We measure the same kind of crossover with both steering and patching, layer by layer, in four model families and on retrieval as well as arithmetic tasks. The crossover takes at most one layer in 30 of 35 pairs. When the shape-change peak is a write event, it falls at the crossover layer: a sharp geometric signature where previous geometric measures saw a

gradual change. In multi-step models, the answer reaches the last token through chat-template relay positions. We use "handoff" in the positional sense of Ma and Rui (2026), not in the sense of the layer after a concept-allocation zone (Henry, 2026b,a).

Closest work. Table 4 summarizes the works closest to ours. The most closely related is concurrent work by Arora et al. (2026). They decompose last-token activations on cyclic analogies with a functional ANOVA. In Llama-3.1-8B, a pairwise relation term dominates around layers 15–17, and a three-way term organized by the answer emerges at layers 17–18, from which point steering with answer vectors becomes effective. Some geometrically structured terms turn out to be causally inert. Our results agree on this transition: our Llama shape peak and handoff for weekday and month fall at layers 17–18. Our simple model also parallels their decomposition. Before the peak, answer-grouped centroids are a blurred image of additive input features. After it, an answer-organized interaction term dominates. Their decomposition requires knowing the task's input factors. Our shape signal needs only answer labels, which lets us cover retrieval and two-hop tasks. It also lets us relate the transition to steering strength, cross-layer transfer, and the handoff between positions.

B Data

Prompts. Cyclic-arithmetic tasks use every combination of start item and offset $k \in \{1, \ldots, 6\}$ under three templates, e.g. for weekdays: "What day is {k} after {x}?", "Today is {x}. What day will it be in {k}?" and "Starting from {x} and counting {k} forward, what day is it?". Retrieval tasks use six or eight cues per answer class, none of which contains the answer word, under three templates. Examples (first template):

- Landmark → city (12 cities): "Which city is home to the Eiffel Tower?"
- Country → continent (6 continents): "Which continent is France in?"
- Capital → continent (6 continents): "Oslo is the capital of a country. On which continent is that country?"
- Object → color (8 colors): "What color is a ripe strawberry?"
- In-context color (8 colors): "Grace likes purple. Leo likes blue. Anna likes red. Emma likes white. What color does Anna like?"
- Story → emotion (8 emotions): "Anna spent the whole afternoon laughing with her best friends at a sunny picnic. How does Anna feel?"

Each prompt is wrapped in the model's chat template with the system instruction "Answer with only the name of the [day / month / . . .]. Do not explain." (Gemma-2 has no system role, so the instruction is prepended to the user turn).

Answers. The model's answer is the class whose surface forms (with and without a leading space, capitalized or not; digits for hours) receive the highest summed probability at the first generated token. Tokens shared by several classes are dropped. Hours in Qwen2.5-7B, Qwen2.5-14B and Gemma are tokenized digit by digit, so interventions on these three models use only targets two to nine; Phi-4 encodes 10, 11 and 12 as single tokens.

Accuracy and sample sizes. Table 5 lists the number of prompts, the accuracy, and the number of correctly answered prompts used for patching. Shape change is computed on all prompts of a task. Mistral on hours is excluded from all analyses. Four further pairs have accuracy below 0.55 (Llama and Mistral on letters, Gemma on hours, Mistral on emotions; none for Qwen2.5-14B or Phi-4); every main result holds when they are removed (Appendix K).

Smaller models. Section 7 and Appendix E also use four smaller instruction-tuned models: Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct (Qwen Team et al., 2024), Qwen3-4B (Yang et al., 2025a) and Llama-3.2-3B-Instruct (Grattafiori et al., 2024). They use the same prompts, on the tasks they answer above chance (9 pairs: months and weekdays, and for Qwen3-4B also the two continent tasks). Gemma-2-2B-it (Gemma Team et al., 2024) was also run on months but shows no peak (Table 18). We have no handoff, steering or transfer experiments for these models, so they enter none of the main statistics.

Models, licenses and compute. Candidate models were chosen by task accuracy alone. Qwen2.5-14B-Instruct (Qwen Team et al., 2024) and Phi-4 (Abdin et al., 2024) were added to the four original models. Llama-2-13B and Mistral-Nemo-12B

Work	Geometric signal	Tasks	Depth	Onset	Transfer	Handoff
Hu et al. (2026b)	Procrustes across tasks	numbers	–	–	–	–
Bhattacharya and Kolli (2026)	adjacent-layer Procrustes, token clouds	generic text	✓	–	–	–
Wurgaft et al. (2026)	answer centroids, one layer	cyclic	–	–	–	–
Billa (2026)	logit-lens accessibility	binary concepts	✓	✓	–	–
Pola and Balasubramanian (2025)	none (patching)	instructions	–	–	–	✓
Haim and McNamee (2026)	participation ratio, angles	cyclic	✓	✓	–	✓
Arora et al. (2026)	functional ANOVA terms	cyclic analogies	✓	✓	–	–
Ours	adjacent-layer Procrustes, answer centroids	cyclic + retrieval	✓	✓	✓	✓

Table 4: Closest prior and concurrent work. "Depth": geometry tracked across layers within a task; "Onset": earliest layer at which steering becomes effective; "Transfer": cross-layer transfer of steering vectors; "Handoff": causal control moving from input to answer positions.

Task	Prompts	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B	Gemma-2-9B	Qwen2.5-14B	Phi-4
Weekday	126	0.98 / 120	0.85 / 91	0.66 / 67	0.91 / 106	0.95 / 115	0.94 / 113
Month	216	1.00 / 216	0.98 / 206	0.88 / 170	0.99 / 213	1.00 / 216	0.97 / 204
Hour	216	0.67 / 91	0.96 / 199	0.06 / –	0.49 / 52	0.62 / 83	0.95 / 198
Letter	468	0.80 / 317	0.45 / 153	0.42 / 148	0.70 / 285	0.89 / 387	0.84 / 330
Landmark → city	216	0.97 / 210	0.99 / 212	0.99 / 214	0.99 / 214	0.99 / 214	0.98 / 211
Country → continent	144	0.99 / 142	0.96 / 138	0.89 / 129	1.00 / 144	1.00 / 144	1.00 / 144
Capital → continent (2-hop)	144	1.00 / 144	0.98 / 141	0.85 / 122	1.00 / 144	1.00 / 144	1.00 / 144
Object → color	144	0.92 / 133	0.83 / 119	0.97 / 140	0.99 / 143	0.97 / 140	0.93 / 134
In-context color	192	1.00 / 192	1.00 / 192	1.00 / 192	1.00 / 192	1.00 / 192	1.00 / 192
Story → emotion	144	0.92 / 99	0.58 / 74	0.52 / 52	0.59 / 69	0.79 / 101	0.74 / 89

Table 5: Tasks, number of prompts, and for each model the accuracy / number of correctly answered prompts with a matched patching partner.

were screened out for low task accuracy before any full run. Gemma-3-12B was run in full but excluded because its shape change at the highest peak stayed below the bootstrap noise floor in all ten tasks (median ratio of peak change to the 95th percentile of the floor: 0.75). All models are publicly available and used for research under their respective licenses: Apache 2.0 (Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-1.5B, Qwen3-4B, Mistral-7B), MIT (Phi-4), the Llama 3.1 and Llama 3.2 Community Licenses, the Gemma Terms of Use, and the Qwen Research License (Qwen2.5-3B). All experiments run inference only, with greedy decoding. The original models ran on four NVIDIA RTX 4090 GPUs (24 GB) for about 20 hours (about 80 GPU-hours in total); Qwen2.5-14B and Phi-4, including their handoff runs, took about 40 additional A100 GPU-hours (NVIDIA A100, 80 GB). We use PyTorch 2.8.0 and Hugging Face Transformers 5.18.0; statistics are computed with NumPy and SciPy. We will release our prompt sets and code.

C Shape change and peaks

Computation. Equation (1) is evaluated in the span of the two centered configurations, which has at most $2K$ dimensions, so the cost does not depend on d. We use centroids rather than all prompts so that the measure reflects the geometry of the answer classes and not the spread within a class.

Peak definition. Peaks are searched from layer $0.25L$ on (rounded), where L is the number of layers; earlier layers carry large changes from embedding and template processing. A peak is a local maximum ($\Delta_\ell \geq \Delta_{\ell\pm1}$) with $\Delta_\ell \geq 1.5\times$ the median over the searched layers. The same definition is used in every section.

Stability. Resampling the prompts within each class 100 times leaves the highest peak within one layer of its original position in a median of 93% of resamples. The highest peak is less stable in 15 of the 59 pairs, mostly because two peaks of similar height compete; this is why our main statistics refer to the three highest peaks. A cross-validated estimate of shape change, which compares unbiased class-distance matrices between disjoint halves of

the prompts, confirms the highest peak as a genuine reorganization in 40 of the 59 pairs; most of the rest have too few prompts per class to decide.

Thresholds. All thresholds were set to round values before the analysis and not tuned. The conclusions of Section 3 hold in all 144 combinations of jump threshold (0.05–0.20), peak threshold (1.25–3 times the median), window (0, ±1, ±2 layers) and start of the search (20–33% of depth). They also hold in all 135 combinations of the rule-specific thresholds: prior patching success for a concentration (0.4–0.8), probe jump (0.10–0.20), probe level (0.7–0.9) and the causal ceiling for a readable reorganization (0.1–0.3). Across these settings, the highest peak is a write in 61–64% of pairs, against 21–24% of other layers.

D Interventions and event types

Targets and partners. All interventions use only prompts that the model answers correctly, and each asks the model to produce a specific target answer. For cyclic arithmetic, each prompt with start s and offset k receives a new start $s' = s + j$, with j cycling over $1, \ldots, n-1$ across prompts; the target is $s'+k$. The patching partner is a correctly answered prompt from the same template with start s' and the same k. For retrieval, the target is the original answer plus j (modulo the number of classes), and the partner is a correctly answered prompt from the same template whose answer is the target and whose cue differs. An intervention succeeds when the model's answer becomes the target. Random directions of equal norm are the control.

Positions. The input token is the last token of the start item or cue. For in-context color it is the color word bound to the queried person. For emotions it is the last token of the story; a variant that steers every story token is reported separately. The answer positions run from the final question mark to the end of the prompt, including the chat-template tokens (two tokens for Mistral, five for Phi-4, six for the other models).

Steering. At the input token, the direction for cyclic arithmetic is the difference between the centroids of the new and the original start items. For retrieval, it is the difference between the target and original answer-class centroids at the cue token, excluding prompts with the same cue. At the answer positions, it is the difference between the target and original answer-class centroids at each position. All centroids use the other two templates and correctly answered prompts only. The strength α is in units of this difference; $\alpha = 1$ moves a state from its class centroid to the target centroid. The strength sweep (Section 5) scales the direction by $\alpha \in \{1, 2, 4, 8\}$, or sets its norm to $c\,\|h\|$ with $c \in \{0.1, 0.2, 0.4, 0.8\}$. Random directions of the same norm are applied at $\alpha = 8$ and $c \in \{0.4, 0.8\}$. For cross-layer transfer (Section 6), the direction built at layer X is normalized and rescaled to the norm of layer Y's own centroid difference.

Boundaries. For each model–task pair we read three boundaries off these interventions, and classify the event at each peak:

- the *handoff layer*, the first layer at which patching the input token falls below half of its maximum success;
- the *steering onset* at strength α, the first layer at which steering the answer positions succeeds on at least half of the prompts;
- the *transfer* of a layer X, the mean success when the centroid-difference vector built at X is rescaled and applied at each layer Y where steering with Y's own vector succeeds (≥ 0.9);
- the *event* at each peak, defined below from which signals jump within one layer of it.

Event types. At each late layer we check which per-layer signals jump within one layer of it. A jump is a rise of at least 0.10 in a success rate between adjacent layers, or of at least 0.15 in probe accuracy. The signals are: patching and steering of all answer positions and of the last position only; a linear identity probe for the answer (cross-validated, grouped by cue for retrieval); the logit-lens probability of the answer; and patching restricted to the subspace spanned by the class centroids. The rules are applied in order:

- *concentration*: the centroid-subspace patching jumps, and patching of the full state already succeeded (≥ 0.6) two layers earlier;
- *write*: any patching or steering signal jumps;
- *readable reorganization*: the probe jumps or is already ≥ 0.8, while every causal signal at the next layer is below 0.2;

- *vocabulary alignment*: only the logit lens jumps;
- otherwise, *no event*.

E Events at Shape-Change Peaks

This appendix gives the per-pair events behind Section 3. Signals and rules are defined in Appendix D. All causal signals use the same prompts, targets and success criterion as the handoff and steering experiments.

E.1 Per-pair events

Table 6 lists the three highest peaks of every model–task pair and the event at each. Figures 7 and 8 show the underlying signals. Three regularities are visible.

- In arithmetic tasks, the highest peak is a write in 18 of 23 pairs, and only three peaks are readable reorganizations (Qwen2.5-14B on weekdays and hours, layer 32; Phi-4 on months, layer 19).
- In retrieval tasks, readable-reorganization peaks typically precede a write peak by two to six layers.
- In every pair with a handoff layer, each concentration peak lies after the handoff (Section 4), that is, after the answer has been written.

E.2 Which signals jump at peaks

For each signal, we collect every jump in the late layers and ask whether it lies within one layer of one of the three highest peaks (Table 7). Jumps of the identity probe and of interventions on all answer positions concentrate at peaks. Interventions on the last position alone concentrate less, because in multi-step models the last position is reached after the write (Section 3.3). Logit-lens jumps are close to the random baseline.

E.3 Readable reorganizations and the logit lens

Among the three highest peaks, 38 are readable reorganizations and 77 are writes. At 35 of the 38 readable-reorganization peaks, the logit-lens probability of the answer is below 0.001, and at all but one it is below 0.01; the logit-lens accuracy A_{lin} is at most 0.1 at all but the same one (Qwen2.5-14B on country $\rightarrow$ continent, layer 32: probability 0.03, $A_{\text{lin}} = 0.30$). Causal success reaches 0.3 a median of 4 layers later (mean 4.7), and the logit lens reaches 0.1 a median of 6 layers later (mean 6.8). At write peaks, causal success rises at the peak itself (median 0 layers) and the logit lens follows after a median of 1 layer. Even at write peaks, the logit-lens probability is still below 0.01 in 32% of cases.

E.4 Threshold sensitivity

We repeat the classification under 144 settings: jump threshold 0.05, 0.10, 0.15 or 0.20; peak threshold 1.25, 1.5, 2 or 3 times the median; window 0, ± 1 or ± 2 layers; and search start at 20%, 25% or 33% of depth. In all 144 settings, the highest peak is a write more often than non-peak layers (by a factor of 2.3 to 8.1, median 2.9), and carries no event less often. Only one highest peak (Qwen2.5-14B, object color) is a pure vocabulary alignment, and only in 48 of the 144 settings. In every setting, at most four readable-reorganization peaks have a logit-lens probability of 0.001 or more.

E.5 An alternative causal protocol and smaller models

An earlier version of this analysis used a different causal signal: patching randomly paired prompts and measuring the normalized logit difference rather than whether the answer switches. With that protocol, over 49 pairs including four smaller models (this earlier analysis also included Mistral on hours), the highest peak was a write in 69% of pairs against 33% of non-peak layers, and no highest peak was a pure vocabulary alignment. For the smaller models (Qwen2.5-1.5B and -3B, Qwen3-4B, Llama-3.2-3B; 9 pairs), only this protocol is available. Their highest peaks are writes in 7 of 9 pairs (all arithmetic pairs), against 27% of non-peak layers.

F Handoff of Causal Control

This appendix gives the per-pair results behind Section 4. Interventions are defined in Appendix D.

F.1 Per-pair results

Table 8 lists, for every pair:

- the three highest peaks;
- the handoff layer measured by patching, with a 95% interval, and by steering;

Task	Qwen2.5-7B	Llama-3.1-8B	Mistral-7B	Gemma-2-9B	Qwen2.5-14B	Phi-4
Weekday	12–, 16–, **23**W	13–, **17**W, 22C	17–, **20**W, 25C	19–, 28W, **34**C	32R, **37**W, 40V	**19**–, 21W, 24W
Month	14–, **23**W, 25V	16–, **18**W, 21C	17–, **20**W, 25W	31C, **33**C, 36W	**37**W, 40W, 42–	19R, 22W, **24**C
Hour	7–, 19–, **23**W	14–, **16**W, 24V	excluded	28W, 31W, **33**W	32R, **37**W, 40C	**21**W, 24C, 28W
Letter	14–, **23**W, 27V	16–, **18**W, 21W	**19**W, 28W, 31W	16–, **27**–, 33W	32–, **37**W, 40W	17–, **19**W, 24C
Landmark → city	**19**R, 21W, 23W	**14**R, 16W, 20W	12–, **15**R, 18W	20R, **25**W, 29C	23–, 31R, **34**W	10–, 17–, **22**W
Country → continent	16R, 18R, **22**W	**14**R, 17W, 20W	14R, **18**W, 31–	19R, **25**W, 32V	26R, 32R, **34**W	16R, 18R, **22**W
Capital → continent	**16**R, 19R, 22W	14R, **16**W, 20V	15R, **18**W, 31–	19R, **25**W, 32V	32W, **34**W, 36C	16R, **22**W, 25–
Object → color	19R, 21W, **24**C	12–, **14**W, 22C	**15**–, 19W, 23W	**29**W, 33W, 41–	23–, **41**V, 43V	13–, **16**–, 23W
In-context color	14–, **19**W, 21W	**12**R, 14W, 16W	13R, 15W, **19**W	**13**–, 18R, 28W	23R, 26–, **28**R	17R, **21**R, 27W
Story → emotion	13R, 18W, **22**C	**13**R, 18C, 20W	**13**R, 15W, 20W	**20**R, 31W, 34W	23R, **28**R, 40–	17R, **20**W, 32–

Table 6: The three highest shape-change peaks of each pair (layer and event; the highest peak in bold). W: write; R: readable reorganization; C: concentration; V: vocabulary alignment; –: no event.

Signal	Near a peak	Jumps
Identity probe	.85	73
Steering, all answer positions	.73	146
Patching, all answer positions	.73	129
Patching, last position	.60	149
Steering, last position	.54	162
Logit lens	.44	186
Random late layer	.30	–

Table 7: Fraction of signal jumps within one layer of one of the three highest shape-change peaks (59 pairs).

- the gap between the handoff and the first layer at which intervening on the answer positions succeeds;
- the number of prompts.

The intervals come from a parametric bootstrap of the per-layer success rates (2,000 draws). Their median width is 0 layers and their maximum width is 3 layers. For Qwen2.5-14B and Phi-4, handoff patching and the full-vocabulary logit lens were run with re-implementations of our scripts written from the description in this paper; on Qwen2.5-7B, these re-implementations reproduce all nine handoff layers exactly and the curves to within 0.03. Figures 9 and 10 show the curves for the cyclic-arithmetic and retrieval pairs.

F.2 Threshold sensitivity

We vary the threshold for input loss (0.3–0.7 of the input's maximum) and for answer success (0.3–0.7), giving 25 combinations. Across them, steering and patching place the handoff at the same layer in 69–82% of pairs, and within one layer in 76–96%. The patching gap is at most one layer in 60–100% of pairs, and the steering gap is two or more layers in 20–61%. The size of the gaps thus depends on the thresholds. Their direction is stable: at every threshold, more pairs have a larger gap under steering than under patching (11–21 versus 3–7 of 33).

F.3 Pairs away from a peak

In nine pairs, the handoff lies two or more layers from the nearest of the three highest peaks(two or three layers in seven of them):

- Gemma, months: handoff at 29; peaks at 31, 33, 36.
- Gemma, letters: handoff at 29; peaks at 16, 27, 33.
- Gemma, object color: handoff at 26; peaks at 29, 33, 41.
- Llama, object color: handoff at 16; peaks at 12, 14, 22.
- Mistral, object color: handoff at 17; peaks at 15, 19, 23.
- Phi-4, letters: handoff at 21; peaks at 17, 19, 24.
- Phi-4, in-context color: handoff at 25; peaks at 17, 21, 27.
- Qwen2.5-14B, in-context color: handoff at 34; peaks at 23, 26, 28.
- Qwen2.5-14B, object color: handoff at 34; peaks at 23, 41, 43.

In Gemma on months and object color, the highest peak lies after the handoff; on months it is a concentration event. In Gemma on letters, the handoff falls between a no-event peak (27) and a write peak (33). In Phi-4 on letters, the handoff falls between a write peak (19) and a concentration peak (24); in Phi-4 on in-context color, between a readable-reorganization peak (21) and a write peak (27). In Qwen2.5-14B on in-context color, all three peaks (two of them readable reorganizations) precede the handoff, and on object color the highest peak (41) is

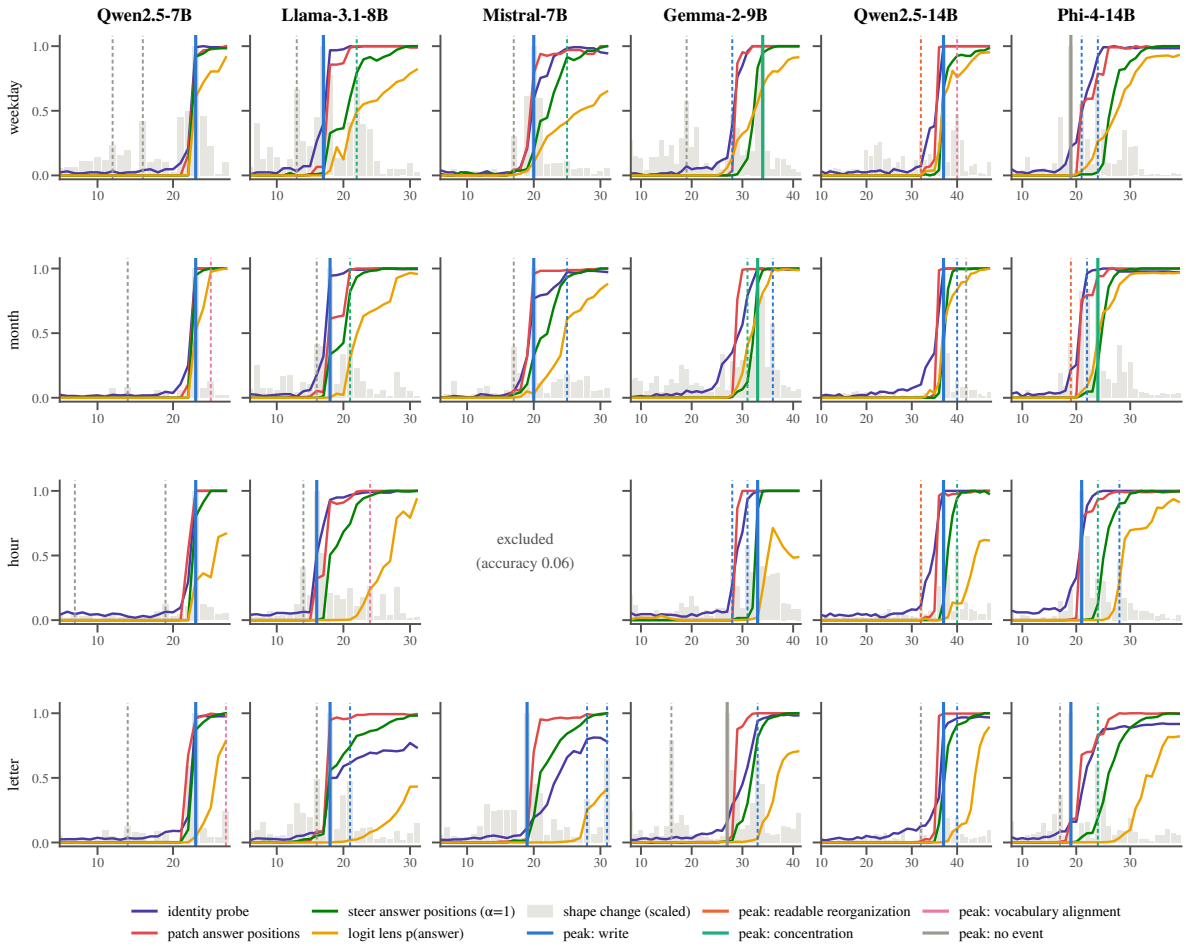


Figure 7: Signals at the answer positions for the arithmetic tasks. Gray bars: adjacent-layer shape change (scaled within each panel). Lines: identity probe, patching and steering success at all answer positions ($\alpha = 1$), and logit-lens probability of the answer. Vertical lines: the three highest peaks, colored by event (solid: highest peak). Mistral on hours is excluded.

a vocabulary alignment seven layers after it. In all four object-color pairs, the input stops controlling the answer gradually, over several layers, rather than at once (Figure 10).

F.4 Steering lags patching in multi-step models

In Qwen2.5-7B, steering and patching of the answer positions take effect within one layer of each other in 8 of 9 pairs. In Llama, Mistral and Gemma, steering the answer positions succeeds two or more layers after the handoff in 12 of 24 pairs with both measurements. Patching reduces the gap to at most one layer in 10 of these 12. Qwen2.5-14B and Phi-4 behave like the multi-step models here: steering succeeds two or more layers after the handoff in 10 of their 18 pairs, and patching reduces the gap to at most one layer in all 10. The answer therefore already reaches the answer positions at the handoff. What steering lacks there is alignment with the direction between class centroids, as the blurred-input model predicts before the answer component dominates (Section 7).

F.5 Decomposing the input

The main experiments intervene on a single input token: the last token of the start item or cue. To check that this token carries the input-side information, we patch four sets of positions separately in four pairs (Llama on weekdays, hours and country → continent; Gemma on weekdays; the offset word exists only in the arithmetic pairs):

- all question tokens before the question mark;
- the start item or cue only;
- the offset word only, using a partner with the same start and a different offset;

- all other question tokens.

Figure 11 shows three results. Patching the whole question and patching the start item alone give nearly identical curves and the same handoff layer. The other question tokens have no effect at any layer (success ≤ 0.03). The offset word hands off earlier than the start item, at layer 15 in Llama on weekdays and hours and at layer 26 in Gemma on weekdays. The handoff measured in the main text is thus the layer at which the last input-side information, the start item, leaves the input.

Agreement with mechanistic studies. For Llama-3.1-8B on cyclic arithmetic, the handoff and the shape-change peaks fall at the layers identified by independent mechanistic analyses (Table 9). Feucht et al. (2026) show that this model copies the offset to the last token by about layer 15, and computes the sum with a small set of MLP neurons at layer 18. Arora et al. (2026) find that a representation organized by the answer emerges at layers 17–18, after which steering with answer vectors becomes effective. Our highest peaks lie at layers 16–18. The start item hands off at layers 16–18, and the offset word hands off earlier, at layer 15 (Figure 11). The shape signal recovers these layers from answer geometry alone.

F.6 Emotion stories

In the emotion task, no single input token controls the answer: steering or patching the last token of the story fails at every layer. We therefore steer every story token with the difference of class centroids of the mean story representation. At natural strength this still fails (maximum success 0.01–0.11). With the norm of the added vector set to the norm of each token's state, a clear handoff appears. The story stops controlling the answer at layer 18 in Qwen2.5-7B, 14 in Llama, 16 in Mistral and 21 in Gemma. In all four models this lies within one layer of one of the three highest peaks (layers 18, 13, 15 and 20, respectively). For Qwen2.5-14B and Phi-4, the last story token likewise does not control the answer (maximum success 0.06), but the all-token variant was not run.

G Strength Sweep

This appendix gives the protocol and per-pair results behind Section 5.

G.1 Protocol

Steering follows Appendix D. It is applied at all answer positions of correctly answered prompts, and centroids are computed on the other two templates. For each pair we sweep every layer Y from the first quarter of the network up to three layers after the highest peak, or one layer after the onset at $\alpha = 1$ if that is later. At each Y we use three directions:

- the centroid difference at Y itself;
- the centroid difference at the last layer;
- the centroid difference at the layer whose vectors transfer best (Section 6).

Each direction is rescaled to the norm of Y's own centroid difference. Strengths are $\alpha \in \{1, 2, 4, 8\}$ in units of that norm, and $c \in \{0.1, 0.2, 0.4, 0.8\}$ in units of the norm of the residual stream at each position. Random directions of the same norm are applied at $\alpha = 8$ and $c \in \{0.4, 0.8\}$. At $c = 0.1$ no pair reaches the success threshold, and at $c = 0.2$ only 14 pairs do; we omit both from the tables.

G.2 Per-pair onsets

Table 10 lists the onset at each strength, taking the earliest layer over the three directions. Figures 12 and 13 show the full success maps for the layer's own direction. In most pairs, the onset either stays at a peak or moves from one peak to an earlier one as the strength grows.

G.3 Choice of direction

Using only the layer's own direction gives onsets that are the same or later. They are identical to the earliest-of-three onset in 83%, 78%, 61% and 37% of pairs at $\alpha = 1, 2, 4, 8$, and in 63% and 34% at $c = 0.4$ and 0.8. With the own direction, the onset precedes the earliest of the three peaks by more than one layer in at most 3 of 59 pairs at any strength. With the earliest-of-three onset used in the main text, this happens in at most 6. The lower bound is therefore not an artifact of the choice of direction.

G.4 Success threshold

With a success threshold of 0.7 instead of 0.5, 56–78% of onsets lie within one layer of one of the three highest peaks, depending on strength. At most 5 pairs per strength have an onset more than one layer before the earliest peak. At $\alpha = 1$, only

one onset lies more than one layer before the highest peak (Qwen2.5-14B on object colors, whose highest peak is a vocabulary alignment).

G.5 Controls

Random directions of the same norm switch at most 11% of prompts at any layer and strength. At the onset, the median probability of answers outside the task's classes is 0.00–0.03, so the model is steered to the target answer rather than off the task. The highest success at any layer before the highest peak has a median of 0.00 at $\alpha = 1$, 0.76 at $\alpha = 8$, and 0.85 at $c = 0.8$. Strong steering can therefore succeed before the highest peak, and when it does, it succeeds at an earlier peak (Section 5.1).

G.6 By event type of the highest peak

Table 11 groups the onsets by the event at the highest peak. Highest peaks that are readable reorganizations are crossed only at the strongest settings, in 2 of 12 pairs at $\alpha = 8$ (Phi-4 on in-context colors and Qwen2.5-14B on emotions) and 1 at $c = 0.8$ (Qwen2.5-14B on emotions): the onset may reach them under strong steering, but at weaker strengths never precedes them. Highest peaks that are concentrations are crossed already at natural strength, because the answer has been written earlier. Highest peaks that are writes are never crossed at $\alpha = 1$; from $\alpha = 2$ on, they are crossed in a growing number of pairs (5, 18 and 26 of 36 at $\alpha = 2, 4, 8$), as the onset moves to an earlier peak. The single vocabulary-alignment highest peak (Qwen2.5-14B on object colors) is crossed at every strength, including $\alpha = 1$.

H Comparison with Logit-Lens Accessibility

H.1 Computing A_{lin}

Following Billa (2026), we apply the final normalization layer and the unembedding to the residual stream at the last token, at every layer, for every prompt. A_{lin} is the fraction of prompts for which the most probable token over the full vocabulary is a first token of the correct answer's surface forms. Tokens shared by several answer classes are excluded, as in our answer criterion (Appendix B). We also report a stricter variant that compares the summed probability of each answer class with the total probability of all other tokens; it gives the same conclusions (Table 12, bottom).

H.2 Predictors and targets

We compare four predictors, none of which uses an intervention:

- the highest shape-change peak;
- the earliest of the three highest peaks;
- the first layer with $A_{\text{lin}} > 0.1$, the threshold for linear accessibility in Billa (2026);
- the layer of maximal A_{lin}, their rule for choosing the steering layer.

The targets are:

- the steering onset at five strengths (Section 5);
- the transfer boundary, the first layer whose vectors transfer with success ≥ 0.5 (Section 6);
- the handoff layer (Section 4).

For each predictor and target we report three numbers:

- the mean absolute error;
- the fraction of pairs within one layer;
- the number of *violations*: pairs whose target lies more than one layer before the predicted layer, so that the predictor fails as a lower bound.

H.3 Paired tests

Table 13 compares the highest shape-change peak with the first layer of $A_{\text{lin}} > 0.1$, pair by pair. For accuracy we use a Wilcoxon signed-rank test on the absolute errors. For lower-bound violations we use an exact McNemar test on the pairs that violate one predictor but not the other. The two signals do not differ significantly in accuracy at natural and moderate strength and for the handoff. The logit lens is more accurate for the transfer boundary, and the shape peak is more accurate at the strongest settings. As lower bounds, the shape peak is violated less often for every steering strength; the difference is significant at all five strengths, and also for the handoff.

H.4 Steering before linear accessibility

At the onset layer, A_{lin} is still at most 0.1 in 23 of 59, 35 of 59, 58 of 59, 38 of 59, 59 of 59, 32 of 53 pairs, for onsets at $\alpha = 1, 2, 8$, $c = 0.4, 0.8$, and for the handoff, respectively. In the regime framework of Billa (2026), a concept that is present but not linearly accessible should not be steerable by a difference of means. In our multi-class tasks it often is. The answer-class structure at these layers is already organized for the model's computation, but not yet aligned with the output vocabulary. This is the situation the readable-reorganization peaks of Section 3.2 identify. Our setting differs from theirs in three respects: we use multi-class rather than binary concepts, measure success by the generated answer rather than a change in target probability, and scale steering by the layer's centroid difference.

H.5 Per-pair values

Table 14 lists the predictors and targets for every pair.

I Cross-Layer Transfer

This appendix gives the protocol and per-pair results behind Section 6.

I.1 Protocol

For every pair of layers (X, Y) we build the centroid-difference direction at X and normalize it. We then rescale it to the norm of Y's own centroid difference ($\alpha = 1$), and add it at all answer positions at layer Y. Centroids use the other two templates, and only correctly answered prompts are steered (Appendix D). The layers Y that enter the transfer of X are those where steering with Y's own direction succeeds on at least 90% of prompts. Two pairs are excluded: Mistral on hours, which is too inaccurate, and Gemma on emotions, where no layer reaches 90%. Figures 14 and 15 show the full $X \times Y$ maps.

I.2 Per-pair results

Table 15 lists, for every pair:

- the highest peak and its event;
- the mean transfer of vectors built in the five layers before the highest peak, and at or after it;
- the transfer of the vector built at the peak itself;
- the number of layers after the peak at which transfer first reaches 0.8;
- the layer of the largest increase in transfer, with the event at that layer if it is one of the three highest peaks;
- the transfer of the vector built at the last layer.

By model, transfer before the peak ranges from 0.02 to 0.20, and after it from 0.67 to 0.82. Transfer reaches 0.8 within 2.4 layers of the highest peak in Qwen2.5-7B, 3.8 in Qwen2.5-14B, 4.9 in Gemma, 5.9 in Phi-4, 7.2 in Llama and 8.1 in Mistral. By the event at the highest peak, it takes 0.8 layers after concentration peaks, 4.5 after write peaks, 8.3 after readable-reorganization peaks and 10.8 after peaks with no event; after the single vocabulary-alignment peak it takes 0. Before a concentration peak, transfer is already 0.33, because the answer was written earlier (0.87 before the single vocabulary-alignment peak); before highest peaks of other types it is at most 0.06.

I.3 Reliable structure before the peak

To check that the boundary is not just the appearance of the answer, we measure how reliably the answer-class structure is present at each layer. We split the prompts into two halves, compute unbiased matrices of distances between class means in each half, and correlate them. Table 16 bins all (pair, layer) combinations by this reliability. At every level of reliability, vectors built after the peak transfer far better than vectors built before it. Before the peak, 230 layers from 36 pairs have reliability of at least 0.5, yet their vectors transfer at a mean of 0.10, and only 7% of them exceed 0.5.

I.4 Candidate boundaries

The transfer boundary is the first layer whose vectors transfer with success of at least 0.5. Table 17 compares candidate markers of it. For each candidate it reports the transfer of vectors built just before and after it, and its distance to the boundary. The highest shape-change peak lies on average 2.6 layers before the boundary, and vectors built before it transfer at 0.08. The first layers with $A_{\text{lin}} > 0.1$ and with logit-lens probability ≥ 0.1 (Section 3) lie closest to the boundary (mean error 1.6 layers each), but vectors built before them still transfer at 0.19 and 0.22. The probe and the reliability of the class structure fire much earlier, 6.4 to 7.6 layers before the boundary. Thus the answer becomes

decodable well before vectors built from it carry over.

I.5 Choosing the layer to build a vector

Mean transfer of the vector built at a layer chosen by different rules:

- five layers before the highest peak: 0.08;
- the highest peak itself: 0.39;
- two layers after it: 0.50;
- the first layer with logit-lens probability $\geq$ 0.1: 0.66;
- 80% of depth: 0.76;
- the first layer where native steering succeeds on 90% of prompts: 0.78;
- the last layer: 0.91;
- the best layer in hindsight: 0.96.

Choosing 80% of depth works poorly in Qwen2.5-7B (0.45), because in Qwen2.5-7B this layer often falls before the peak.

I.6 Permutation test and cumulative shape change

The largest increase in transfer falls exactly on one of the three highest peaks in 39 of 58 pairs, within one layer of one in 48, and within one layer of the highest peak in 26. To test these counts, we circularly shift the three peaks of each pair within its late layers, keeping their spacing (10,000 shifts). The shifted peaks give 6.6, 18.8 and 6.7 pairs on average, and no shift reaches the observed counts ($p < 10^{-4}$).

For the cumulative-shape analysis of Section 6.3, we use the 58 pairs of Table 15 and group all (X, Y) with $X \neq Y$ by distance $|X - Y|$ and by direction ($X < Y$ or $X > Y$). Within each group we correlate transfer with the cumulative shape change $\sum_\ell \Delta_\ell$ between X and Y, and average the Spearman correlations over groups. Over all pairs of layers, transfer correlates with layer distance at a median $\rho = -0.72$ and with cumulative shape change at -0.81. The second is stronger in 41 of 58 pairs. Within groups of fixed distance and direction, the median correlation with cumulative shape change is -0.50, and it is negative in 54 of 58 pairs.

J The Blurred-Input Model

This appendix gives the derivation behind Section 7, the regime after the answer is computed, the extension to tasks without arithmetic structure, the estimation procedure, and per-pair results.

J.1 Derivation

Let s be uniform on $\mathbb{Z}_N$, k uniform on $\mathcal{K} \subset \mathbb{Z}_N$, and $D = s + k$. Under the additive model of Equation (2), $\mathbb{E}[b(k) \mid D] = \mathbb{E}[b(k)] = 0$, because k is independent of D when s is uniform. Hence $c(D) = \mathbb{E}[a(D - k)] = (a * p_\mathcal{K})(D)$. Writing $\hat{f}(\omega) = \sum_x f(x) e^{-2\pi i \omega x / N}$, the convolution theorem gives

$$\begin{aligned} \hat{c}(\omega) &= \hat{a}(\omega)\, \hat{p}_\mathcal{K}(\omega), \\ \hat{p}_\mathcal{K}(\omega) &= \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} e^{-2\pi i \omega k / N}. \end{aligned} \tag{4}$$

For our tasks $\mathcal{K} = \{1, \ldots, 6\}$.

- *Weekdays* ($N = 7$): $\mathcal{K}$ contains every nonzero residue, so $\sum_{k=0}^{6} e^{-2\pi i \omega k/7} = 0$ gives $\hat{p}_\mathcal{K}(\omega) = -1/6$ for all $\omega \neq 0$. The blur is uniform: $c = -a/6$, a reflected and shrunken copy of the start-item configuration.
- *Months and hours* ($N = 12$): $|\hat{p}_\mathcal{K}(\omega)| = 0.64, 0, 0.24, 0, 0.17, 0$ for $\omega = 1, \ldots, 6$. All even frequencies vanish, so the blurred configuration keeps only the odd Fourier components of a and has a different shape from it.
- *Letters* ($N = 26$): $\hat{p}_\mathcal{K}$ is a low-pass filter $(0.92, 0.69, 0.39, \ldots)$, and the blurred configuration stays close to a smoothed copy of a at every layer.

Procrustes disparity ignores scale and reflection. For weekdays, the blurred configuration therefore has the same shape as the start-item configuration, and the shape changes only when the answer-organized component $u(D)$ takes over. For months and hours, the shape also changes because the blur removes some frequencies.

After the answer is computed. Once the model computes the answer, a component organized by D itself enters the state,

$$\begin{aligned} h(s, k) &= \mu + a(s) + b(k) + u(D) + \varepsilon, \\ c(D) &= (a * p_\mathcal{K})(D) + u(D). \end{aligned} \tag{5}$$

While the blurred term dominates, a centroid difference $c(D') - c(D)$ is a difference of blurred start-item features, so adding it moves the representation of the inputs rather than of the answer. When u dominates, centroid differences point along the answer, and steering and transfer can work. This decomposition parallels the functional ANOVA of Arora et al. (2026): the additive terms correspond to their main and pairwise terms, and $u(D)$ to their answer-organized three-way term. Unlike that analysis, the shape signal needs no knowledge of the inputs.

General tasks. For an answer $Y = g(X)$, the law of total variance gives $\mathrm{Var}(h) = \mathbb{E}[\mathrm{Var}(h \mid Y)] + \mathrm{Var}(\mathbb{E}[h \mid Y])$. If $h = f(X) + \varepsilon$ does not yet depend on Y beyond X, then $\mathbb{E}[h \mid Y]$ averages f over the inputs with answer Y. The class-separation ratio $\eta^2/(1-\eta^2)$, estimated by the ratio of between-class to within-class variance (the Fisher ratio), is then small, and so is the signal-to-noise ratio of a centroid difference; steering then requires more strength. The quantitative tests below use cyclic arithmetic, where $a(s)$ and $b(k)$ can be estimated from a balanced design; for the other tasks we test only this qualitative prediction (*Separability and steering strength* below).

J.2 Estimation

At each layer we fit $h = \mu + a(s) + b(k) + t(\text{template}) + r$ by the main effects of the balanced design ($s \times k \times$ template), and split each answer centroid into the convolved part $(a * p_{\mathcal{K}})(D)$ and the residual part $\mathbb{E}[r \mid D]$. The *shares* are the fractions of the centroid energy $\sum_D \|c(D)\|^2$ carried by each part. The *crossover* is the first late layer after which the residual part stays above the convolved part.

Noise correction for the weekday prediction. Each class has only 18 prompts, so sampling noise inflates centroid norms. Each template is a complete $s \times k$ design. We therefore estimate c from one template and a from another, and average over the six ordered pairs of templates. Cross-template inner products such as $\langle c_i, c_j \rangle$ and $\langle c_i, -a_j \rangle$ are unbiased by the noise. Before the highest peak, this correction moves the median cosine between c and $-a$ from 0.68 to 0.78, and the median size ratio $\|c\|/\|a\|$ from 0.27 to 0.23 (predicted: 1 and 0.17). The remaining gap from the prediction indicates a small answer-organized component already present before the peak. After the peak, the size ratio rises to 0.3–1.6, as the answer component dominates.

Separability and steering strength. This tests the prediction for general tasks: steering should need less strength where the answer classes are more separable. For each layer Y in the strength sweep (Section 5), we take the smallest strength at which steering with Y's own direction succeeds on half of the prompts. Layers where no strength succeeds are assigned twice the largest strength. Across layers, this required strength correlates negatively with the Fisher ratio in all 59 pairs (median Spearman $\rho = -0.83$ for α, -0.81 for c). Both quantities also change with depth. Controlling for the layer index, the partial rank correlation remains negative in 54 of 59 pairs for α and 57 of 59 for c (median -0.13 and -0.13).

J.3 Per-pair results

Table 18 lists the per-pair estimates and Figure 16 shows four examples. The crossover lies within one layer of a peak in 19 of 25 pairs. Five of the six exceptions are weekday tasks, where the crossover precedes the highest peak: in four it lies two layers after an earlier peak, and in Qwen2.5-14B it lies five layers before the earliest of the three highest peaks. This is expected from the uniform blur: the energy shifts toward the answer component before the configuration changes shape. Letter tasks show no crossover in Qwen2.5-7B, Llama, Mistral and Gemma: with 26 classes and offsets up to six, the blurred component dominates at every layer. In Qwen2.5-14B and Phi-4, a late crossover appears on letters (layers 38 and 30); in Phi-4 it lies six layers after the last of the three highest peaks, the sixth exception.

K Results Without Low-Accuracy Pairs

Four model–task pairs have accuracy below 0.55: Llama and Mistral on letters, Gemma on hours, and Mistral on emotions (Table 5). Table 19 repeats the main statistics of Sections 3–6 without them. Every result keeps its direction and remains significant where a test applies. The largest change in significance is in the comparison with logit-lens accessibility: lower-bound violations at $\alpha = 1$ become 3 against 15 ($p = 0.002$, from 0.001). The permutation test for the handoff uses 5,000 circular shifts here.

L Figures for the Other Models

The main text illustrates each result with Qwen2.5-7B, the model that writes the answer in one step. This appendix gives the counterparts of Figures 2–6 for Llama-3.1-8B, Mistral-7B, Gemma-2-9B, Qwen2.5-14B and Phi-4. Overview, event and handoff figures use a compact layout and show the tasks where each model's events are clearest; onset and transfer figures show all ten tasks, as in Figures 5 and 6. All pairs appear in Appendices E, F, G and I. Panel (c) of each handoff figure aggregates over all pairs and is the same for every model. In multi-step models, steering and transfer rise more gradually after the peak than in Qwen2.5-7B, as described in Sections 3.3 and 6.2.

L.1 Llama-3.1-8B

Llama writes the answer in several steps. On months, the highest peak (write, layer 18) coincides with the handoff, and steering and transfer rise over the following layers (Figure 17). On most retrieval tasks, a readable reorganization precedes the write peak by two to three layers (Figure 18a), and the handoff falls on or one layer after the write peak (Figure 19). Across all ten tasks, the steering onset at $\alpha = 2$ and the largest rise in transfer each lie within one layer of a peak in 8 tasks (Figures 20 and 21).

L.2 Mistral-7B

Mistral also writes in several steps, and in some tasks the last position lags all answer positions by up to eight layers (Figure 23d). On months, the highest peak (write, layer 20) lies one layer after the handoff (Figure 22); on weekdays and capital $\rightarrow$ continent, handoff and highest peak coincide (Figure 24). The steering onset at $\alpha = 2$ lies within one layer of a peak in 8 of 9 tasks, and the largest rise in transfer in all 9 (Figures 25 and 26); hours are excluded.

L.3 Gemma-2-9B

Gemma is the model where peaks and boundaries agree least. On months, the handoff (layer 29) precedes the three highest peaks, and the highest peak is a concentration of an answer that has already been written (Figure 27). On weekdays, a write peak (layer 28) precedes the concentration at the highest peak (layer 34); the handoff follows the write peak (Figures 28b and 29a). On retrieval tasks, the pattern matches the other models: a readable reorganization followed by a write peak at the handoff (Figures 28a and 29b). The steering onset at $\alpha = 2$ lies within one layer of a peak in 6 of 10 tasks, and the largest rise in transfer in 7 of 9 (Figures 30 and 31).

L.4 Qwen2.5-14B

Qwen2.5-14B writes most answers in one step, like Qwen2.5-7B: patching the last position takes effect at the same layer as patching all answer positions in 8 of 10 tasks. On landmark $\rightarrow$ city, a readable reorganization (layer 31) precedes the handoff (layer 32) and the highest peak (write, layer 34), where steering at $\alpha = 2$ and transfer set in (Figure 32). On weekdays, the handoff (layer 36) falls one layer before the highest peak (write, layer 37); on country $\rightarrow$ continent, handoff and highest peak coincide at layer 34 (Figure 34). The steering onset at $\alpha = 2$ and the largest rise in transfer each lie within one layer of a peak in 7 of 10 tasks; the exceptions are the two color tasks, where both lie seven or eight layers from a peak, and emotions (Figures 35 and 36).

L.5 Phi-4

Phi-4 writes in several steps: patching the last position takes effect at the same layer as patching all answer positions in only 5 of 10 tasks. On landmark $\rightarrow$ city, the highest peak (write, layer 22) coincides with the handoff, the steering onset and the rise in transfer (Figure 37). On weekdays, the highest peak (layer 19) has no event, and the handoff falls on the write peak at layer 21 (Figure 39a). The steering onset at $\alpha = 2$ lies within one layer of a peak in 9 of 10 tasks, and the largest rise in transfer in 7 of 10 (Figures 40 and 41).

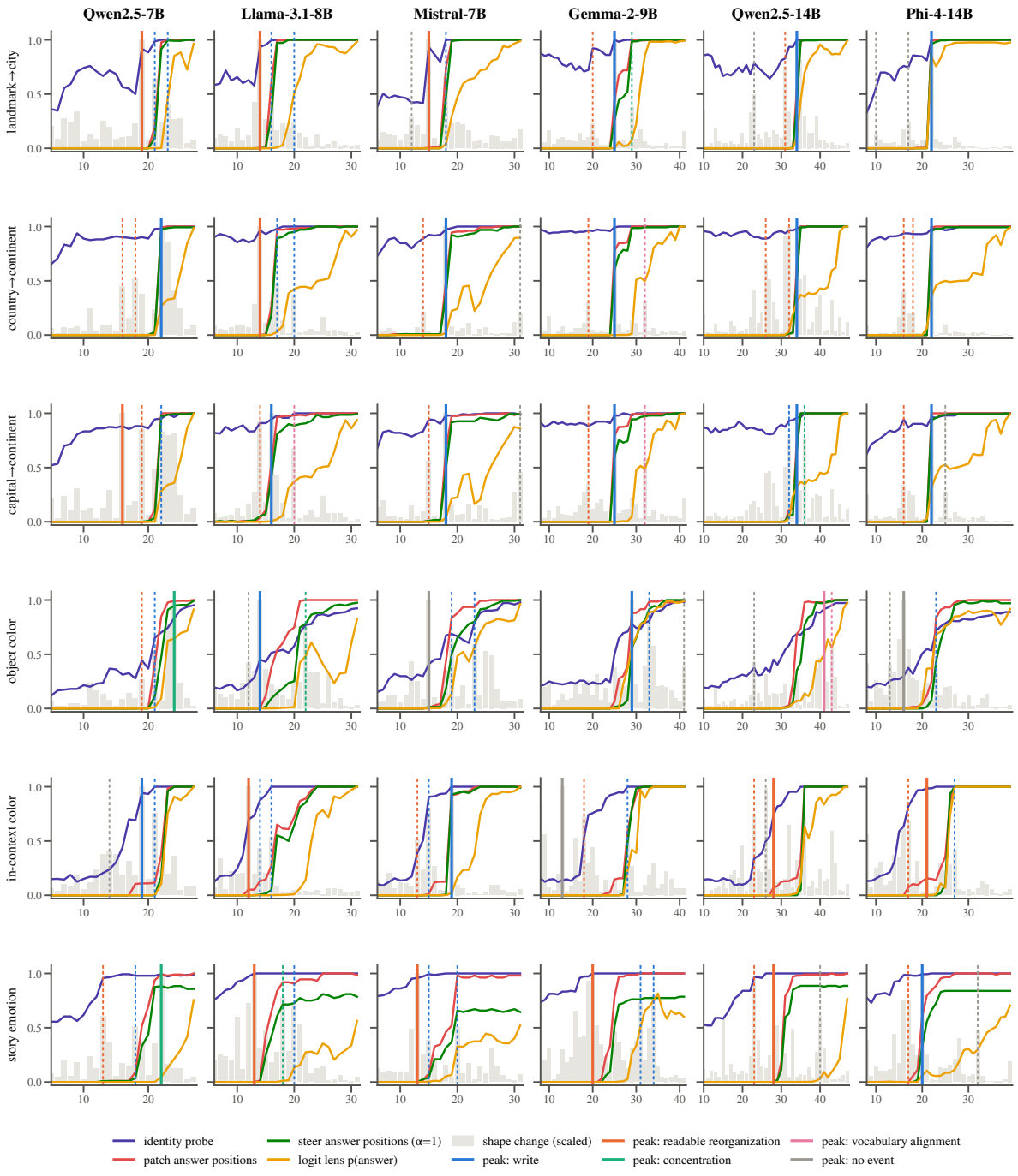


Figure 8: Signals at the answer positions for the retrieval tasks; layout as in Figure 7.

Task	Model	Top-3 peaks	Handoff patch [95%]	Handoff steer	Gap patch	Gap steer	n
Weekday	Qwen	12, 16, **23**	23 [23, 23]	23	0	0	120
	Llama	13, **17**, 22	18 [18, 18]	18	0	3	91
	Mistral	17, **20**, 25	20 [20, 20]	20	0	3	67
	Gemma	19, 28, **34**	29 [29, 29]	29	0	4	106
	Qwen-14B	32, **37**, 40	36 [36, 36]	36	0	1	115
	Phi-4	**19**, 21, 24	21 [21, 21]	21	0	6	113
Month	Qwen	14, **23**, 25	23 [23, 23]	23	0	0	216
	Llama	16, **18**, 21	18 [18, 18]	18	0	3	206
	Mistral	17, **20**, 25	19 [19, 19]	19	1	4	170
	Gemma	31, **33**, 36	29 [29, 29]	29	0	4	213
	Qwen-14B	**37**, 40, 42	36 [36, 36]	36	0	2	216
	Phi-4	19, 22, **24**	21 [21, 21]	21	0	4	204
Hour	Qwen	7, 19, **23**	22 [22, 22]	22	0	1	91
	Llama	14, **16**, 24	16 [16, 16]	16	2	2	199
	Gemma	28, 31, **33**	29 [29, 29]	29	0	4	52
	Qwen-14B	32, **37**, 40	36 [35, 36]	35	0	3	83
	Phi-4	**21**, 24, 28	21 [21, 21]	21	0	5	198
Letter	Qwen	14, **23**, 27	22 [22, 22]	22	0	1	317
	Llama	16, **18**, 21	18 [18, 18]	18	0	0	153
	Mistral	**19**, 28, 31	20 [20, 20]	20	0	1	148
	Gemma	16, **27**, 33	29 [29, 29]	29	0	3	285
	Qwen-14B	32, **37**, 40	36 [36, 36]	36	0	2	387
	Phi-4	17, **19**, 24	21 [20, 21]	20	0	6	330
Landmark → city	Qwen	**19**, 21, 23	21 [19, 21]	21	1	1	210
	Llama	**14**, 16, 20	16 [15, 16]	16	0	1	212
	Mistral	12, **15**, 18	18 [16, 18]	18	0	1	214
	Gemma	20, **25**, 29	25 [25, 25]	25	1	3	214
	Qwen-14B	23, 31, **34**	32 [31, 33]	34	2	1	214
	Phi-4	10, 17, **22**	22 [22, 22]	22	0	0	211
Country → continent	Qwen	16, 18, **22**	22 [22, 22]	22	0	0	142
	Llama	**14**, 17, 20	17 [16, 17]	16	0	1	138
	Mistral	14, **18**, 31	18 [18, 18]	18	1	1	129
	Gemma	19, **25**, 32	25 [25, 25]	25	0	0	144
	Qwen-14B	26, 32, **34**	34 [34, 34]	34	0	0	144
	Phi-4	16, 18, **22**	22 [22, 22]	22	0	0	144
Capital → continent	Qwen	**16**, 19, 22	22 [21, 22]	22	0	0	144
	Llama	14, **16**, 20	16 [16, 16]	15	0	2	141
	Mistral	15, **18**, 31	18 [18, 18]	18	0	1	122
	Gemma	19, **25**, 32	25 [25, 25]	25	0	0	144
	Qwen-14B	32, **34**, 36	34 [32, 34]	32	0	2	144
	Phi-4	16, **22**, 25	22 [22, 22]	22	0	0	144
Object → color	Qwen	19, 21, **24**	21 [21, 21]	21	1	2	133
	Llama	12, **14**, 22	16 [15, 16]	–	1	–	119
	Mistral	**15**, 19, 23	17 [15, 18]	–	2	–	140
	Gemma	**29**, 33, 41	26 [26, 27]	25	3	4	143
	Qwen-14B	23, **41**, 43	34 [33, 34]	33	0	3	140
	Phi-4	13, **16**, 23	23 [22, 23]	21	0	3	134
In-context color	Qwen	14, **19**, 21	21 [21, 22]	23	2	0	192
	Llama	**12**, 14, 16	16 [15, 17]	20	1	-3	192
	Mistral	13, 15, **19**	16 [14, 17]	19	3	0	192
	Gemma	**13**, 18, 28	27 [26, 28]	29	1	0	192
	Qwen-14B	23, 26, **28**	34 [32, 34]	36	2	0	192
	Phi-4	17, **21**, 27	25 [24, 25]	26	1	0	192
Story → emotion	Qwen	13, 18, **22**	–	–	–	–	99
	Llama	**13**, 18, 20	–	–	–	–	74
	Mistral	**13**, 15, 20	–	–	–	–	52
	Gemma	**20**, 31, 34	–	–	–	–	69
	Qwen-14B	23, **28**, 40	–	–	–	–	101
	Phi-4	17, **20**, 32	–	–	–	–	89

Table 8: Handoff of causal control for every pair. Top-3 peaks: the highest in bold. Handoff: first layer at which intervening on the input token falls below half of its maximum success. Gap: layers from the handoff to the first layer at which intervening on all answer positions succeeds on half of the prompts. –: no single input token controls the answer (emotion stories), or steering of the input never reaches 0.2. n: prompts with a matched patching partner. Mistral on hours is excluded.

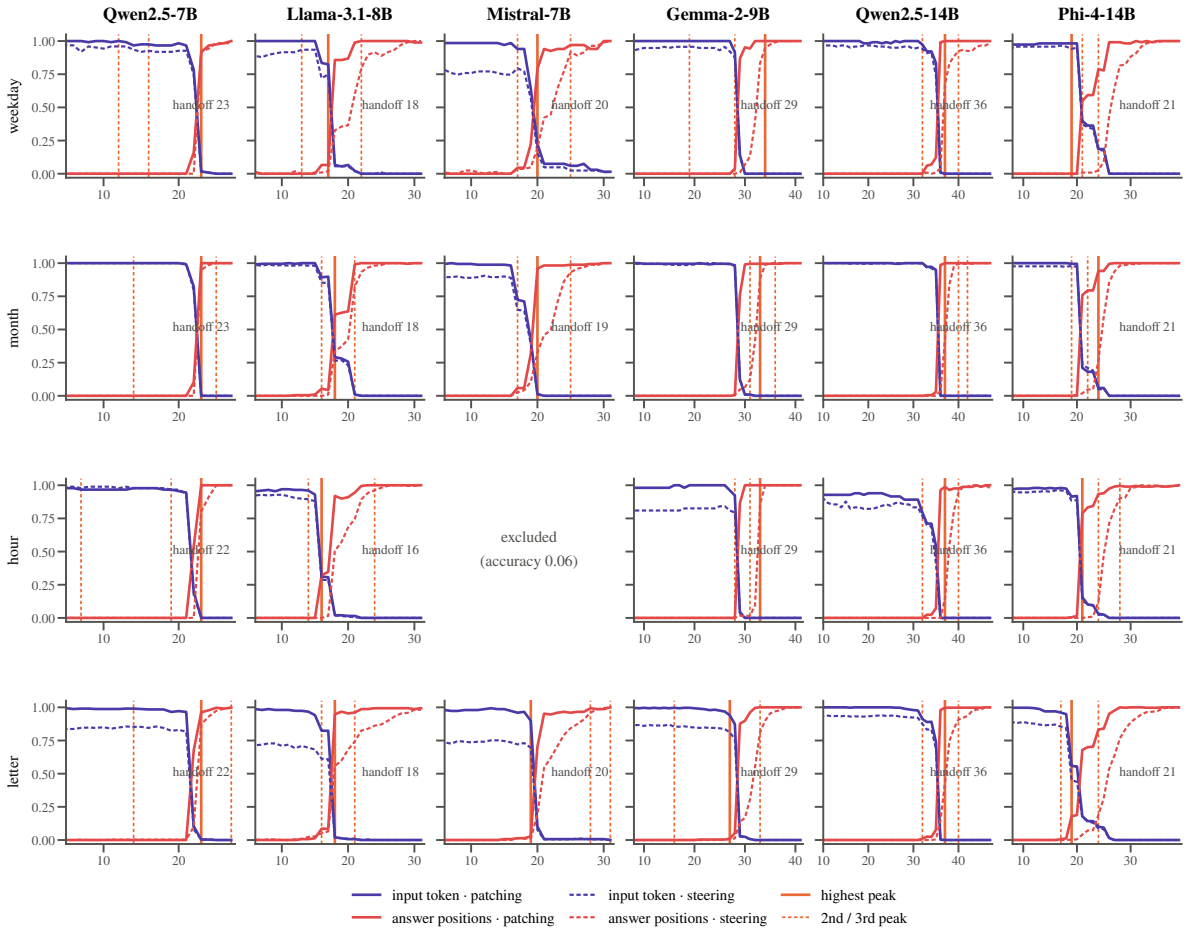


Figure 9: Handoff curves for the cyclic-arithmetic pairs: patching (solid) and steering (dashed) of the input token (violet) and of the answer positions (red). Orange lines mark the three highest peaks (solid: highest). Each panel gives the handoff layer.

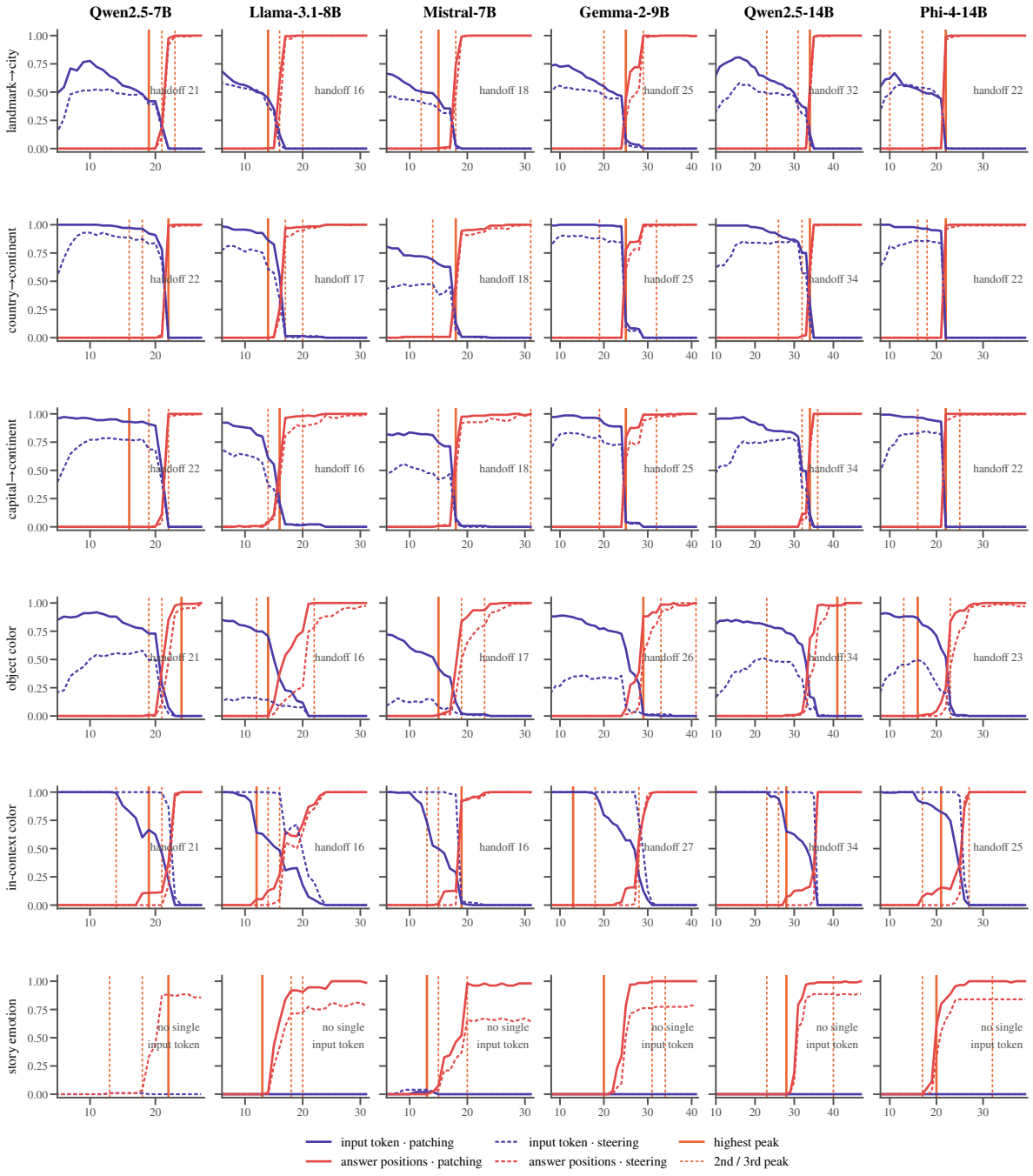


Figure 10: Handoff curves for the retrieval pairs, as in Figure 9. Emotion stories have no single input token that controls the answer, so no handoff layer is defined.

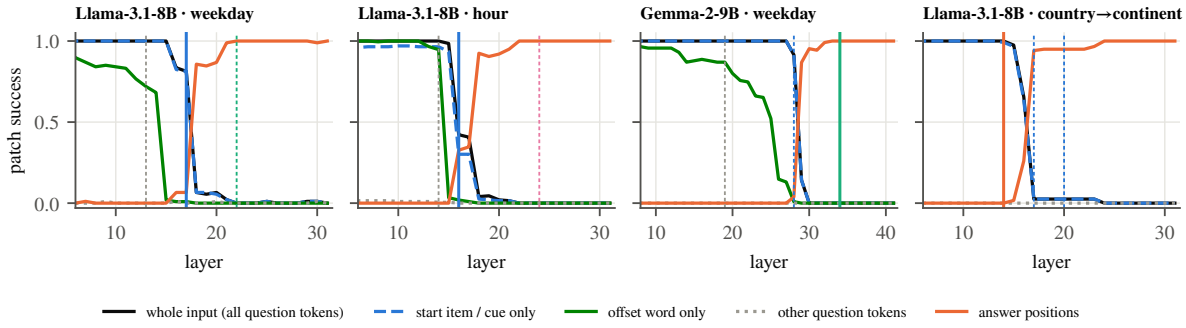


Figure 11: Patching different parts of the input. The whole question (black) and the start item or cue alone (blue, dashed) coincide; the offset word (green) hands off one to three layers earlier; other question tokens (gray, dotted) never change the answer. Orange: answer positions. Vertical lines: the three highest peaks.

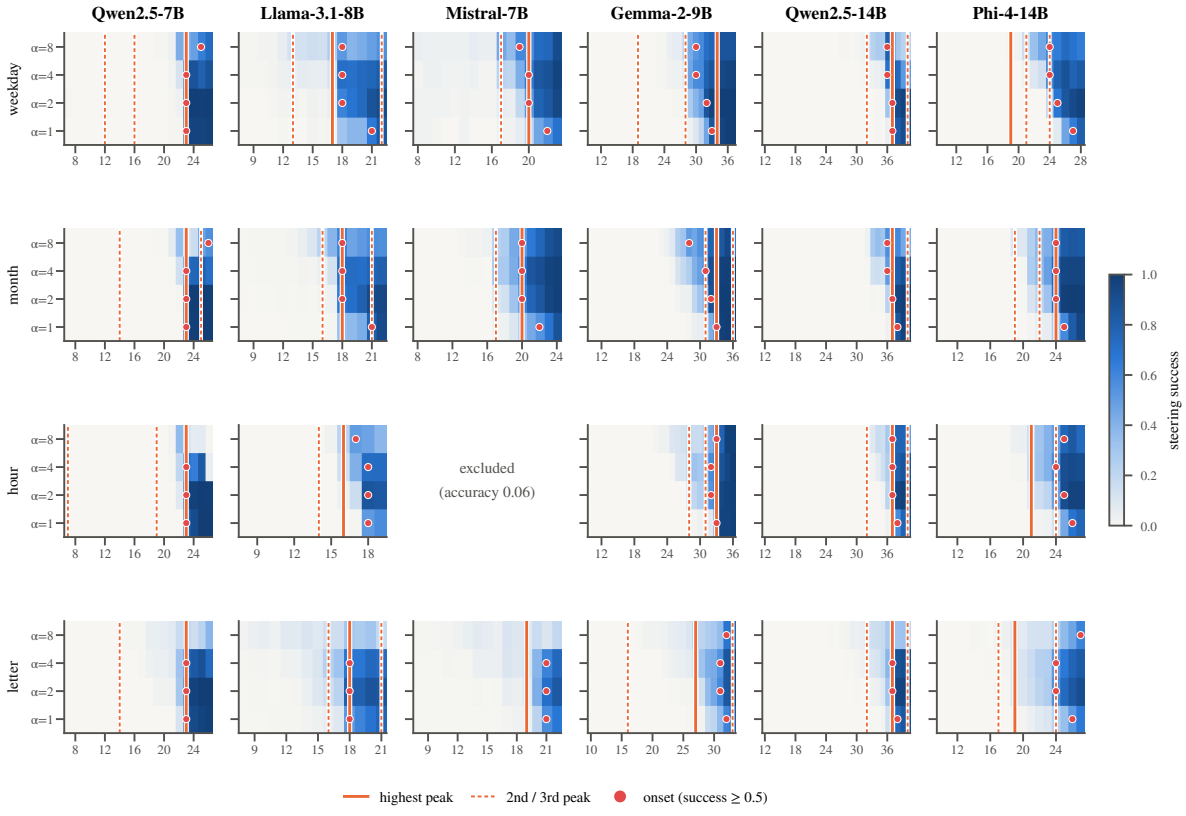


Figure 12: Steering success at the answer positions for the cyclic-arithmetic pairs, by layer (horizontal) and strength α (vertical), using each layer's own centroid-difference direction. Red dots: onset at each strength. Orange lines mark the three highest peaks (solid: highest).

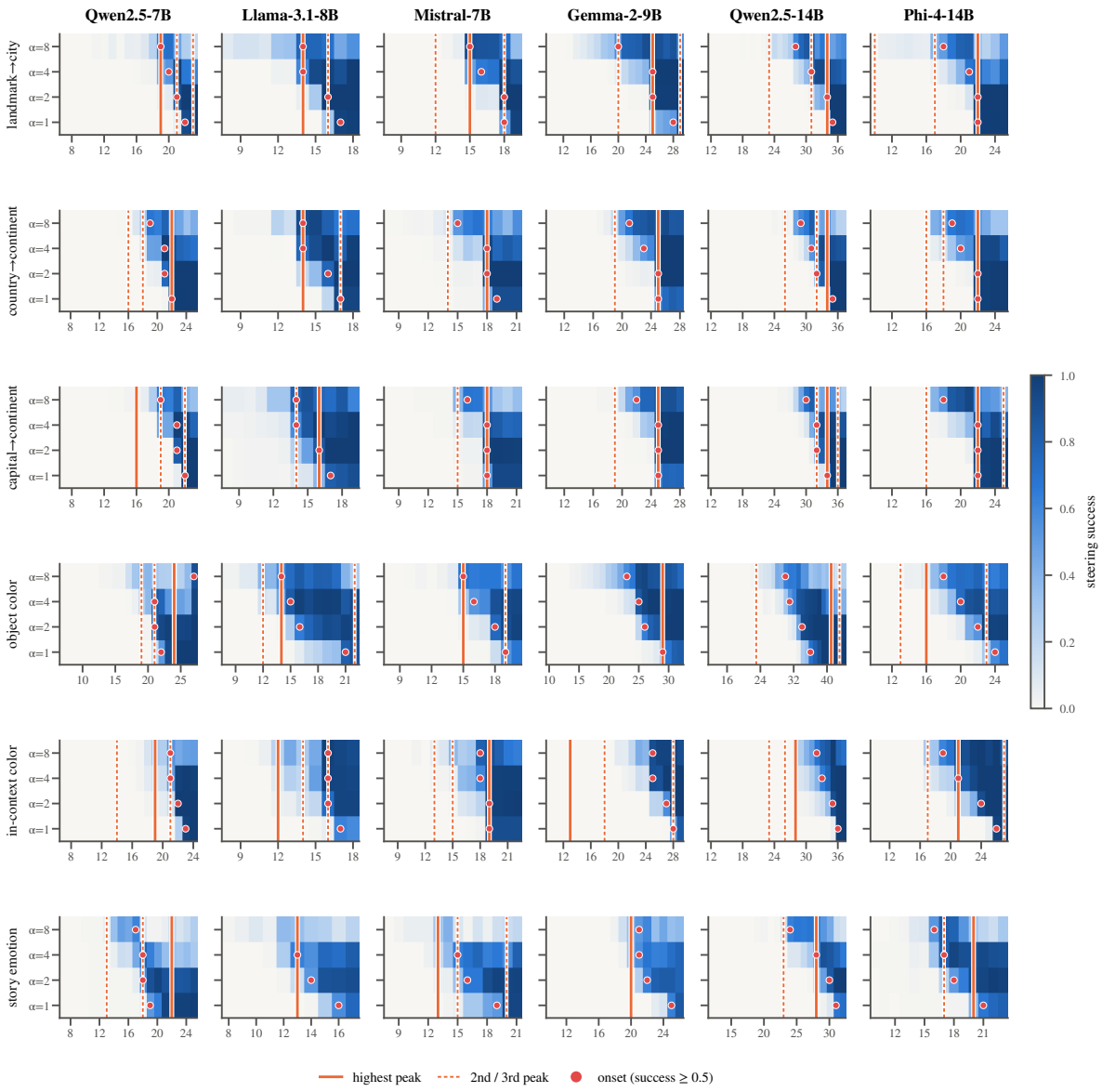


Figure 13: Steering success for the retrieval pairs, as in Figure 12.

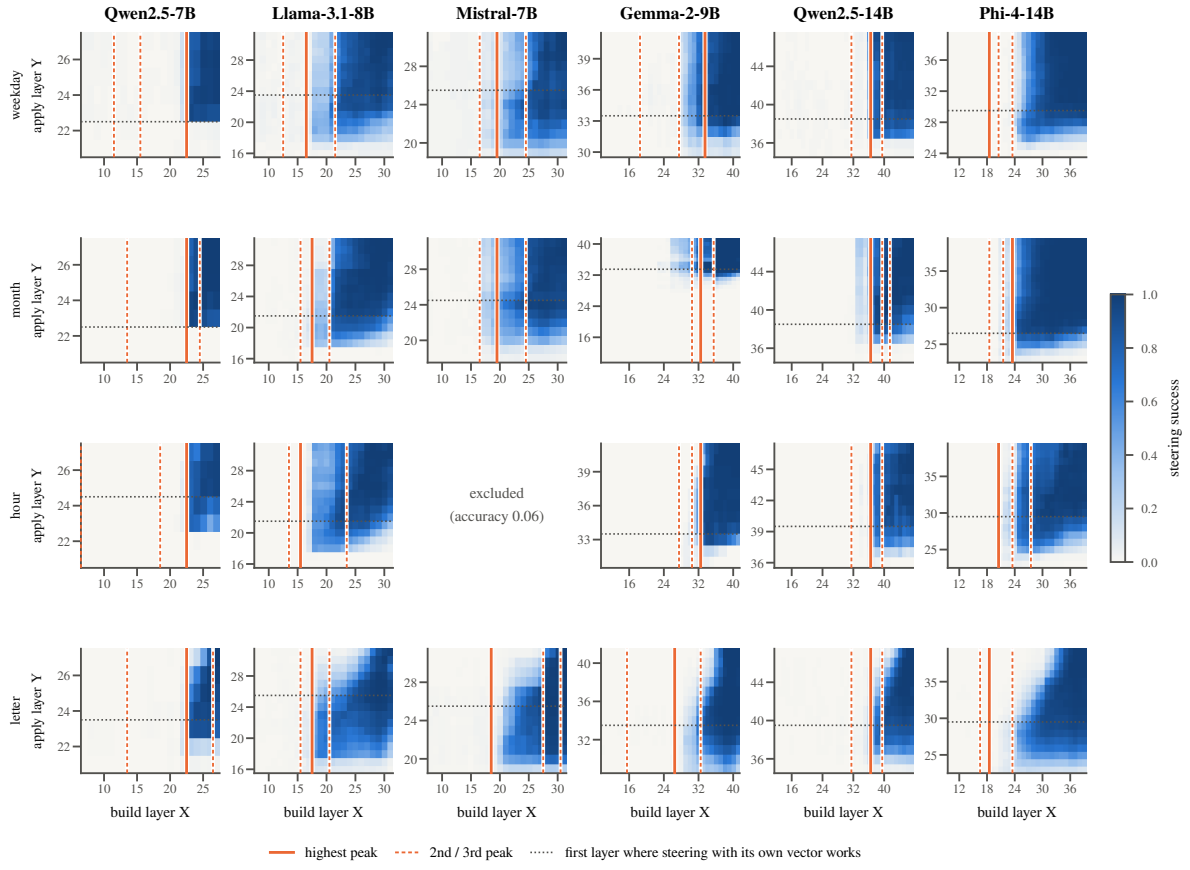


Figure 14: Cross-layer transfer for the cyclic-arithmetic pairs: success when the vector built at layer X (horizontal) is applied at layer Y (vertical). Orange lines mark the three highest peaks (solid: highest); below the dotted line, steering fails even with Y's own vector.

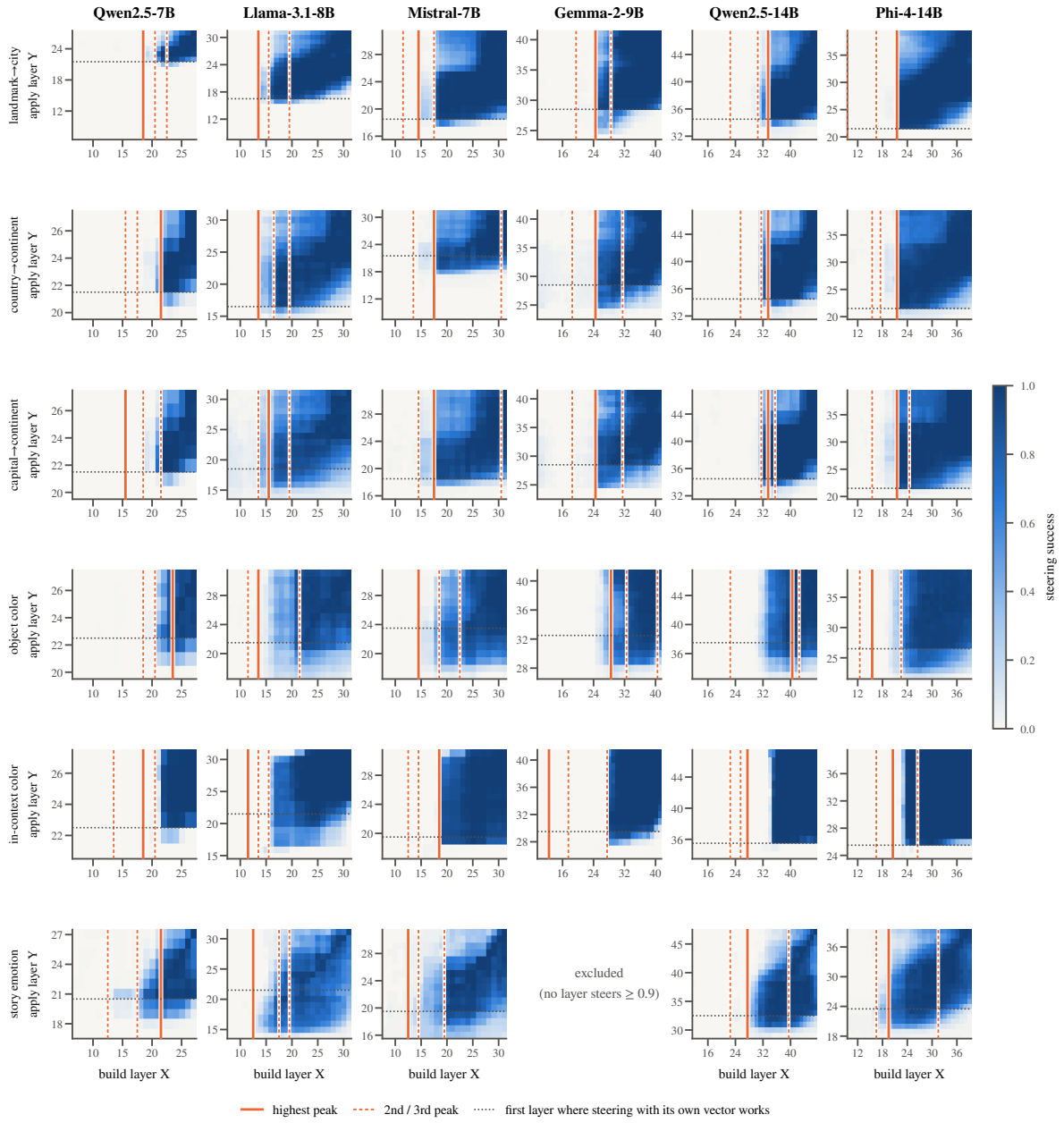


Figure 15: Cross-layer transfer for the retrieval pairs, as in Figure 14.

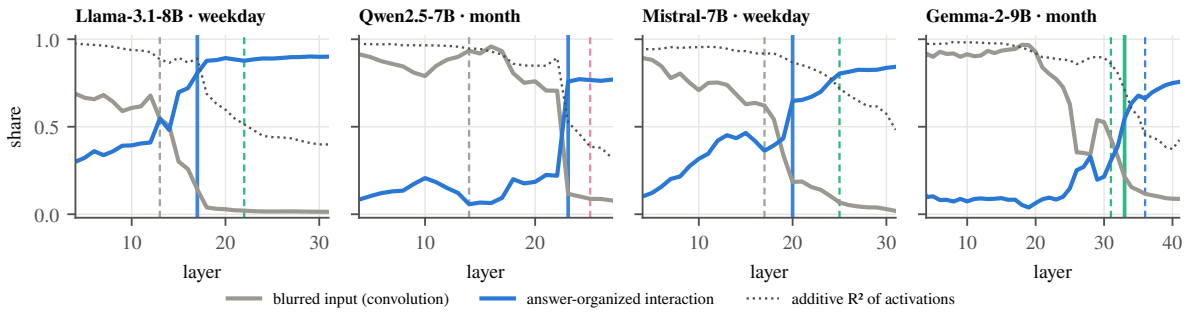


Figure 16: Share of the answer-centroid energy carried by the blurred input (gray) and by the answer-organized component (blue), and the additive R^2 of the activations (dotted), across layers. Vertical lines mark the three highest shape-change peaks (solid, highest peak).

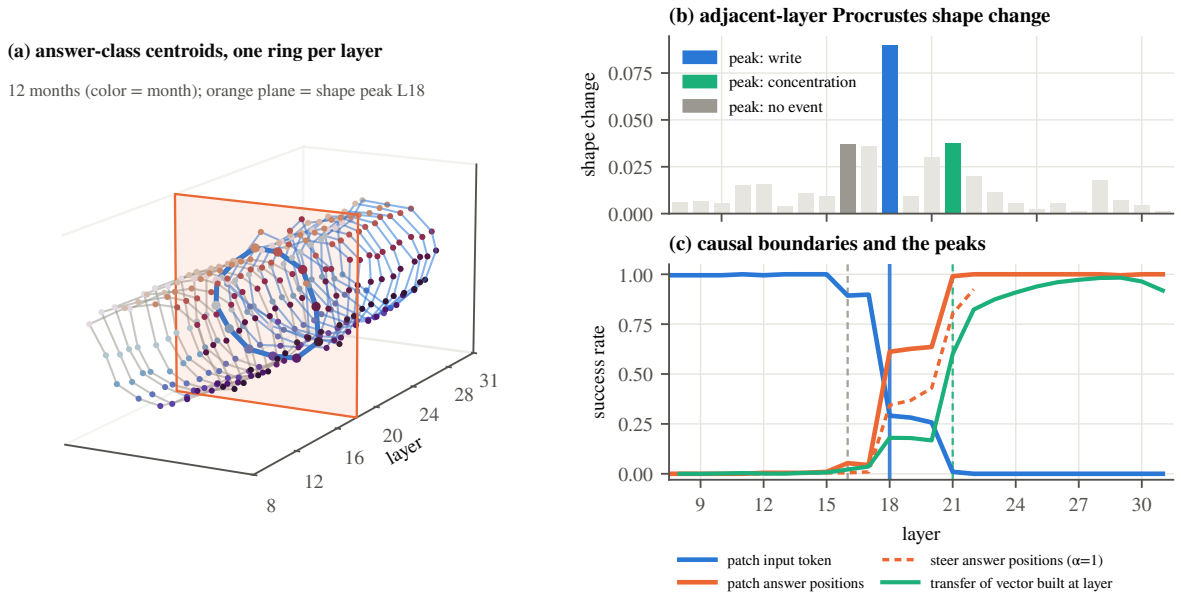


Figure 17: Overview for Llama-3.1-8B (counterpart of Figure 2). Month: the highest peak (write, layer 18) coincides with the handoff (layer 18). Steering and transfer rise over the following layers, because Llama relays the answer through chat-template positions before it reaches the last token.

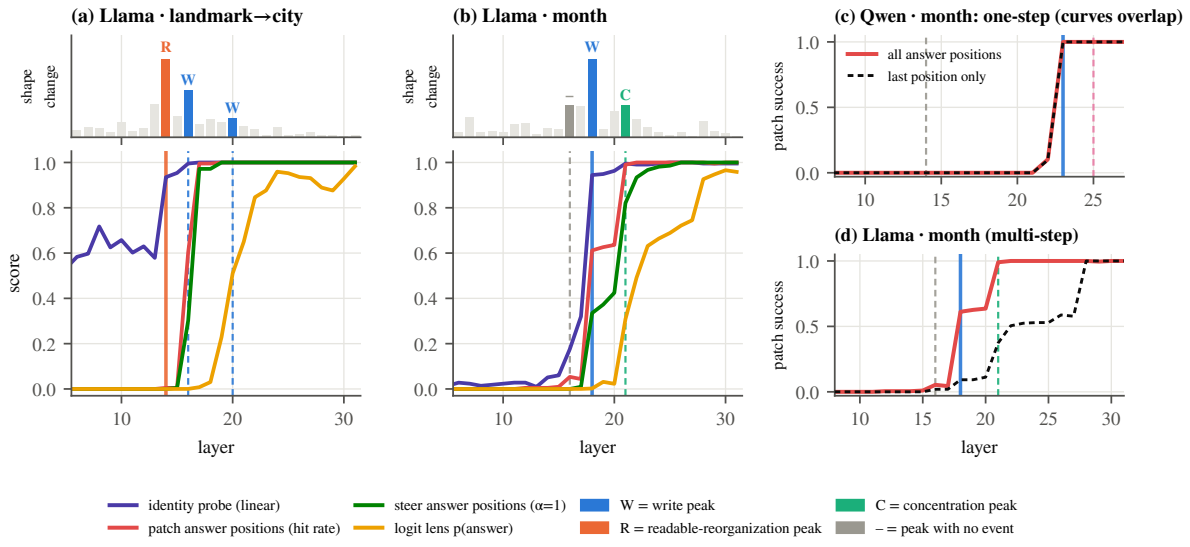


Figure 18: Events at peaks for Llama-3.1-8B (counterpart of Figure 3). (a) Landmark → city: readable reorganization at layer 14, write at layer 16. (b) Month: write at layer 18. (c, d) One-step (Qwen2.5-7B) versus multi-step (Llama) writing.

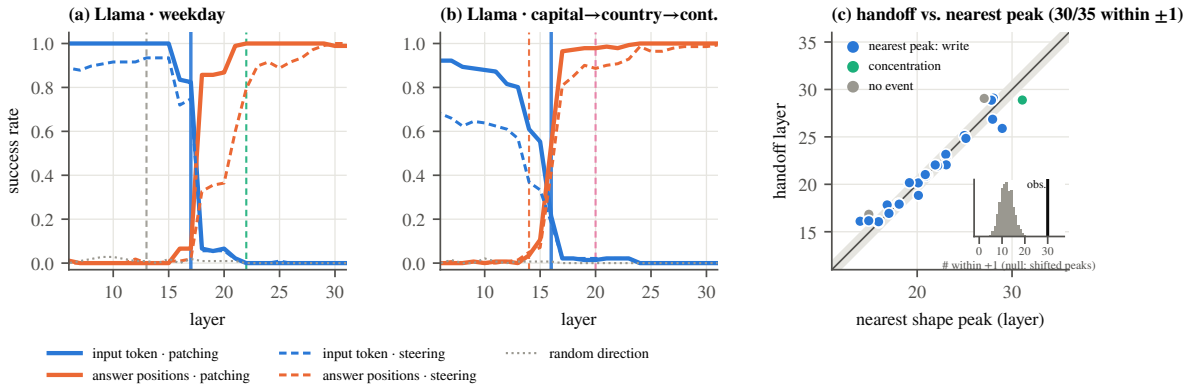


Figure 19: Handoff for Llama-3.1-8B (counterpart of Figure 4). (a) Weekday: the handoff (layer 18) falls one layer after the highest peak (write, layer 17). (b) Capital → continent: the handoff falls on the write peak at layer 16, after a readable reorganization at layer 14.

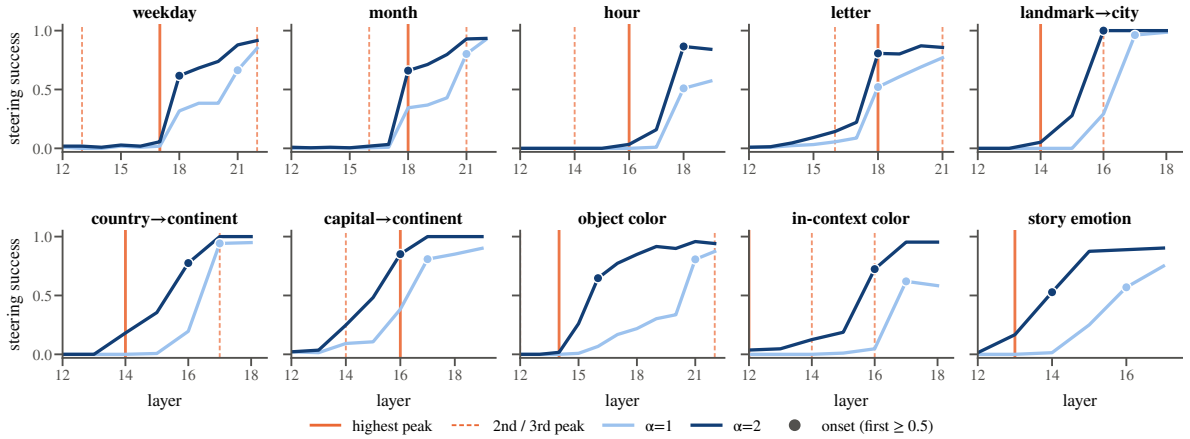


Figure 20: Steering onset for Llama-3.1-8B, all tasks (as Figure 5). The onset at $\alpha = 2$ lies within one layer of a peak in 8 of 10 tasks.

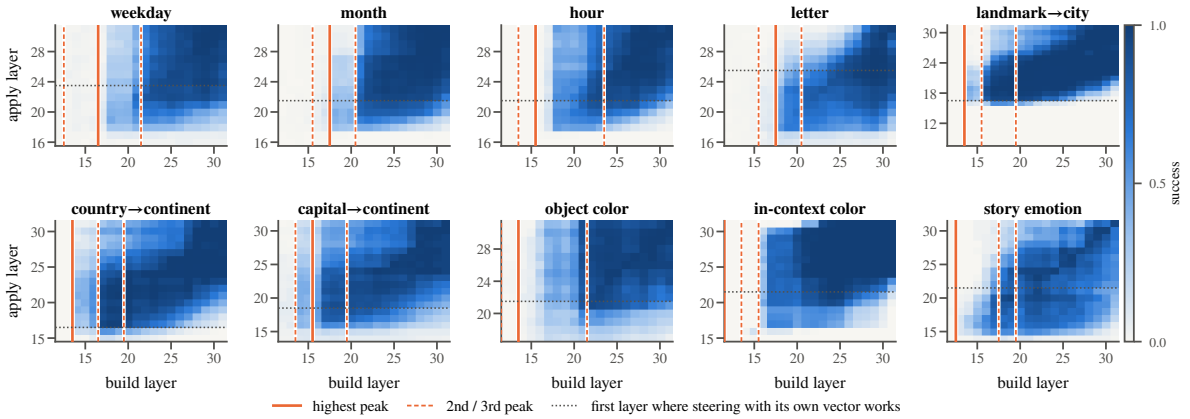


Figure 21: Transfer for Llama-3.1-8B, all tasks (as Figure 6). The largest one-layer rise in transfer lies within one layer of a peak in 8 of 10 tasks.

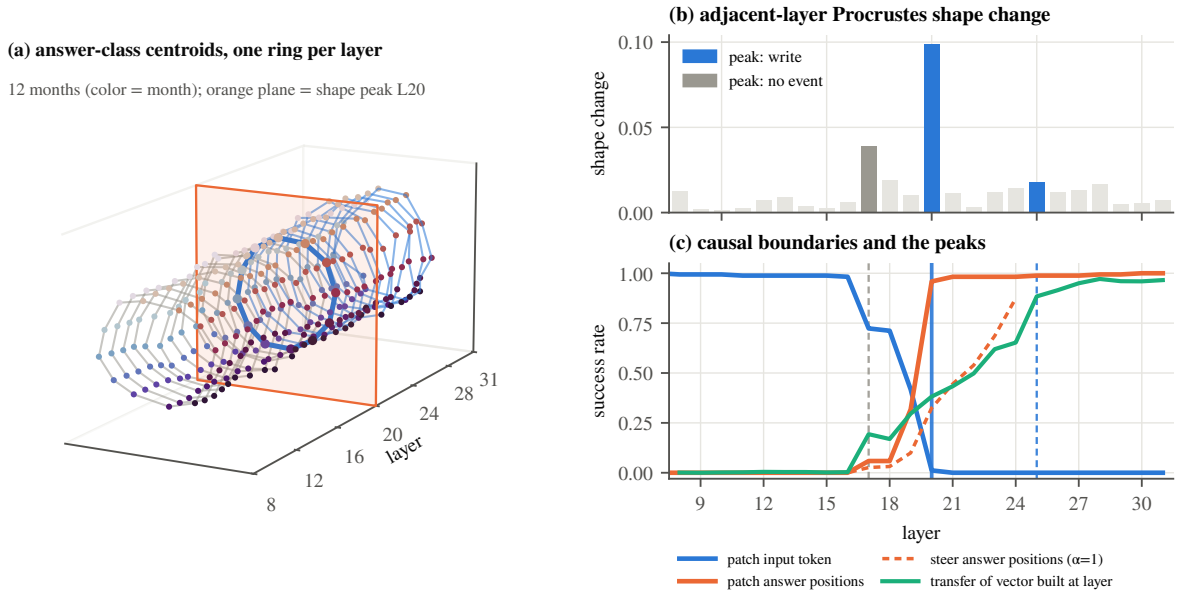


Figure 22: Overview for Mistral-7B (counterpart of Figure 2). Month: the highest peak (write, layer 20) lies one layer after the handoff (layer 19).

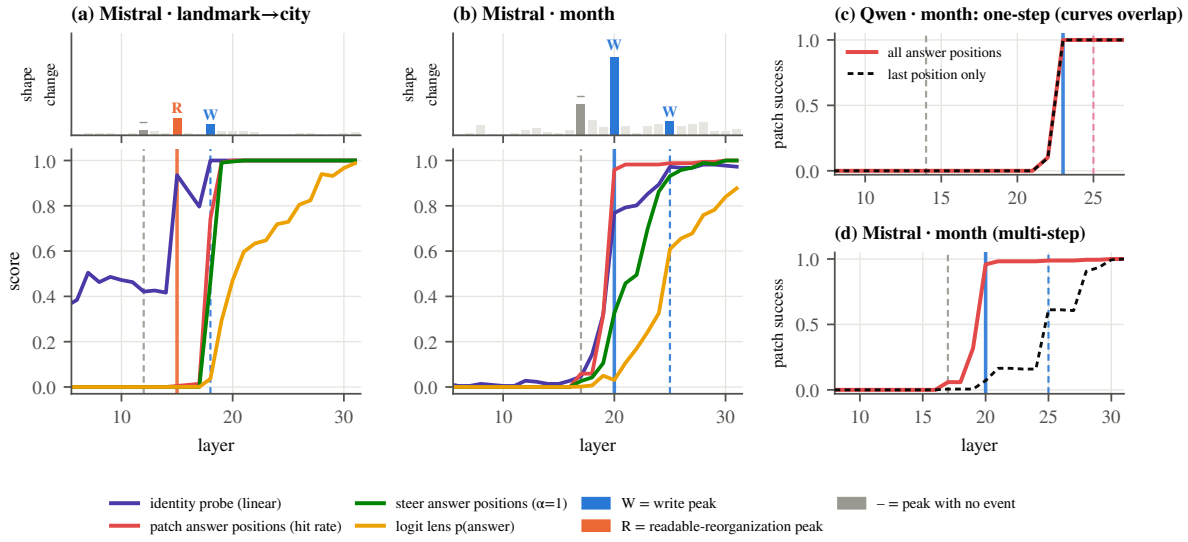


Figure 23: Events at peaks for Mistral-7B (counterpart of Figure 3). (a) Landmark → city: readable reorganization at layer 15, write at layer 18. (b) Month: write at layer 20. (d) In Mistral, patching the last position alone takes effect several layers after patching all answer positions.

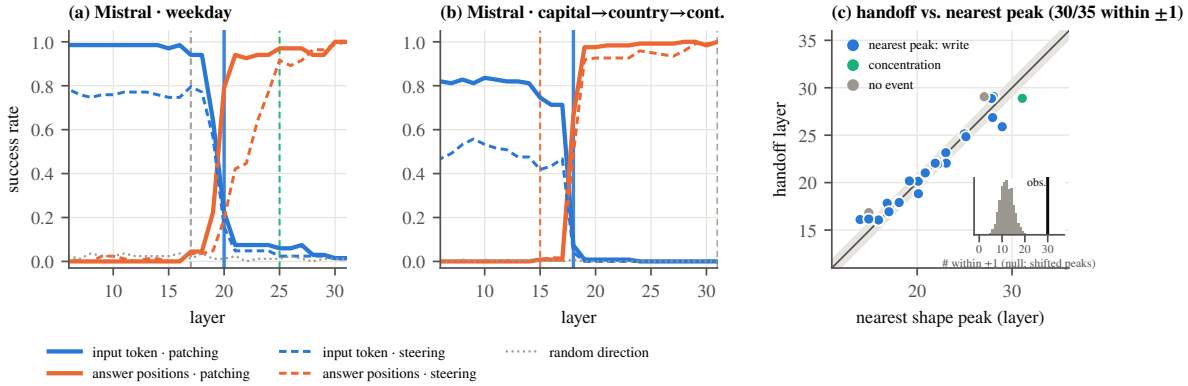


Figure 24: Handoff for Mistral-7B (counterpart of Figure 4). (a) Weekday: handoff and highest peak at layer 20. (b) Capital → continent: handoff and highest peak at layer 18.

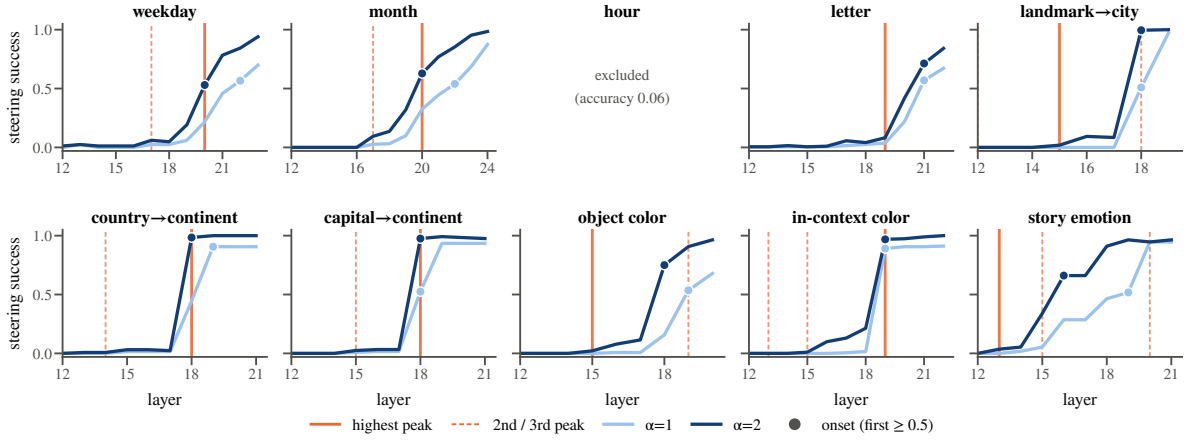


Figure 25: Steering onset for Mistral-7B, all tasks (as Figure 5). The onset at $\alpha = 2$ lies within one layer of a peak in 8 of 9 tasks; hours are excluded.

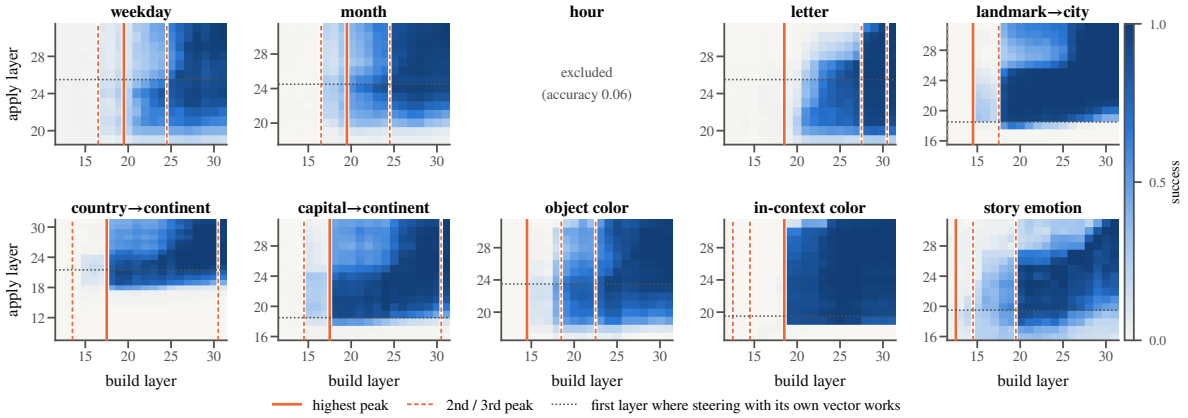


Figure 26: Transfer for Mistral-7B, all tasks (as Figure 6). The largest one-layer rise in transfer lies within one layer of a peak in all 9 tasks; hours are excluded.

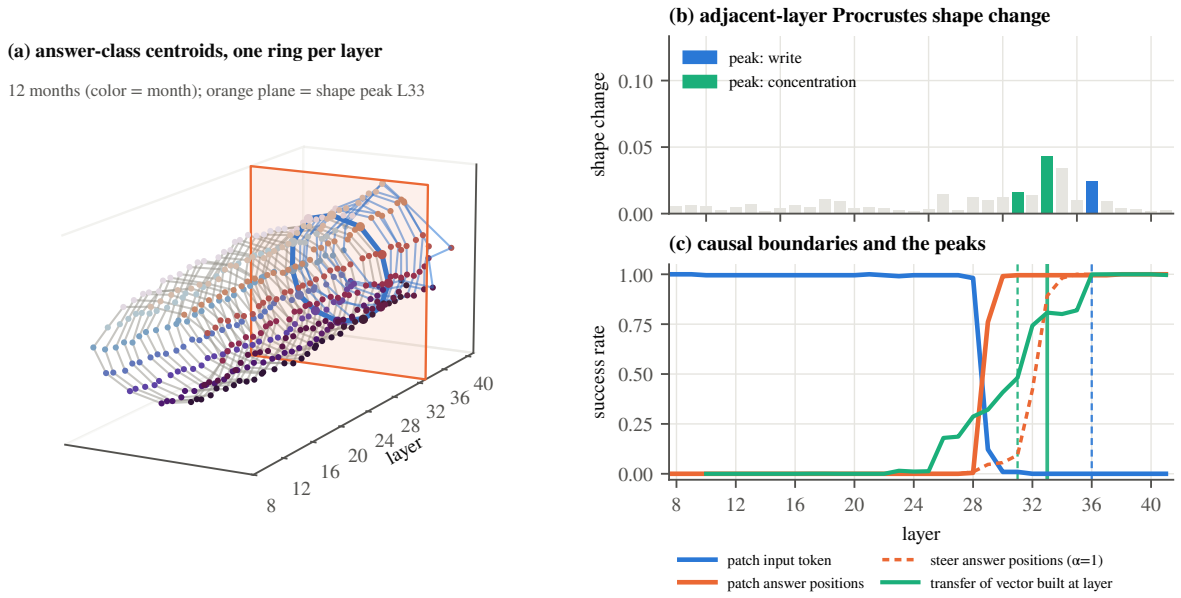


Figure 27: Overview for Gemma-2-9B (counterpart of Figure 2). Month: the handoff (layer 29) precedes the three highest peaks (31, 33, 36); the highest peak (layer 33) is a concentration of the answer that has already been written. This is one of the five pairs whose handoff lies two or three layers from a peak (Appendix F).

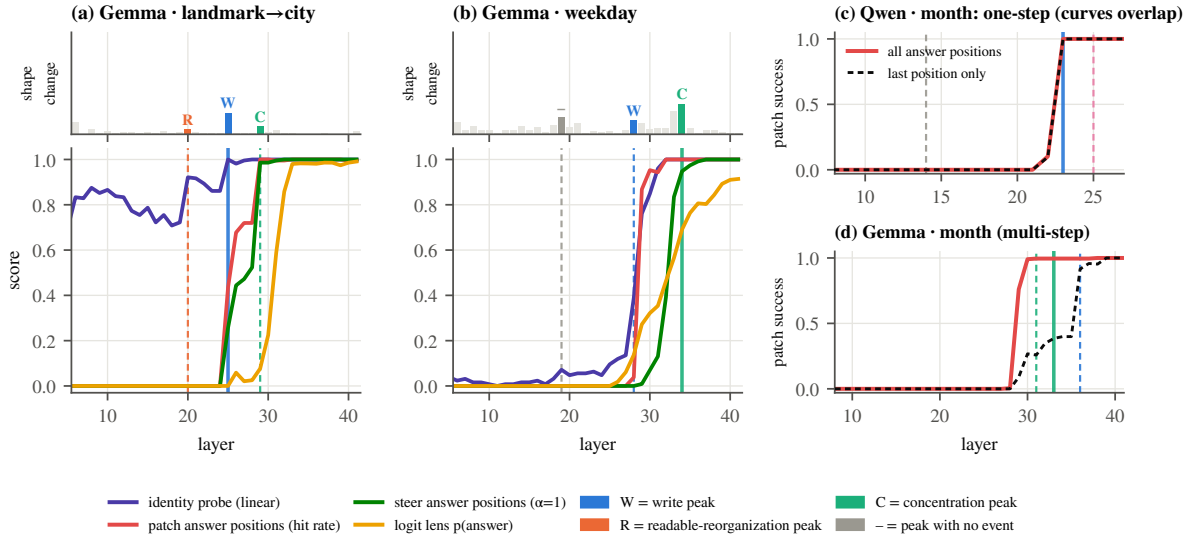


Figure 28: Events at peaks for Gemma-2-9B (counterpart of Figure 3). (a) Landmark → city: readable reorganization at layer 20, write at layer 25. (b) Weekday: write at layer 28, concentration at the highest peak (layer 34). (d) In Gemma, the last position lags all answer positions by the largest margin among our models.

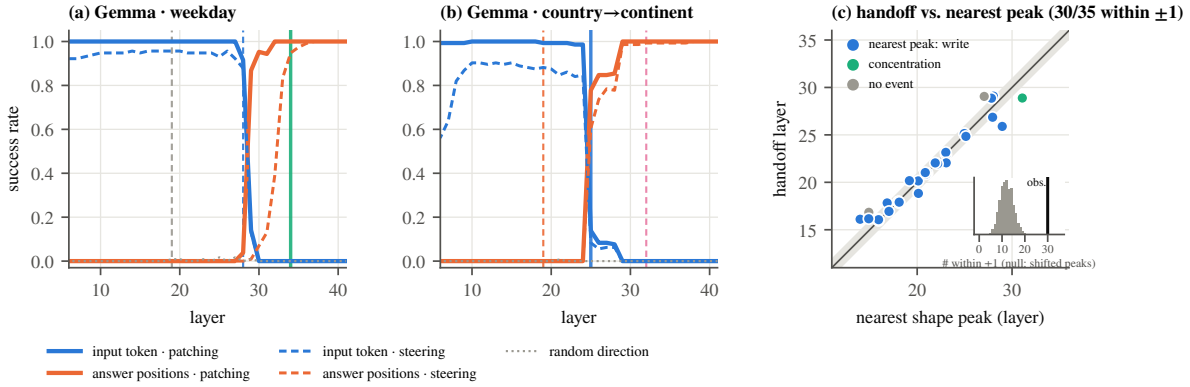


Figure 29: Handoff for Gemma-2-9B (counterpart of Figure 4). (a) Weekday: the handoff (layer 29) falls within one layer of the write peak at layer 28, before the concentration peak at layer 34. (b) Country → continent: handoff and highest peak at layer 25.

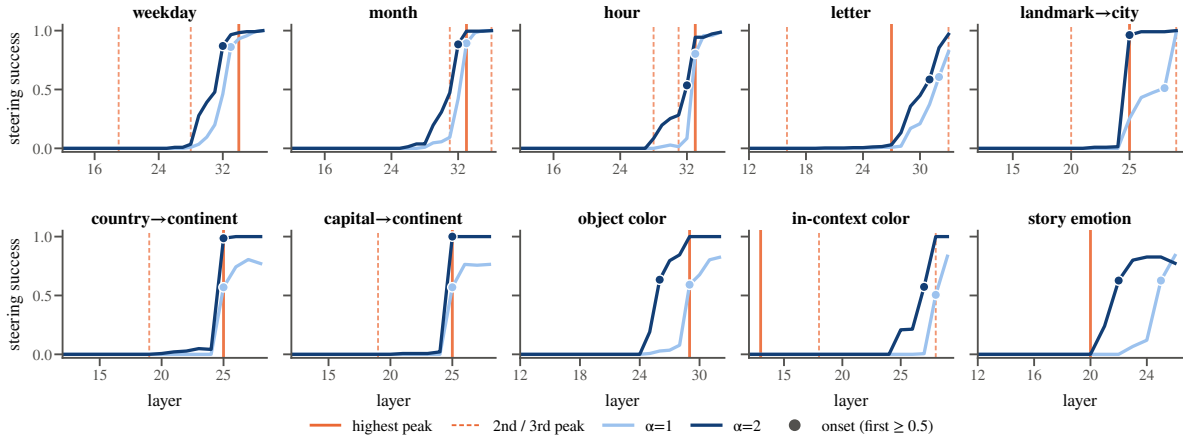


Figure 30: Steering onset for Gemma-2-9B, all tasks (as Figure 5). The onset at $\alpha = 2$ lies within one layer of a peak in 6 of 10 tasks; the other four lie two to three layers from one.

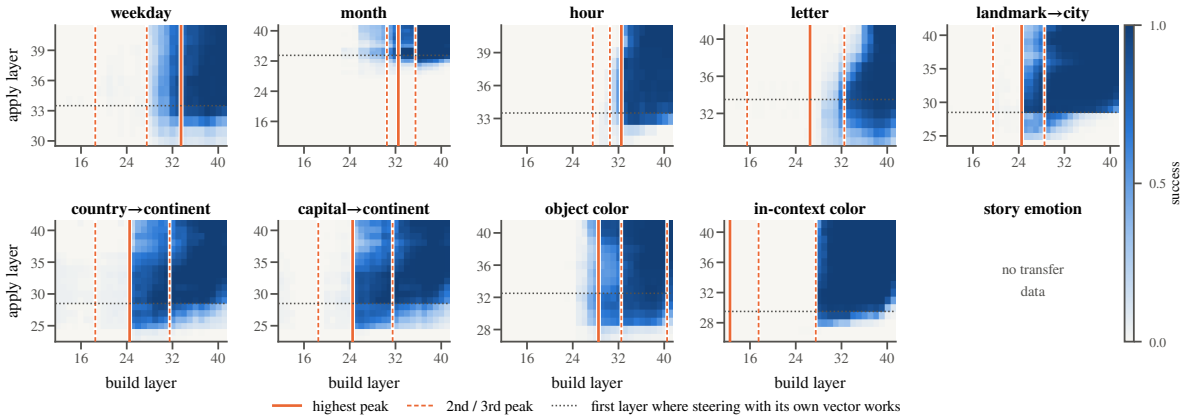


Figure 31: Transfer for Gemma-2-9B, all tasks (as Figure 6). The largest one-layer rise in transfer lies within one layer of a peak in 7 of 9 tasks; there is no transfer data for emotions.

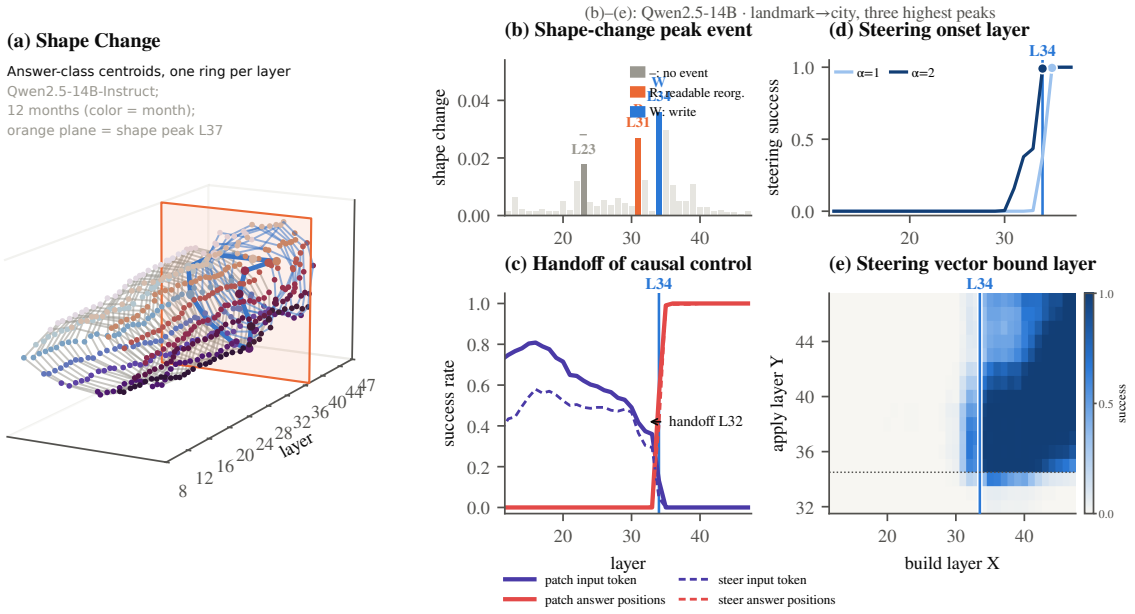


Figure 32: Overview for Qwen2.5-14B (counterpart of Figure 2). (a) Month: highest peak at layer 37. (b–e) Landmark → city: a readable reorganization at layer 31 precedes the handoff (layer 32); steering at $\alpha = 2$ succeeds and transfer sets in at the highest peak (write, layer 34).

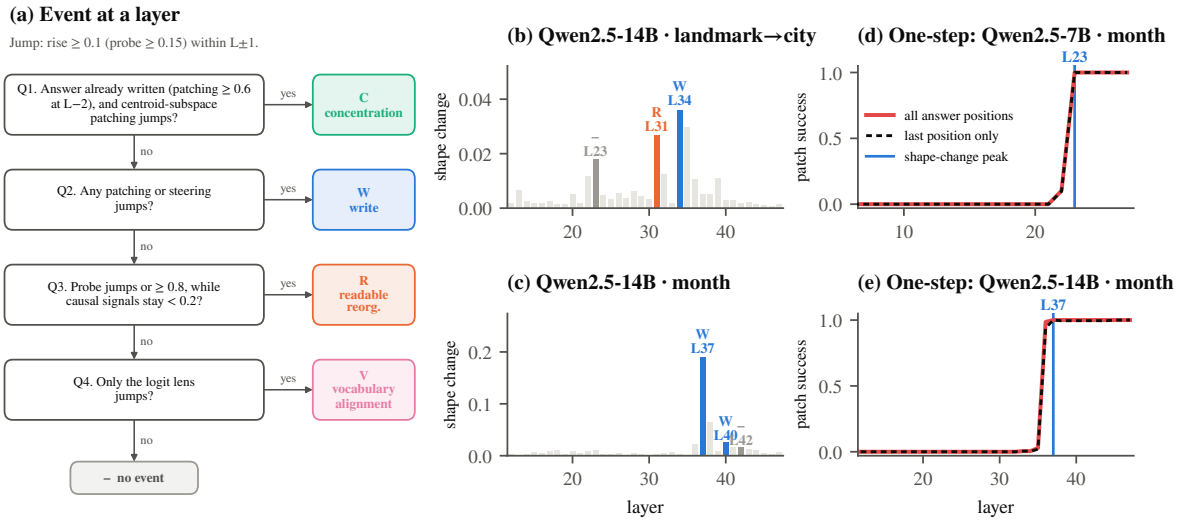


Figure 33: Events at peaks for Qwen2.5-14B (counterpart of Figure 3; (a) is the decision tree and (d) the Qwen2.5-7B reference). (b) Landmark → city: readable reorganization at layer 31, write at layer 34. (c) Month: write at layer 37. (e) As in Qwen2.5-7B, patching the last position takes effect at the same layer as patching all answer positions (layer 36).

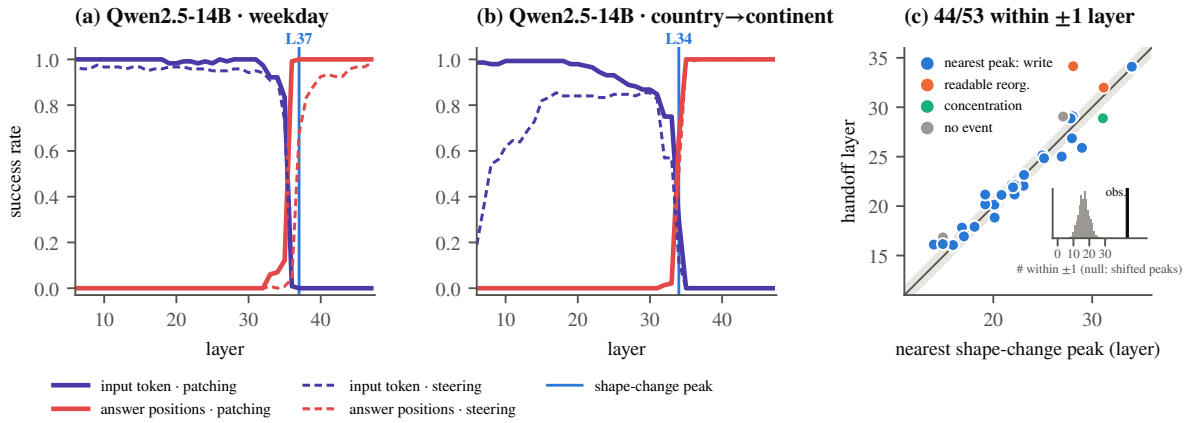


Figure 34: Handoff for Qwen2.5-14B (counterpart of Figure 4). (a) Weekday: the handoff (layer 36) falls one layer before the highest peak (write, layer 37). (b) Country → continent: handoff and highest peak (write) at layer 34, after readable reorganizations at layers 26 and 32.

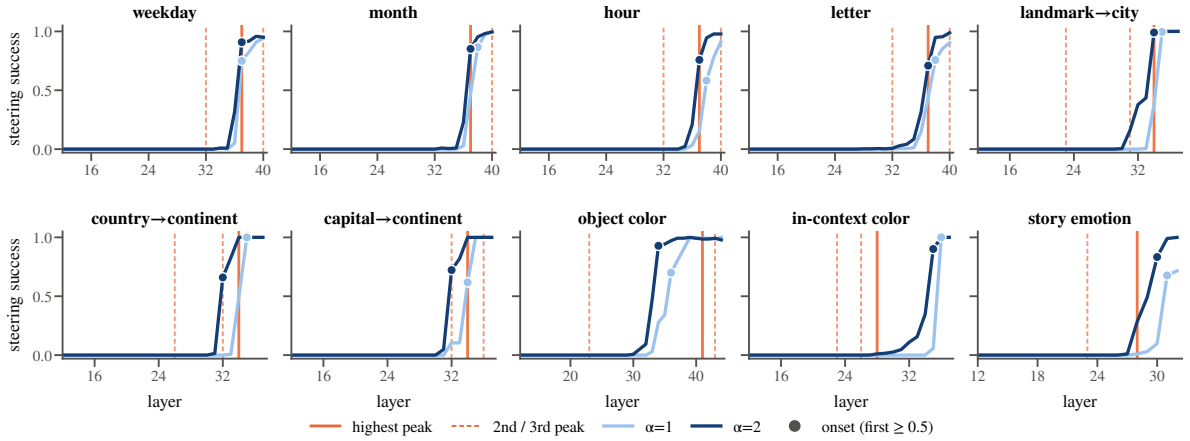


Figure 35: Steering onset for Qwen2.5-14B, all tasks (as Figure 5). The onset at $\alpha = 2$ lies within one layer of a peak in 7 of 10 tasks; in the two color tasks it lies seven layers from one.

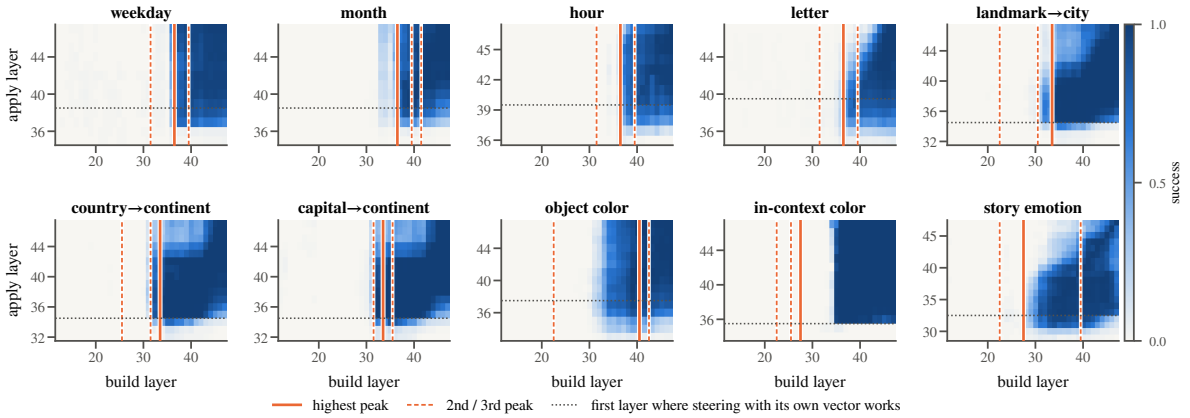


Figure 36: Transfer for Qwen2.5-14B, all tasks (as Figure 6). The largest one-layer rise in transfer lies within one layer of a peak in 7 of 10 tasks.

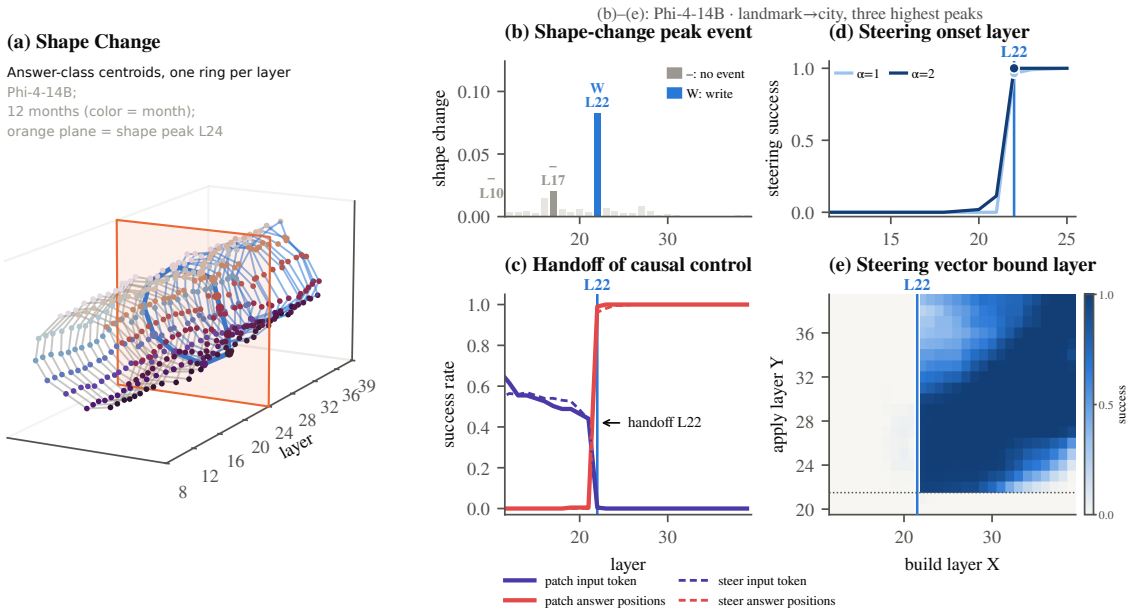


Figure 37: Overview for Phi-4 (counterpart of Figure 2). (a) Month: highest peak at layer 24. (b–e) Landmark → city: the highest peak (write, layer 22) coincides with the handoff, the steering onset and the rise in transfer (all layer 22).

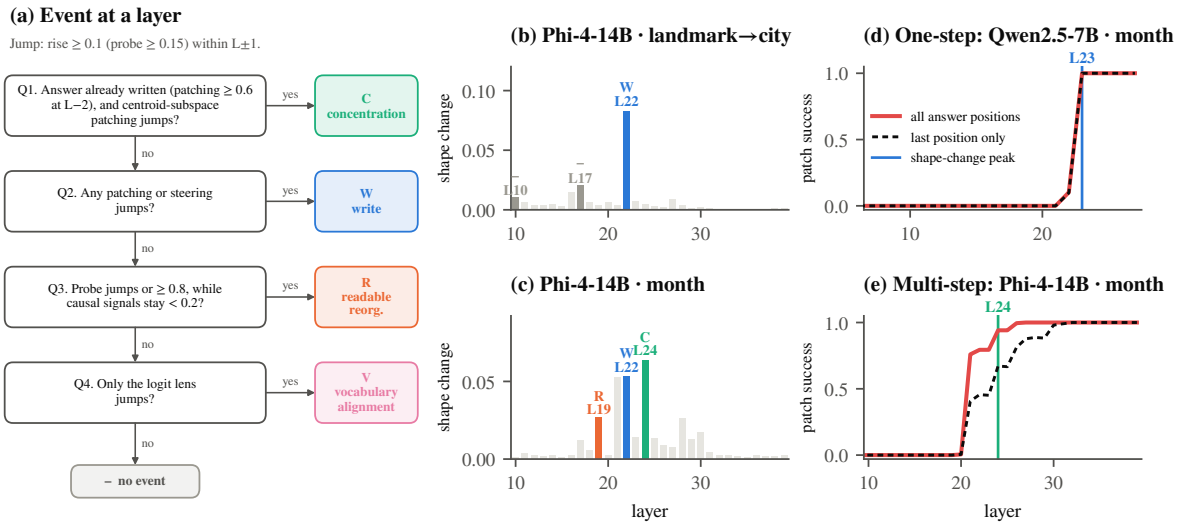


Figure 38: Events at peaks for Phi-4 (counterpart of Figure 3; (a) is the decision tree and (d) the Qwen2.5-7B reference). (b) Landmark → city: write at layer 22; the earlier peaks (10, 17) have no event. (c) Month: readable reorganization at layer 19, write at layer 22, concentration at the highest peak (layer 24). (e) In Phi-4, patching the last position alone takes effect three layers after patching all answer positions on months (layers 24 and 21).

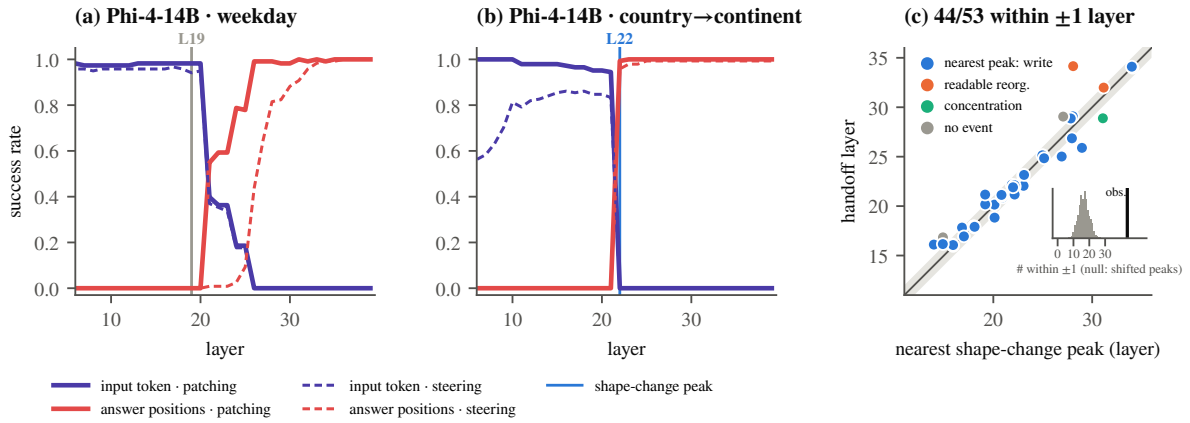


Figure 39: Handoff for Phi-4 (counterpart of Figure 4). (a) Weekday: the handoff (layer 21) falls on a write peak, two layers after the highest peak (layer 19, no event). (b) Country → continent: handoff and highest peak (write) at layer 22, after readable reorganizations at layers 16 and 18.

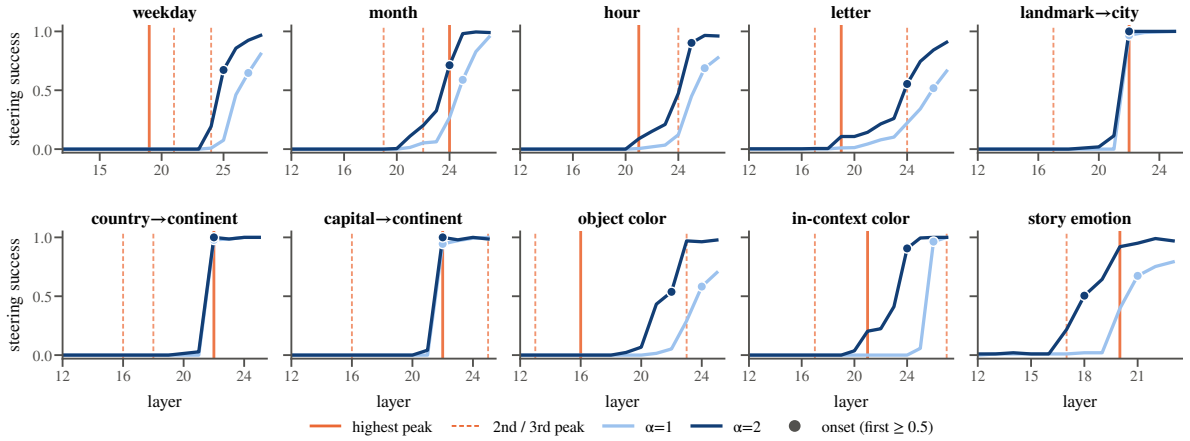


Figure 40: Steering onset for Phi-4, all tasks (as Figure 5). The onset at $\alpha = 2$ lies within one layer of a peak in 9 of 10 tasks; in-context color is three layers from one.

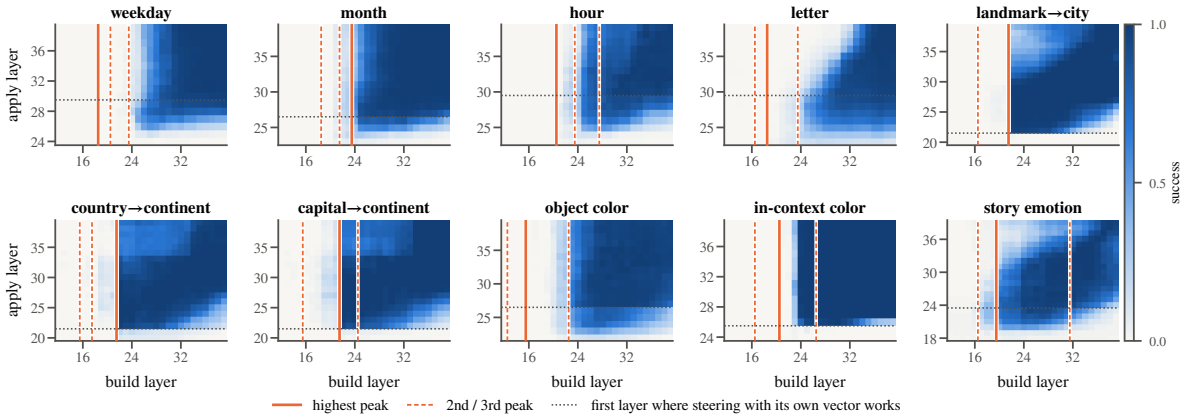


Figure 41: Transfer for Phi-4, all tasks (as Figure 6). The largest one-layer rise in transfer lies within one layer of a peak in 7 of 10 tasks.

	Wkday	Month	Hour
Highest peak	17	18	16
Handoff (start)	18	18	16
Handoff (offset)	15	–	15

Table 9: Shape-change peak and handoff layers in Llama-3.1-8B on cyclic arithmetic. Feucht et al. (2026): offset copied ≈15, sum at 18. Arora et al. (2026): answer-organized term at 17–18. The offset handoff is from the input decomposition above (not run for months).

Model	Task	Top-3 peaks	α=1	α=2	α=4	α=8	c=0.4	c=0.8
Qwen2.5-7B	Weekday	12–, 16–, **23**W	23	23	23	22	22	20
Llama-3.1-8B	Weekday	13–, **17**W, 22C	21	18	18	16	19	16
Mistral-7B	Weekday	17–, **20**W, 25C	21	20	18	17	18	15
Gemma-2-9B	Weekday	19–, 28W, **34**C	33	31	29	28	31	27
Qwen2.5-14B	Weekday	32R, **37**W, 40V	37	37	36	34	36	31
Phi-4	Weekday	**19**–, 21W, 24W	27	25	24	22	25	22
Qwen2.5-7B	Month	14–, **23**W, 25V	23	23	22	22	22	20
Llama-3.1-8B	Month	16–, **18**W, 21C	20	18	18	17	19	17
Mistral-7B	Month	17–, **20**W, 25W	21	19	17	17	18	16
Gemma-2-9B	Month	31C, **33**C, 36W	32	29	28	26	29	25
Qwen2.5-14B	Month	**37**W, 40W, 42–	38	37	36	33	35	32
Phi-4	Month	19R, 22W, **24**C	25	24	21	20	24	20
Qwen2.5-7B	Hour	7–, 19–, **23**W	23	23	23	22	23	21
Llama-3.1-8B	Hour	14–, **16**W, 24V	18	18	18	16	18	17
Gemma-2-9B	Hour	28W, 31W, **33**W	33	31	29	28	32	27
Qwen2.5-14B	Hour	32R, **37**W, 40C	38	37	36	35	36	33
Phi-4	Hour	**21**W, 24C, 28W	26	25	23	21	25	23
Qwen2.5-7B	Letter	14–, **23**W, 27V	23	22	22	21	21	20
Llama-3.1-8B	Letter	16–, **18**W, 21W	18	18	16	14	18	15
Mistral-7B	Letter	**19**W, 28W, 31W	20	19	17	16	19	16
Gemma-2-9B	Letter	16–, **27**–, 33W	30	29	27	26	28	26
Qwen2.5-14B	Letter	32–, **37**W, 40W	37	36	35	33	35	31
Phi-4	Letter	17–, **19**W, 24C	25	23	21	19	23	20
Qwen2.5-7B	Landmark → city	**19**R, 21W, 23W	22	21	20	19	21	19
Llama-3.1-8B	Landmark → city	**14**R, 16W, 20W	17	16	14	14	16	14
Mistral-7B	Landmark → city	12–, **15**R, 18W	18	18	16	15	18	15
Gemma-2-9B	Landmark → city	20R, **25**W, 29C	28	25	25	20	25	21
Qwen2.5-14B	Landmark → city	23–, 31R, **34**W	35	34	31	28	32	28
Phi-4	Landmark → city	10–, 17–, **22**W	22	22	20	18	22	18
Qwen2.5-7B	Country → cont.	16R, 18R, **22**W	22	21	20	19	21	18
Llama-3.1-8B	Country → cont.	**14**R, 17W, 20W	17	16	14	14	16	14
Mistral-7B	Country → cont.	14R, **18**W, 31–	19	18	18	15	18	15
Gemma-2-9B	Country → cont.	19R, **25**W, 32V	25	25	23	21	25	21
Qwen2.5-14B	Country → cont.	26R, 32R, **34**W	35	32	31	29	32	30
Phi-4	Country → cont.	16R, 18R, **22**W	22	22	20	18	22	19
Qwen2.5-7B	Capital → cont.	**16**R, 19R, 22W	22	21	21	19	21	19
Llama-3.1-8B	Capital → cont.	14R, **16**W, 20V	17	16	14	14	16	14
Mistral-7B	Capital → cont.	15R, **18**W, 31–	18	18	18	16	18	15
Gemma-2-9B	Capital → cont.	19R, **25**W, 32V	25	25	25	22	25	21
Qwen2.5-14B	Capital → cont.	32W, **34**W, 36C	34	32	32	30	32	29
Phi-4	Capital → cont.	16R, **22**W, 25–	22	22	19	17	20	17
Qwen2.5-7B	Object → color	19R, 21W, **24**C	22	21	21	19	21	20
Llama-3.1-8B	Object → color	12–, **14**W, 22C	20	16	14	13	19	15
Mistral-7B	Object → color	**15**–, 19W, 23W	19	18	16	15	18	16
Gemma-2-9B	Object → color	**29**W, 33W, 41–	29	26	25	23	28	24
Qwen2.5-14B	Object → color	23–, **41**V, 43V	35	33	31	29	33	29
Phi-4	Object → color	13–, **16**–, 23W	24	21	19	18	24	19
Qwen2.5-7B	In-context color	14–, **19**W, 21W	23	22	21	19	22	20
Llama-3.1-8B	In-context color	**12**R, 14W, 16W	17	16	15	14	17	15
Mistral-7B	In-context color	13R, 15W, **19**W	19	18	16	15	18	16
Gemma-2-9B	In-context color	**13**–, 18R, 28W	28	27	25	24	27	25
Qwen2.5-14B	In-context color	23R, 26–, **28**R	36	34	32	29	33	30
Phi-4	In-context color	17R, **21**R, 27W	26	24	21	18	24	20
Qwen2.5-7B	Story → emotion	13R, 18W, **22**C	19	18	18	16	19	16
Llama-3.1-8B	Story → emotion	**13**R, 18C, 20W	16	14	13	13	17	14
Mistral-7B	Story → emotion	**13**R, 15W, 20W	19	16	15	14	20	15
Gemma-2-9B	Story → emotion	**20**R, 31W, 34W	25	22	21	21	25	21
Qwen2.5-14B	Story → emotion	23R, **28**R, 40–	31	30	28	24	30	25
Phi-4	Story → emotion	17R, **20**W, 32–	21	18	17	16	23	18

Table 10: Steering onset (first layer with success ≥ 0.5; earliest over three directions) at each strength. Top-3 peaks: layer and event (W: write; R: readable reorganization; C: concentration; V: vocabulary alignment; –: no event), highest peak in bold. A dash in a strength column means no layer reaches 0.5.

Highest peak	n	α=1	α=8	c=0.8
Write	36	+0.5 (0)	-3 (26)	-3 (29)
Readable reorg.	12	+4 (0)	+0 (2)	+0.5 (1)
Concentration	5	-1 (2)	-6 (5)	-6 (5)
Vocab. alignment	1	-6 (1)	-12 (1)	-12 (1)
No event	5	+8 (0)	+2 (0)	+3 (0)

Table 11: Onset minus highest peak (median), and in parentheses the number of pairs whose onset lies more than one layer before the highest peak.

Target	n	**Highest peak**	**Earliest peak**	$A_{\text{lin}} > 0.1$	$\arg\max A_{\text{lin}}$
Onset α=1	59	2.5 / 0.53 / 3	6.5 / 0.07 / 0	2.1 / 0.51 / 17	9.2 / 0.03 / 57
Onset α=2	59	2.0 / 0.47 / 10	5.3 / 0.08 / 2	2.4 / 0.53 / 25	10.5 / 0.00 / 59
Onset α=8	59	3.0 / 0.34 / 34	3.1 / 0.41 / 6	4.8 / 0.12 / 52	13.4 / 0.00 / 59
Onset c=0.4	59	2.5 / 0.37 / 14	5.5 / 0.08 / 2	2.3 / 0.47 / 27	10.4 / 0.00 / 59
Onset c=0.8	59	3.4 / 0.25 / 36	3.3 / 0.36 / 6	4.8 / 0.03 / 57	13.4 / 0.00 / 59
Transfer boundary	58	3.2 / 0.41 / 5	7.1 / 0.07 / 0	1.6 / 0.57 / 11	8.6 / 0.00 / 58
Handoff	53	2.0 / 0.53 / 9	5.3 / 0.08 / 2	2.2 / 0.53 / 21	10.3 / 0.00 / 53
Stricter logit-lens variant (class probability against all other tokens):					
Onset α=1	57	2.5 / 0.53 / 3	6.6 / 0.05 / 0	2.3 / 0.51 / 20	9.4 / 0.04 / 55
Onset α=2	57	2.0 / 0.49 / 9	5.4 / 0.09 / 1	3.0 / 0.40 / 32	10.6 / 0.00 / 57
Onset α=8	57	2.9 / 0.33 / 33	3.0 / 0.40 / 5	5.6 / 0.09 / 52	13.5 / 0.00 / 57
Onset c=0.4	57	2.5 / 0.37 / 14	5.6 / 0.07 / 2	2.9 / 0.35 / 34	10.5 / 0.00 / 57
Onset c=0.8	57	3.4 / 0.25 / 35	3.3 / 0.35 / 5	5.6 / 0.04 / 55	13.5 / 0.00 / 57
Transfer boundary	57	3.2 / 0.40 / 5	7.2 / 0.05 / 0	1.8 / 0.61 / 15	8.8 / 0.00 / 57
Handoff	52	2.0 / 0.54 / 8	5.3 / 0.08 / 1	2.8 / 0.40 / 30	10.4 / 0.00 / 52

Table 12: Shape-change peaks versus logit-lens accessibility as predictors of seven boundaries. Each cell: mean absolute error (layers) / fraction within one layer / number of violations of a lower bound. Top: A_{lin} as defined by Billa (2026); bottom: a stricter variant. The shape-peak columns are identical in both blocks.

	Absolute error			**Violations**		
Target	**Peak**	A_{lin}	p	**Peak only**	A_{lin} **only**	p
Onset α=1	2.49	2.14	0.542	1	15	0.001
Onset α=2	2.02	2.37	0.427	6	21	0.006
Onset α=8	2.97	4.85	0.001	3	21	<0.001
Onset c=0.4	2.51	2.25	0.342	8	21	0.024
Onset c=0.8	3.37	4.81	0.006	1	22	<0.001
Transfer boundary	3.17	1.62	0.004	4	10	0.180
Handoff	2.00	2.17	0.648	4	16	0.012

Table 13: Paired comparison of the highest shape-change peak and the first layer with $A_{\text{lin}} > 0.1$. Absolute error: mean, with Wilcoxon p. Violations: pairs violating only one of the two lower bounds, with exact McNemar p.

Task	Model	Predictors				Targets			
		λ_1	$\lambda_{\min}$	$A_{\text{lin}} > 0.1$	$\arg\max A_{\text{lin}}$	Onset α=1	Onset α=8	Transfer	Handoff
Weekday	Qwen	23	12	23	27 (0.98)	23	22	23	23
	Llama	17	13	18	31 (0.85)	21	16	22	18
	Mistral	20	17	19	30 (0.67)	21	17	25	20
	Gemma	34	19	28	41 (0.91)	33	28	32	29
	Qwen2.5-14B	37	32	36	44 (0.96)	37	34	37	36
	Phi-4	19	19	23	35 (0.95)	27	22	27	21
Month	Qwen	23	14	23	26 (1.00)	23	22	23	23
	Llama	18	16	19	30 (0.98)	20	17	21	18
	Mistral	20	17	18	31 (0.88)	21	17	23	19
	Gemma	33	31	30	36 (1.00)	32	26	32	29
	Qwen2.5-14B	37	37	34	47 (1.00)	38	33	37	36
	Phi-4	24	19	23	31 (0.98)	25	20	25	21
Hour	Qwen	23	7	23	26 (0.68)	23	22	23	22
	Llama	16	14	22	31 (0.96)	18	16	21	16
	Gemma	33	28	34	36 (0.73)	33	28	33	29
	Qwen2.5-14B	37	32	39	45 (0.69)	38	35	37	36
	Phi-4	21	21	28	38 (0.95)	26	21	25	21
Letter	Qwen	23	14	24	27 (0.80)	23	21	24	22
	Llama	18	16	24	30 (0.48)	18	14	26	18
	Mistral	19	19	27	31 (0.42)	20	16	27	20
	Gemma	27	16	34	41 (0.71)	30	26	35	29
	Qwen2.5-14B	37	32	39	47 (0.89)	37	33	40	36
	Phi-4	19	17	31	39 (0.84)	25	19	31	21
Landmark → city	Qwen	19	19	23	27 (0.97)	22	19	22	21
	Llama	14	14	17	30 (1.00)	17	14	17	16
	Mistral	15	12	18	29 (0.99)	18	15	18	18
	Gemma	25	20	30	41 (0.99)	28	20	25	25
	Qwen2.5-14B	34	23	34	40 (1.00)	35	28	34	32
	Phi-4	22	10	22	28 (0.98)	22	18	22	22
Country → continent	Qwen	22	16	22	27 (0.99)	22	19	22	22
	Llama	14	14	17	29 (0.98)	17	14	17	17
	Mistral	18	14	18	30 (0.96)	19	15	18	18
	Gemma	25	19	30	40 (1.00)	25	21	25	25
	Qwen2.5-14B	34	26	32	45 (1.00)	35	29	32	34
	Phi-4	22	16	22	38 (1.00)	22	18	22	22
Capital → continent	Qwen	16	16	22	27 (1.00)	22	19	22	22
	Llama	16	14	16	29 (0.98)	17	14	17	16
	Mistral	18	15	18	30 (0.97)	18	16	18	18
	Gemma	25	19	30	41 (1.00)	25	22	25	25
	Qwen2.5-14B	34	32	32	45 (1.00)	34	30	32	34
	Phi-4	22	16	22	39 (1.00)	22	17	22	22
Object → color	Qwen	24	19	22	27 (0.93)	22	19	22	21
	Llama	14	12	21	31 (0.85)	20	13	20	16
	Mistral	15	15	18	31 (0.97)	19	15	21	17
	Gemma	29	29	27	37 (1.00)	29	23	29	26
	Qwen2.5-14B	41	23	38	47 (0.97)	35	29	34	34
	Phi-4	16	13	21	39 (0.94)	24	18	23	23
In-context color	Qwen	19	14	22	24 (1.00)	23	19	22	21
	Llama	12	12	19	23 (1.00)	17	14	17	16
	Mistral	19	13	20	24 (1.00)	19	15	19	16
	Gemma	13	13	28	31 (1.00)	28	24	28	27
	Qwen2.5-14B	28	23	35	39 (1.00)	36	29	35	34
	Phi-4	21	17	26	27 (1.00)	26	18	24	25
Story → emotion	Qwen	22	13	23	27 (0.79)	19	16	21	–
	Llama	13	13	20	31 (0.60)	16	13	20	–
	Mistral	13	13	18	31 (0.53)	19	14	20	–
	Gemma	20	20	29	35 (0.83)	25	21	–	–
	Qwen2.5-14B	28	23	41	47 (0.80)	31	24	33	–
	Phi-4	20	17	27	39 (0.74)	21	16	23	–

Table 14: Per-pair predictors and targets (layers). λ_1: highest peak; $\lambda_{\min}$: earliest of the three highest peaks; $\arg\max A_{\text{lin}}$ is followed by the maximal value. Transfer: first build layer whose vectors transfer with success ≥ 0.5. –: undefined (no transfer data for Gemma on emotions; no single input token for emotion stories). Mistral on hours is excluded.

Task	Model	Highest peak	Before	After	At peak	To 0.8	Largest jump	Last layer
Weekday	Qwen	23 W	0.04	0.92	0.87	+0	23 W	0.96
	Llama	17 W	0.01	0.67	0.03	+6	22 C	0.93
	Mistral	20 W	0.07	0.63	0.12	+7	25 C	0.96
	Gemma	34 C	0.55	0.96	0.86	+0	32†	1.00
	Qwen2.5-14B	37 W	0.10	0.92	0.79	+1	36†	0.96
	Phi-4	19 –	0.00	0.59	0.00	+10	26†	1.00
Month	Qwen	23 W	0.01	0.94	0.86	+0	23 W	0.96
	Llama	18 W	0.01	0.75	0.18	+4	21 C	0.92
	Mistral	20 W	0.13	0.77	0.38	+5	25 W	0.97
	Gemma	33 C	0.45	0.94	0.81	+0	32†	1.00
	Qwen2.5-14B	37 W	0.16	0.92	0.75	+1	37 W	0.93
	Phi-4	24 C	0.08	0.91	0.43	+2	24 C	0.98
Hour	Qwen	23 W	0.01	0.87	0.76	+2	23 W	0.96
	Llama	16 W	0.00	0.69	0.01	+7	18†	0.86
	Gemma	33 W	0.09	0.94	0.78	+1	33 W	0.98
	Qwen2.5-14B	37 W	0.02	0.85	0.58	+3	37 W	0.97
	Phi-4	21 W	0.00	0.71	0.00	+8	25†	0.98
Letter	Qwen	23 W	0.01	0.76	0.48	+2	23 W	0.94
	Llama	18 W	0.01	0.50	0.07	+10	28†	0.90
	Mistral	19 W	0.00	0.47	0.01	+9	28 W	0.99
	Gemma	27 –	0.00	0.49	0.00	+10	36†	0.99
	Qwen2.5-14B	37 W	0.02	0.68	0.20	+6	40 W	0.97
	Phi-4	19 W	0.00	0.44	0.00	+14	31†	0.98
Landmark → city	Qwen	19 R	0.00	0.60	0.07	+5	21 W	0.91
	Llama	14 R	0.00	0.72	0.15	+8	16 W	0.75
	Mistral	15 R	0.00	0.71	0.12	+9	18 W	0.93
	Gemma	25 W	0.01	0.90	0.58	+4	25 W	0.94
	Qwen2.5-14B	34 W	0.17	0.86	0.67	+6	34 W	0.95
	Phi-4	22 W	0.01	0.83	0.71	+5	22 W	0.76
Country → continent	Qwen	22 W	0.05	0.83	0.71	+3	22 W	0.89
	Llama	14 R	0.00	0.71	0.24	+6	17 W	0.73
	Mistral	18 W	0.03	0.79	0.57	+7	18 W	0.98
	Gemma	25 W	0.02	0.86	0.54	+5	25 W	0.94
	Qwen2.5-14B	34 W	0.29	0.85	0.73	+1	32 R	0.91
	Phi-4	22 W	0.04	0.86	0.86	+0	22 W	0.79
Capital → continent	Qwen	16 R	0.00	0.46	0.00	+7	22 W	0.89
	Llama	16 W	0.13	0.77	0.33	+8	17†	0.86
	Mistral	18 W	0.09	0.81	0.67	+7	18 W	0.91
	Gemma	25 W	0.02	0.85	0.58	+5	25 W	0.90
	Qwen2.5-14B	34 W	0.34	0.87	0.79	+1	32 W	0.91
	Phi-4	22 W	0.07	0.87	0.89	+0	22 W	0.81
Object → color	Qwen	24 C	0.34	0.94	0.92	+0	23†	0.97
	Llama	14 W	0.00	0.68	0.03	+7	21†	0.91
	Mistral	15 –	0.00	0.62	0.03	+10	18†	1.00
	Gemma	29 W	0.16	0.88	0.63	+4	29 W	0.96
	Qwen2.5-14B	41 V	0.87	0.97	0.97	+0	33†	0.98
	Phi-4	16 –	0.00	0.67	0.00	+9	23 W	0.92
In-context color	Qwen	19 W	0.00	0.66	0.00	+3	22†	0.96
	Llama	12 R	0.00	0.68	0.00	+9	17†	0.89
	Mistral	19 W	0.00	0.92	0.77	+2	19 W	0.98
	Gemma	13 –	0.00	0.46	0.00	+15	28 W	0.85
	Qwen2.5-14B	28 R	0.00	0.65	0.00	+7	35†	0.99
	Phi-4	21 R	0.00	0.84	0.00	+3	24†	0.95
Story → emotion	Qwen	22 C	0.25	0.82	0.72	+2	22 C	0.83
	Llama	13 R	0.00	0.51	0.00	–	18 C	0.64
	Mistral	13 R	0.01	0.54	0.03	+17	20 W	0.85
	Qwen2.5-14B	28 R	0.00	0.62	0.07	+12	31†	0.72
	Phi-4	20 W	0.05	0.75	0.37	+8	20 W	0.75

Table 15: Cross-layer transfer per pair. Highest peak: layer and event (W: write; R: readable reorganization; C: concentration; V: vocabulary alignment; –: no event). Before / After: mean transfer of vectors built in the five layers before the highest peak / at or after it. At peak and Last layer: transfer of vectors built there. To 0.8: layers after the peak at which transfer first reaches 0.8 (–: never). Largest jump: layer of the largest one-layer increase in transfer, with the event there; †: not one of the three highest peaks. Mistral on hours and Gemma on emotions are excluded.

	Before peak		At / after peak	
Reliability	**Transfer**	**Layers**	**Transfer**	**Layers**
< 0.3	0.01	370	0.07	19
0.3–0.5	0.01	188	0.07	14
0.5–0.7	0.04	88	0.13	19
0.7–0.9	0.04	61	0.32	75
≥ 0.9	0.20	81	0.82	699

Table 16: Mean transfer of vectors built at layers binned by the split-half reliability of the answer-class structure at that layer.

Candidate	**Before**	**After**	**Error**	**Bias**
Highest shape peak	0.08	0.75	3.2	-2.6
$A_{\text{lin}} > 0.1$	0.19	0.85	1.6	+0.0
Logit-lens probability ≥ 0.1	0.22	0.86	1.6	+0.5
Probe ≥ 0.5	0.02	0.61	7.6	-7.6
Reliability ≥ 0.5	0.03	0.64	6.4	-6.4

Table 17: Candidate markers of the transfer boundary (58 pairs). Before / After: mean transfer of vectors built in the five layers before the candidate / at or after it. Error: mean absolute distance to the boundary (layers); Bias: mean signed distance (negative: earlier). A_{lin} follows Billa (2026); the logit-lens probability of the answer is the signal of Section 3. Probe and reliability are restricted to the late layers.

Model	Task	Peak	Top-3	Cross.	Blur (pre)	Answer (post)	R^2 (pre)
Gemma-2-9B	hour	33	28, 31, 33	33	0.71	0.71	0.90
Gemma-2-9B	letter	27	16, 27, 33	–	0.95	0.14	0.87
Gemma-2-9B	month	33	31, 33, 36	32	0.43	0.68	0.87
Gemma-2-9B	weekday	34	19, 28, 34	21	0.12	0.91	0.83
Llama-3.1-8B	hour	16	14, 16, 24	17	0.88	0.66	0.88
Llama-3.1-8B	letter	18	16, 18, 21	–	0.89	0.30	0.87
Llama-3.1-8B	weekday	17	13, 17, 22	15	0.46	0.88	0.89
Llama-3.2-3B	month	16	10, 16, 19	16	0.85	0.59	0.88
Llama-3.2-3B	weekday	16	10, 16, 21	12	0.41	0.87	0.88
Mistral-7B	letter	19	19, 28, 31	–	0.98	0.09	0.89
Mistral-7B	month	20	17, 20, 25	24	0.76	0.36	0.91
Mistral-7B	weekday	20	17, 20, 25	19	0.55	0.72	0.92
Qwen2.5-1.5B	month	22	15, 22, 24	23	0.92	0.52	0.92
Qwen2.5-3B	month	28	21, 28, 33	29	0.88	0.55	0.88
Qwen2.5-3B	weekday	28	21, 24, 28	28	0.52	0.86	0.85
Qwen2.5-7B	hour	23	7, 19, 23	23	0.86	0.72	0.85
Qwen2.5-7B	letter	23	14, 23, 27	–	0.96	0.46	0.87
Qwen2.5-7B	month	23	14, 23, 25	23	0.75	0.77	0.86
Qwen2.5-7B	weekday	23	12, 16, 23	18	0.30	0.90	0.85
Qwen3-4B	month	24	21, 24, 29	24	0.87	0.54	0.88
Qwen3-4B	weekday	24	21, 24, 30	24	0.51	0.83	0.89
Phi-4	hour	21	21, 24, 28	22	0.92	0.69	0.90
Phi-4	letter	19	17, 19, 24	30	0.96	0.11	0.91
Phi-4	month	24	19, 22, 24	24	0.68	0.59	0.88
Phi-4	weekday	19	19, 21, 24	18	0.55	0.69	0.92
Qwen2.5-14B	hour	37	32, 37, 40	37	0.82	0.66	0.87
Qwen2.5-14B	letter	37	32, 37, 40	38	0.89	0.49	0.88
Qwen2.5-14B	month	37	37, 40, 42	37	0.68	0.64	0.86
Qwen2.5-14B	weekday	37	32, 37, 40	27	0.27	0.88	0.88

Table 18: Blurred-input model on cyclic-arithmetic pairs. Peak: highest shape-change peak; Cross.: crossover layer. Shares are fractions of the answer-centroid energy, averaged over the five layers before (pre) or after (post) the highest peak. Additive R^2 is the variance of the activations explained by the additive model in the five layers before the highest peak. Gemma-2-2B on months has no peak and is omitted; for Llama-3.1-8B on months the activations were not stored. Mistral-7B on hours is excluded throughout.

Section	Result	All pairs	Without the four pairs
Sec. 3	Highest peak is a write vs. non-peak layers	0.61 vs 0.22 (n=59)	0.60 vs 0.22 (n=55)
	Highest peaks that are vocabulary alignment only	1	1
	Readable-reorganization peaks with logit lens < 0.001	35/38	34/37
Sec. 4	Handoff within at most one layer (patching gap ≤ 1)	46/53	43/50
	Handoff within ± 1 of a top-3 peak	44/53 (null 16.7, p<0.001)	41/50 (null 15.7, p<0.001)
	Write highest peaks at the handoff (± 1)	28/35	26/32
Sec. 5	Onset at α=1 no earlier than highest peak $- 1$	56/59	52/55
	Onset within ± 1 of a top-3 peak (α=1 / α=8 / c=0.8)	0.81 / 0.71 / 0.66	0.80 / 0.73 / 0.65
	Onsets > 1 layer before the earliest peak (α=1 / α=8 / c=0.8)	0 / 6 / 6 of 59	0 / 4 / 5 of 55
	Lower-bound violations at α=1: highest peak vs $A_{\text{lin}} > 0.1$	3 vs 17 (p=0.001)	3 vs 15 (p=0.002)
Sec. 6	Transfer before / at-or-after the highest peak	0.08 / 0.75 (n=58)	0.09 / 0.76 (n=54)
	Largest transfer jump exactly on a top-3 peak	39/58	36/54
	Transfer decreases with cumulative shape change at fixed distance	54/58 (median $\rho = -0.50$)	50/54 (median $\rho = -0.48$)

Table 19: Main results with all pairs and without the four pairs whose accuracy is below 0.55. Pair counts differ across rows because each analysis has its own exclusions (Sections 4 and 6).